Lecture Notes of MAT2040 Linear Algebra

Fall 2019

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Preface

Fall 2019

In this semester, I teach this course with Prof. Simon Pun. We plan to accelerate the pace and fit in a little bit more content and add certain programming projects. We further refer to [1, 4].

Spring 2019

In this semester, I teach the course with Dr. Guang (Grant) Rao and Prof, Tzongyi Lee. We further refer to [2].

Fall 2018

This is the set of notes of my lectures on a linear algebra courses in Autumn 2018 (MAT2040). I use some content from the slides of Prof. Dongdong He, Dr. Guang (Grant) Rao, and Prof. Frans Schalekamp, and the textbook [3]. I also refer to Mr. Jie Wang's notes of Prof. Tom Luo and Ruoyu Sun's lecture [5].

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Part I

Systems of Linear Equations and Matrices

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Lecture 1

Linear Systems

1.1 Systems of Linear Equations

Examples of Linear and Nonlinear Equations

- linear equations:
  1. $x + 3y - 2z = 8$
  2. $4x_1 + 5x_2 - x_3 + x_4 + x_5 = 0$
  3. $9a - 2b + 7c + 2d = -7$
- nonlinear equations:
  1. $xy + 5yz = 13$
  2. $x_1 + x_2^3 - x_3 x_4 x_5^2 = 0$
  3. $\tan(a) + \log(b) - cd = 0$

Example: Input-Output Analysis

- A society's economy has three industries: Wood, Rice and Labor.
- Each industry uses the inputs from the other two:
  - 1 unit of Wood needs $\frac{1}{2}$ units of Rice and $\frac{1}{4}$ units of Labor.
  - 1 unit of Rice needs $\frac{1}{2}$ units of Wood and $\frac{1}{2}$ units of Labor.
  - 1 unit of Labor needs $\frac{1}{8}$ units of Wood and $\frac{1}{4}$ units of Rice.
- The outputs of each industry are consumed either as inputs to other industry or as the final demand.
  - Suppose that the final demands of the three industries are 0, 1 and 2, respectively.
- Let x_1, x_2, x_3 be the output of Wood, Rice and Labor industries, respectively.

We have the following three equations:

$$\begin{aligned} x_1 &= \frac{1}{2}x_2 + \frac{1}{8}x_3 \\ x_2 &= \frac{1}{2}x_1 + \frac{1}{4}x_3 + 1 \\ x_3 &= \frac{1}{4}x_1 + \frac{1}{2}x_2 + 2 \end{aligned} \tag{1.1}$$

The LHS of each equation is the input of an industry and the RHS is the consumption. We will use this example throughout this lecture. Harvard Professor Wassily

Leontief first developed such a model for analysis national/regional economies, and earned the 1973 Nobel Prize in Economics.

System of Linear Equations

A linear system of m equations in n unknowns (also called an $m \times n$ system):

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

A solution of the system is an assignment of $(x_1, x_2, \ldots, x_n)$ that satisfies all the equations.

The system of linear equations (1.1) has three unknowns and three equations, i.e., $m = 3$ and $n = 3$, and can be written in the form:

$$\begin{aligned} x_1 - \frac{1}{2}x_2 - \frac{1}{8}x_3 &= 0 \\ -\frac{1}{2}x_1 + x_2 - \frac{1}{4}x_3 &= 1 \\ -\frac{1}{4}x_1 - \frac{1}{2}x_2 + x_3 &= 2 \end{aligned} \tag{1.2}$$

1.2 Solution Set

We are interested in the following questions about a system of linear equations:

- Does a solution exists? A linear system with a solution is called *consistent*.
- How many solutions does a linear system have?
- How to find the solutions?

Solution Set

- The set of all solutions of a linear system is called the *solution set* of the system.
- The solution set can have 0, 1 or infinite elements (to be proved the next lecture).

1×1 System

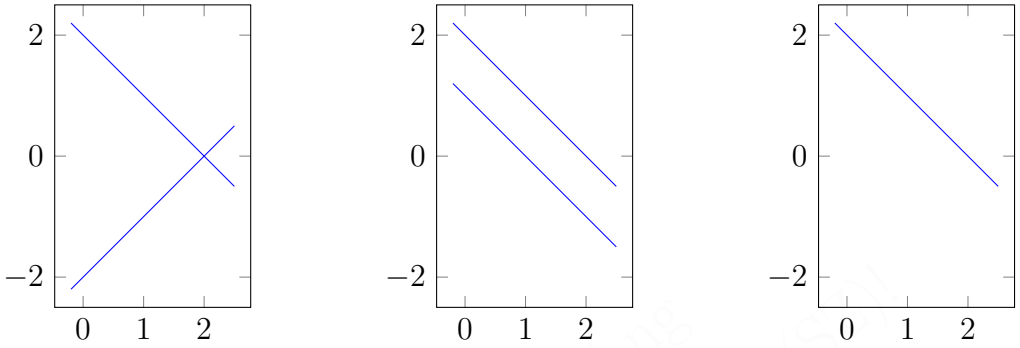
$$ax = b$$

Thought considered trivial by many, it has some properties of a general one:

- $a \neq 0$: exactly one solution $x = b/a$.
- $a = 0, b \neq 0$: no solution.
- $a = 0, b = 0$: infinitely many solutions.

The solution set of these three cases can be denoted as $\{b/a\}$, $\emptyset$ and $\mathbb{R}$, respective.

$$x_1 + x_2 = 2 \qquad x_1 + x_2 = 2 \qquad x_1 + x_2 = 2$$
$$x_1 - x_2 = 2 \qquad x_1 + x_2 = 1 \qquad -x_1 - x_2 = -2$$



(a) exactly one solution (b) no solution (c) many solutions

■ **Example 1.1** Consider the following three 2×2 systems:

The solution sets of these three examples are $\{(2, 0)\}$, $\emptyset$, $\{(x, 2 - x) : x \in \mathbb{R}\}$, respectively. ■

How do we solve the first 2×2 system in the above example? Your approach can be summerized as two "equation operations":

- Add to one equation a multiple of another one.
- Multiply a nonzero number on both sides of a equation.

Now let's see how to solve the 3×3 system (1.2).

An Example of Solving a 3×3 System

- First, from the third equation

$$x_1 = \frac{1}{2}x_2 + \frac{1}{8}x_3. \tag{1.3}$$

- Substituting this formula into the other two equations, we obtain (after combining the terms of the same variable)

$$\begin{aligned} \frac{3}{4}x_2 - \frac{5}{16}x_3 &= 1 \\ -\frac{5}{8}x_2 + \frac{31}{32}x_3 &= 2 \end{aligned}$$

  which is a 2×2 system.
- We see that $x_2 = 3$ and $x_3 = 4$ solves the above system. Substituting this solution to the expression of x_1 in (1.3), we get $x_1 = 2$.

The substitution of (1.3) into two other equations (called forward substitution) can be formulated as adding multiples of the first equation to the other two equations. The substitution of $x_2 = 3$ and $x_3 = 4$ back to (1.3) is called backward substitution

A general idea to solve an $n \times n$ system is to first reduce it to a $(n-1) \times (n-1)$ system by forward substitution, and then back substituting the solution of the $(n-1) \times (n-1)$ system to solve the remaining variable, which forms a recursion.

Back to 1949, Leontief had a 500×500 system, which cannot be solved by any computers at that time, so he managed to reduce the system to a 42×42 one, which takes the largest computer at that time 56 hours to solve.

1.3 Equivalent Systems and Equation Operations

When solving the examples of the linear systems in the last section, we used some equation operations. Let's formally define these operations and justify the completeness of our solutions.

Equivalent Systems

Definition 1.1 Two systems of linear equations involving the same variables are said to be equivalent if they have the same solution set.

Definition 1.2 Three equation operations for a linear system:

I. Swap the order of two equations.
II. Multiply by the same nonzero real number on both sides of an equation.
III. Add a multiple of one equation to another.

Proposition 1.1 If we apply one of the three equation operations to a system of linear equations, then the original system and the transformed one are equivalent.

R Not that we do not use the first equation operation in the previous examples. Show that equation operation I can be obtained by combining equation operation II and equation operation III. For the convenience of discussing Gaussian Elimination, we add this operation here.

Proof. Let S and S' be the solution sets of the original system and the transformed system, respectively.

- When Operation I is applied, the set of equations are the same for both systems, and hence $S = S'$.
- When Operation II is applied, suppose that the first equation is multiplied by $\alpha \neq 0$ on both sides. (If it is not the first equation, we can apply Operation I to change it to the first.) So the original system and the transformed system are different only on the ith equation: The first equation of the original system is

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

and the ith equation of the transformed system is

$$\alpha a_{11}x_1 + \alpha a_{12}x_2 + \cdots + \alpha a_{1n}x_n = \alpha b_1.$$

For $(\beta_1, \ldots, \beta_n) \in S$, we can verify that $(\beta_1, \ldots, \beta_n)$ satisfies all the 2nd to the last equations in the transformed system as they are the same as the ones in the original system. As $(\beta_1, \ldots, \beta_n)$ satisfies the first equation in the original system, we know that $\sum_{j=1}^{n} a_{ij}\beta_j = b_i$, which implies $\alpha \sum_{j=1}^{n} a_{ij}\beta_j = \alpha b_i$,

i.e., $(\beta_1, \ldots, \beta_n)$ also satisfies the first equation in the transformed system. Therefore, $(\beta_1, \ldots, \beta_n) \in S'$.
For $(\beta_1, \ldots, \beta_n) \in S'$, we can verify that $(\beta_1, \ldots, \beta_n)$ satisfies all the 2nd to the last equations in the original system as they are the same as the ones in the transformed system. As $(\beta_1, \ldots, \beta_n)$ satisfies the first equation in the transformed system, we know that $\sum_{j=1}^{n} \alpha a_{ij}\beta_j = \alpha b_i$, which implies $\sum_{j=1}^{n} a_{ij}\beta_j = b_i$ by multiplying $1/\alpha$ on both sides (which is feasible as $\alpha \neq 0$, i.e., $(\beta_1, \ldots, \beta_n)$ also satisfies the first equation in the original system. Therefore, $(\beta_1, \ldots, \beta_n) \in S$.

- (The proof of operation III is left as an exercise.)

■

1.4 Augmented Matrix and Row Operations

Matrices

- A *matrix* is a rectangular array of numbers.
- For an $m \times n$ system of linear equations, the $m \times n$ matrix

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & & & \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix}$$

  is called the *coefficient matrix* of the system.
- For an $m \times n$ system of linear equations, the $m \times (n+1)$ matrix

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \cdots & & & & \\ a_{m1} & a_{m2} & \ldots & a_{mn} & b_m \end{array}\right]$$

  is called the *augmented matrix* of the system.

An augmented matrix has the complete information to specify a system of linear equations. The augmented matrix of system (1.2) is

$$\begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{8} & 0 \\ -\frac{1}{2} & 1 & -\frac{1}{4} & 1 \\ -\frac{1}{4} & -\frac{1}{2} & 1 & 2 \end{bmatrix}.$$

Definition 1.3 Elementary Row Operations:

I. Interchange two rows.
II. Multiply a row by a nonzero real number.
III. Replace a row by its sum with a multiple of another row.

These three row operations correspond to the three equation operations, respectively, and hence are reversible.

Definition 1.4 Two matrices are said to be *row equivalent* if one can be obtained from the other by a sequence of elementary row operations.

Theorem 1.2 Consider two linear systems. If the augmented matrix of one system can be obtained from the one of the other system by a sequence of elementary row operations, then the two linear systems are equivalent.

From now on, we use R_i to denote the ith equation of a system of linear equations or the ith row in the corresponding augmented matrix.

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Lecture 2

Row Echelon Form

2.1 Solving $n \times n$ Systems

Basic Idea

- Forward substitution:
  - Use the x_1 term in the first equation to eliminate the x_1 terms in the other equations.
  - Use the x_2 term in the second equation to eliminate the x_2 terms in the other equations, and so on.
- If we can repeat the above steps until the last equation, we obtain a system like

$$\begin{aligned} 2x_1 - x_2 + 3x_3 - 2x_4 &= 1 \\ x_2 - 2x_3 + 3x_4 &= 2 \\ 4x_3 + 3x_4 &= 3 \\ 4x_4 &= 4 \end{aligned} \qquad \begin{bmatrix} 2 & -1 & 3 & -2 & 1 \\ 0 & 1 & -2 & 3 & 2 \\ 0 & 0 & 4 & 3 & 3 \\ 0 & 0 & 0 & 4 & 4 \end{bmatrix}$$

- The method of solving the above triangular type system is called *back substitution.*

Question: What shall we do if there is no x_1 term in the first equation?

In general, given a system of n linear equations in n unknowns, we can try to use equation operations to obtain an equivalent system that is of a triangular type.

■ **Example 2.1** Solve

$$\begin{aligned} x_1 + 2x_2 &= 5 \\ 4x_1 + 5x_2 &= 14 \end{aligned} \qquad \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 4 & 5 & 14 \end{array}\right]$$

- Adding (-4) times R_1 to R_2 yields

$$\begin{aligned} x_1 + 2x_2 &= 5 \\ -3x_2 &= -6 \end{aligned} \qquad \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -3 & -6 \end{array}\right]$$

- Multiplying $(-\frac{1}{3})$ to R_2 yields

$$\begin{aligned} x_1 + 2x_2 &= 5 \\ x_2 &= 2 \end{aligned} \qquad \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 2 \end{array}\right]$$

- Adding (-2) times R_2 to R_1 gives

$$\begin{aligned} x_1 &= 1 \\ x_2 &= 2 \end{aligned} \qquad \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right]$$

■

The above process can be expressed by the following row operations:

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 4 & 5 & 14 \end{array}\right] \xrightarrow{R_2 - 4R_1} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & -3 & -6 \end{array}\right] \xrightarrow{-\frac{1}{3}R_2} \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 1 & 2 \end{array}\right] \xrightarrow{R_1 - 2R_2} \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right].$$

■ **Example 2.2** The system of equations

$$\begin{aligned} x_1 + 2x_2 + 2x_3 &= 4 \\ x_1 + 3x_2 + 3x_3 &= 5 \\ 2x_1 + 6x_2 + 5x_3 &= 6 \end{aligned}$$

can be solved using the following steps:

1. Form the augmented matrix

$$\left[\begin{array}{ccc|c} \boxed{1} & 2 & 2 & 4 \\ 1 & 3 & 3 & 5 \\ 2 & 6 & 5 & 6 \end{array}\right].$$

■

2. Use the first row to eliminate the elements in the first column of the remaining rows (row operation III):
   - Add -1 times R_1 to R_2.
   - Add -2 times R_1 to R_3.

$$\left[\begin{array}{ccc|c} \boxed{1} & 2 & 2 & 4 \\ 0 & \boxed{1} & 1 & 1 \\ 0 & 2 & 1 & -2 \end{array}\right]$$

3. Use the second row to eliminate the last element in the second column (row operation III):
   - Add 2 times R_2 to R_3.

$$\left[\begin{array}{ccc|c} \boxed{1} & 2 & 2 & 4 \\ 0 & \boxed{1} & 1 & 1 \\ 0 & 0 & \boxed{-1} & -4 \end{array}\right]$$

4. Multiply -1 to the third row and perform back substitution:
   - Multiply R_3 by -1.
   - Add -1 times R_3 to R_2.
   - Add -2 times R_3 to R_1.

$$\left[\begin{array}{ccc|c} \boxed{1} & 2 & 0 & -4 \\ 0 & \boxed{1} & 0 & -3 \\ 0 & 0 & \boxed{1} & 4 \end{array}\right]$$

5. Back substitution: add -1 times R_2 to R_1.

$$\left[\begin{array}{ccc|c} \boxed{1} & 0 & 0 & 2 \\ 0 & \boxed{1} & 0 & -3 \\ 0 & 0 & \boxed{1} & 4 \end{array}\right]$$

Solving a System of Linear Equations:

1. Perform a sequence of elementary row operations to reduce the augmented matrix to one of a triangular type.
2. Solve the corresponding triangular-type system.

$$\left[\begin{array}{cccc|c} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{array}\right] \to \left[\begin{array}{cccc|c} \blacksquare & * & * & * & * \\ 0 & * & * & * & * \\ 0 & * & * & * & * \\ 0 & * & * & * & * \end{array}\right] \to \left[\begin{array}{cccc|c} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & * & * & * \\ 0 & 0 & * & * & * \end{array}\right]$$
$$\to \left[\begin{array}{cccc|c} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & \blacksquare & * & * \\ 0 & 0 & 0 & \blacksquare & * \end{array}\right]$$

Figure 2.1: Here $\blacksquare$ represents a nonzero number, and $*$ represents an arbitrary number.

$$\left[\begin{array}{cccc|c} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & \blacksquare & * & * \\ 0 & 0 & 0 & \blacksquare & * \end{array}\right] \to \left[\begin{array}{cccc|c} \blacksquare & * & * & 0 & * \\ 0 & \blacksquare & * & 0 & * \\ 0 & 0 & \blacksquare & 0 & * \\ 0 & 0 & 0 & 1 & * \end{array}\right] \to \left[\begin{array}{cccc|c} \blacksquare & * & 0 & 0 & * \\ 0 & \blacksquare & 0 & 0 & * \\ 0 & 0 & 1 & 0 & * \\ 0 & 0 & 0 & 1 & * \end{array}\right]$$
$$\to \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & * \\ 0 & 1 & 0 & 0 & * \\ 0 & 0 & 1 & 0 & * \\ 0 & 0 & 0 & 1 & * \end{array}\right]$$

Figure 2.2: Here $\blacksquare$ represents a nonzero number, and $*$ represents an arbitrary number.

2.2 Row-Echelon Form

In this section, we study how to transform a general $m \times n$ matrix to a triangular form.

- A *nonzero* row (resp. column) in a matrix means a row (resp. column) that contains a least one nonzero entry.
- A *leading entry* of a nonzero row refers to the *leftmost* nonzero entry.
- A zero entry in a row in called a leading zero entry if all the entries on the left of the zero entry are zero.

$$\begin{bmatrix} 0 & 1 & 1 & 2 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

There is only one leading entry of a nonzero row, but there can be multiple leading zeros.

Definition 2.1 A matrix is said to be in *row echelon form* if

(i) If row k is nonzero, the number of leading zero entries of row $k+1$ is greater than the number of leading zero entries in row k.

(ii) If there are rows whose entries are all zero, they are below any nonzero rows.

■ **Example 2.3** Matrices are in echelon form

$$\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \end{bmatrix}$$

■

Definition 2.2 If a matrix in row echelon form satisfies the following additional condition, the matrix is said to be in *reduced row echelon form*:

(ii) The leading entry in each nonzero row is 1.

(iv) Each leading 1 is the only nonzero entry in its column.

■ **Example 2.4** Matrices are in reduced row echelon form

$$\begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

■

■ **Example 2.5** The following matrices are in row echelon form, but not reduced row echelon form:

$$\begin{bmatrix} 2 & 4 & 6 \\ 0 & 3 & 5 \\ 0 & 0 & 4 \end{bmatrix}, \quad \begin{bmatrix} 1 & 4 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

■

■ **Example 2.6** The following matrices are not in row echelon form:

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

■

We can use Elementary Row Operation I and III to transform any matrix to row echelon form. Row operation I, interchanging two rows, can transform the matrices in Example 2.6 to (reduced) row echelon form. As another example,

$$\begin{bmatrix} 0 & 0 & 4 \\ 0 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 2 & 4 & 6 \\ 0 & 3 & 5 \\ 0 & 0 & 4 \end{bmatrix}$$

■ **Example 2.7** Reduce the following matrix to row echelon form:

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & 0 & 0 & 1 \\ -2 & -2 & 0 & 0 & 3 \\ 0 & 0 & 1 & 1 & 3 \\ 1 & 1 & 2 & 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} \boxed{1} & 1 & 1 & 1 & 1 \\ -1 & -1 & 0 & 0 & 1 \\ -2 & -2 & 0 & 0 & 3 \\ 0 & 0 & 1 & 1 & 3 \\ 1 & 1 & 2 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} \boxed{1} & 1 & 1 & 1 & 1 \\ 0 & 0 & \boxed{1} & 1 & 2 \\ 0 & 0 & 2 & 2 & 5 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 1 & 3 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} \boxed{1} & 1 & 1 & 1 & 1 \\ 0 & 0 & \boxed{1} & 1 & 2 \\ 0 & 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} \boxed{1} & 1 & 1 & 1 & 1 \\ 0 & 0 & \boxed{1} & 1 & 2 \\ 0 & 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

■

Gaussian Elimination for an $m \times n$ Matrix

- Initialize $i = 1$.
- For each j from 1 to n, repeat the following operations:
  - Find a nonzero element as the *pivot* from the entries in the jth column, rows i through m, of the matrix.
  - If there exits no such a pivot found, continue with the next value of j (i.e., $j + 1$).
  - If necessary, we interchange rows so that the pivotal row (the row containing the pivot) is the ith row.
  - Multiples of the pivotal row are then added to the rows $i + 1$ through m so as to eliminate all entries below the pivot (in the jth column).
  - Increase i by 1. If $i > m$, halt the process.
- After the process ends, the matrix is reduced to row echelon form.

Try the above algorithm for the matrix in Example 2.7.

Gauss-Jordan Elimination

- Reduce an $m \times n$ matrix to row echelon form.
- For i from m down to 1, repeat the following operations:
  - Find the pivot in row i, i.e., the leading nonzero element. If all entries in row i are 0, continue with the next value of i (i.e., $i - 1$).
  - Multiply the ith row by a constant so that the pivot is 1.
  - Multiples of row i is added to row 1 through row $i - 1$ so as to eliminate all the elements above the pivot.
- When the process stops, the matrix is in reduced row echelon form.

Try the above algorithm for the matrix in Example 2.7.

Uniqueness of Reduced Row Echelon Form

- A matrix A is row equivalent to a unique matrix U of reduced row echelon form.
- A matrix A can be row equivalent to multiple matrices of row echelon form.

2.3 Solution Types of System of Linear Equations

Suppose that the augmented matrix of a system of linear equations is row equivalent to the following matrix in row echelon form:

$$\left[\begin{array}{ccccc|c} \boxed{1} & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & \boxed{1} & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} & 3 \\ 0 & 0 & 0 & 0 & 0 & \boxed{1} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

This augmented matrix corresponds to the following system

$$\begin{aligned} x_1 + x_2 + x_3 + x_4 + x_5 &= 1 \\ x_3 + x_4 + 2x_5 &= 0 \\ x_5 &= 3 \\ 0 &= 1 \\ 0 &= 0 \end{aligned}$$

which has no solution as the second last equation cannot be satisfied.

Consider the augmented matrix

$$\left[\begin{array}{ccccc|c} \boxed{1} & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & \boxed{1} & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \tag{2.1}$$

The matrix can be further reduced by *backward substitution* to reduced row echelon form:

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 4 \\ 0 & 0 & 1 & 1 & 0 & -6 \\ 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

which is the augmented matrix of the following system

$$\begin{aligned} x_1 + x_2 &= 4 \\ x_3 + x_4 &= -6 \\ x_5 &= 3 \\ 0 &= 0. \end{aligned}$$

Here x_2 and x_4 are called free variables and x_1, x_3 and x_5 are called lead (dependent) variables. The solution set of the system can be written as

$$\begin{aligned} x_1 &= 4 - x_2 \\ x_2 &= x_2 \text{ (free variable)} \\ x_3 &= -6 - x_4 \\ x_4 &= x_4 \text{ (free variable)} \\ x_5 &= 3. \end{aligned}$$

Thus for any real number x_2 and x_4, the 5-tuple

$$(4 - x_2, x_2, -6 - x_4, x_4, 3)$$

is a solution of the system corresponding to the augmented matrix in (2.1).

Free Variables and Dependent Variables

For an augmented matrix in (reduced) row echelon form,

- x_i is a lead (dependent) variable if column i has a pivot;
- x_i is a free variable if column i does not have a pivot.

Three Possible Cases

Unique solution:
$$\left[\begin{array}{ccc|c} \boxed{1} & 0 & 0 & -3 \\ 0 & \boxed{1} & 0 & 5 \\ 0 & 0 & \boxed{1} & 2 \end{array}\right]$$

Infinitely many solutions:
$$\left[\begin{array}{ccc|c} \boxed{1} & 0 & 1 & 3 \\ 0 & \boxed{1} & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

No solution:
$$\left[\begin{array}{cccc|c} \boxed{1} & 0 & 3 & -2 & 0 \\ 0 & \boxed{1} & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{array}\right]$$

- A system of linear equations is *consistent* if it has at least one solution.
- Check the last non-zero row of the augmented matrix in row echelon form:
  - If it is of the form $(0, \dots, 0, *)$, where $*$ is nonzero, then the system has no solution (not consistent).
  - Otherwise, the system is consistent.
- Consider the augmented matrix in row echelon form of a consistent system.
  - The number r of nonzero rows is the number of lead variables.
  - $n - r$ is the number of free variables.
  - If $n - r > 0$, a consistent system has infinitely many solutions.

Note that $r \leq \min\{m, n\}$. The above statement implies that if $m < n$, a consistent system has infinitely many solutions.

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Homogeneous System

- A system of linear equations is said to be *homogeneous* if the constants on the right-hand side are all zero.
- Homogeneous systems are always consistent.
- $(0, 0, \dots, 0)$ is always a solution of a homogeneous system, and is call the *trivial* solution.
- An $m \times n$ homogeneous system of linear equations has a nontrivial solution if $n > m$.

Exercise 2.1 Find the solutions to the following system of equations by Gauss-Jordan elimination

$$\begin{aligned} 2x_1 + x_2 + 7x_3 - 7x_4 &= 2 \\ -3x_1 + 4x_2 - 5x_3 - 6x_4 &= 3 \\ x_1 + x_2 + 4x_3 - 5x_4 &= 2. \end{aligned}$$

■

Answer of exercise 2.1

Form the augmented matrix and perform row operations:

$$\left[\begin{array}{cccc|c} 2 & 1 & 7 & -7 & 2 \\ -3 & 4 & -5 & -6 & 3 \\ 1 & 1 & 4 & -5 & 2 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{cccc|c} 1 & 1 & 4 & -5 & 2 \\ -3 & 4 & -5 & -6 & 3 \\ 2 & 1 & 7 & -7 & 2 \end{array}\right]$$

$$\xrightarrow[R_3 \to -2R_1+R_3]{R_2 \to 3R_1+R_2} \left[\begin{array}{cccc|c} \boxed{1} & 1 & 4 & -5 & 2 \\ 0 & 7 & 7 & -21 & 9 \\ 0 & -1 & -1 & 3 & -2 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{cccc|c} \boxed{1} & 1 & 4 & -5 & 2 \\ 0 & -1 & -1 & 3 & -2 \\ 0 & 7 & 7 & -21 & 9 \end{array}\right]$$

$$\xrightarrow{R_2 \to -R_3} \left[\begin{array}{cccc|c} \boxed{1} & 1 & 4 & -5 & 2 \\ 0 & 1 & 1 & -3 & 2 \\ 0 & 7 & 7 & -21 & 9 \end{array}\right]$$

$$\xrightarrow{R_3 \to -7R_2+R_3} \left[\begin{array}{cccc|c} \boxed{1} & 1 & 4 & -5 & 2 \\ 0 & \boxed{1} & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 & -5 \end{array}\right]$$

$$\xrightarrow{R_1 \to -R_2+R_1} \left[\begin{array}{cccc|c} \boxed{1} & 0 & 3 & -2 & 0 \\ 0 & \boxed{1} & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 & -5 \end{array}\right].$$

The equivalent system is

$$\begin{aligned} x_1 + 3x_3 - 2x_4 &= 0, \\ x_2 + x_3 - 3x_4 &= 2, \\ 0 &= -5. \end{aligned}$$

As the last equality never holds, the system has no solution.

Exercise 2.2 Find the solutions to the following system of equations using Gauss-Jordan elimination

$$\begin{aligned} -7x_1 - 6x_2 - 12x_3 &= -33 \\ 5x_1 + 5x_2 + 7x_3 &= 24 \\ x_1 + x_2 + 4x_3 &= 10. \end{aligned}$$

■

Answer of exercise 2.2

Apply the elementary row operations to the augmented matrix:

$$\left[\begin{array}{ccc|c} -7 & -6 & -12 & -33 \\ 5 & 5 & 7 & 24 \\ 1 & 1 & 4 & 10 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 4 & 10 \\ 5 & 5 & 7 & 24 \\ -7 & -6 & -12 & -33 \end{array}\right]$$

$$\xrightarrow[R_3 \to 7R_1+R_3]{R_2 \to -5R_1+R_2} \left[\begin{array}{ccc|c} \boxed{1} & 1 & 4 & 10 \\ 0 & 0 & -13 & -26 \\ 0 & 1 & 16 & 37 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_3} \left[\begin{array}{ccc|c} \boxed{1} & 1 & 4 & 10 \\ 0 & 1 & 16 & 37 \\ 0 & 0 & -13 & -26 \end{array}\right]$$

$$\xrightarrow{R_3 \to -\frac{1}{13}R_3} \left[\begin{array}{ccc|c} \boxed{1} & 1 & 4 & 10 \\ 0 & \boxed{1} & 16 & 37 \\ 0 & 0 & \boxed{1} & 2 \end{array}\right]$$

$$\xrightarrow[R_2 \to -16R_3+R_2]{R_1 \to -4R_3+R_1} \left[\begin{array}{ccc|c} \boxed{1} & 1 & 0 & 2 \\ 0 & \boxed{1} & 0 & 5 \\ 0 & 0 & \boxed{1} & 2 \end{array}\right]$$

$$\xrightarrow{R_1 \to -R_2+R_1} \left[\begin{array}{ccc|c} \boxed{1} & 0 & 0 & -3 \\ 0 & \boxed{1} & 0 & 5 \\ 0 & 0 & \boxed{1} & 2 \end{array}\right]$$

The equivalent system is

$$\begin{aligned} x_1 &= -3 \\ x_2 &= 5 \\ x_3 &= 2 \end{aligned}$$

which is the unique solution.

Exercise 2.3 Find the solution set of the following system of equations using Gauss-Jordan elimination

$$\begin{aligned} x_1 - x_2 + 2x_3 &= 1 \\ 2x_1 + x_2 + x_3 &= 8 \\ x_1 + x_2 &= 5. \end{aligned}$$

■

Answer of exercise 2.3

Apply the elementary row operations to the augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 1 \\ 2 & 1 & 1 & 8 \\ 1 & 1 & 0 & 5 \end{array}\right] \xrightarrow[R_3 \to -R_1+R_3]{R_2 \to -2R_1+R_2} \left[\begin{array}{ccc|c} \boxed{1} & -1 & 2 & 1 \\ 0 & 3 & -3 & 6 \\ 0 & 2 & -2 & 4 \end{array}\right]$$

$$\xrightarrow{R_2 \to \frac{1}{3}R_2} \left[\begin{array}{ccc|c} \boxed{1} & -1 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 2 & -2 & 4 \end{array}\right]$$

$$\xrightarrow{R_3 \to -2R_2+R_3} \left[\begin{array}{ccc|c} \boxed{1} & -1 & 2 & 1 \\ 0 & \boxed{1} & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_2+R_1} \left[\begin{array}{ccc|c} \boxed{1} & 0 & 1 & 3 \\ 0 & \boxed{1} & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The equivalent system is

$$\begin{aligned} x_1 + x_3 &= 3, \\ x_2 - x_3 &= 2, \end{aligned}$$

or

$$\begin{aligned} x_1 &= 3 - x_3, \\ x_2 &= 2 + x_3. \end{aligned}$$

Here x_3 is the free variable. Thus, for any real number α, the 3-tuple

$$(3 - \alpha, 3 + \alpha, \alpha)$$

is a solution.

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Lecture 3

Matrix Arithmetic

3.1 Matrix Operations

Matrix Notation

- An $m \times n$ matrix is an array of numbers with m rows and n columns in the form:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix},$$

  where a_{ij} is the entry of the matrix in the ith row and jth column, also known as the (i, j) entry.
- $A = (a_{ij}) = (a_{ij})_{m\times n}$
- $(A)_{ij}$ is the (i, j) entry of A.
- When $m = n$, the matrix is a *square* matrix.
- Matrices are usually denoted by uppercase letters, e.g., $A, B, C, \cdots$.
- When all the entries of a matrix are zeros, the matrix is called **zero matrix**, denoted by $\mathbf{0}$.

We may use $\mathbb{R}^{m\times n}$ to denote the collection of all $m \times n$ matrices.

Definition 3.1 The *transpose* of an $m \times n$ matrix $A = (a_{ij})$ is the $n \times m$ matrix $B = (b_{ij})$ such that

$$b_{ji} = a_{ij}, \quad \forall i = 1, \ldots, m, j = 1, \ldots, n.$$

The transpose of A is denoted by A^{T}.

■ **Example 3.1** Transpose of couple matrices. ■

With the help of the transpose, we only need to focus on the row operations, row equivalence, etc. The similar column operations/equivalence is just the row operations/equivalence on the transpose.

Vectors

- A *column* vector is a matrix of one column, which is usually denoted by a **boldface** lowercase letter, e.g., $\mathbf{x}$.
- The set of all column vectors of n real numbers is denoted by $\mathbb{R}^n$.
- A *row* vector is a matrix of one row, which is the transpose of a column vector.

Matrix Operations

Definition 3.2 — Equality. Two $m \times n$ matrices $A = (a_{ij})$ and $B = (b_{ij})$ are said to be *equal* (written as $A = B$) if $a_{ij} = b_{ij}$ for each i and j.

Definition 3.3 — Matrix Addition. If $A = (a_{ij})$ and $B = (b_{ij})$ are two $m \times n$ matrices, then the *sum* of A and B (written as $A + B$) is the $m \times n$ matrix with the (i, j) entry $a_{ij} + b_{ij}$ for each ordered pair (i, j).

Definition 3.4 — Scalar Multiplication. Let $A = (a_{ij})_{m\times n}$ be a matrix and α be a scalar (any real number). The *scalar product* of α and A (written as αA) is the $m \times n$ matrix with the (i, j) entry αa_{ij}.

We write $-A = (-1)A$, then $A - B$ is $A + (-1)B$. The matrix addition and scalar multiplication operations can be combined to form more complicated matrix operations:

Properties of Matrices Operations

Consider matrices A, B, C of the same dimension, and scalars α and β.

1. $A + B = B + A$.
2. $A + (B + C) = (A + B) + C$. (We will therefore write $A + B + C$.)
3. $A + \mathbf{0} = \mathbf{0} + A = A$.
4. $\alpha(\beta A) = (\alpha\beta)A$.
5. $\alpha(A + B) = \alpha A + \alpha B$.
6. $(\alpha + \beta)A = \alpha A + \beta A$.

These properties can be proved using the similar properties of real number operations.

Proof of $\alpha(A + B) = \alpha A + \alpha B$. Fix a row number i and a column number j.

$$\begin{aligned}(\alpha(A + B))_{ij} &= \alpha(A + B)_{ij} && \text{(definition of scalar multiplication)}\\ &= \alpha(a_{ij} + b_{ij}) && \text{(definition of matrix addition).}\end{aligned}$$

Similarly,

$$\begin{aligned}(\alpha A + \alpha B)_{ij} &= (\alpha A)_{ij} + (\alpha B)_{ij} && \text{(definition of matrix addition)}\\ &= \alpha a_{ij} + \alpha b_{ij} && \text{(definition of scalar multiplication).}\end{aligned}$$

Now we are just comparing real (or complex) numbers, and you know the rules for comparing these:

$$\alpha(a_{ij} + b_{ij}) = \alpha a_{ij} + \alpha b_{ij} \qquad \text{(distributive property).}$$

We see that the (i, j)th entry of $\alpha(A + B)$ equals the (i, j)th entry of $\alpha A + \alpha B$ for all i and j, which is exactly the definition of two matrices being equal. ■

3.2 Matrix-Vector Multiplication

Let $A = (a_{ij})_{m\times n}$ be a matrix.

- The jth column of A is denoted by $\mathbf{a}_j$, i.e.,

$$\mathbf{a}_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}.$$

- The ith row of A is denoted by $\vec{\mathbf{a}}_i$, i.e.,

$$\vec{\mathbf{a}}_i = (a_{i1}, a_{i2}, \ldots, a_{in}).$$

- Matrix A can be represented in terms of either its columns and rows:

$$A = [\mathbf{a}_1, \cdots, \mathbf{a}_n] = \begin{bmatrix} \vec{\mathbf{a}}_1 \\ \vec{\mathbf{a}}_2 \\ \vdots \\ \vec{\mathbf{a}}_m \end{bmatrix}$$

Definition 3.5 — Matrix-vector multiplication. For an $m \times n$ matrix A with the ith column $\mathbf{a}_i$, and a vector $\mathbf{u} = (u_1, u_2, \ldots, u_n)^{\mathrm{T}}$, the multiplication of A and $\mathbf{u}$ is defined as

$$A\mathbf{u} = u_1\mathbf{a}_1 + u_2\mathbf{a}_2 + \cdots + u_n\mathbf{a}_n.$$

■ **Example 3.2**

$$A = \begin{bmatrix} 1 & 4 & 2 & 3 & 5 \\ -2 & 1 & 3 & 0 & -1 \\ 0 & 7 & -1 & -2 & 4 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 5 \\ -1 \end{bmatrix},$$

what is $A\mathbf{u}$? ■

Inner Product

- Given a vector $\mathbf{a} = (a_1, \ldots, a_n)^{\mathrm{T}}$ and a vector $\mathbf{b} = (b_1, \ldots, b_n)^{\mathrm{T}}$, following the rule of matrix-vector product, we have

$$\mathbf{a}^{\mathrm{T}}\mathbf{b} = a_1b_1 + a_2b_2 + \cdots a_nb_n.$$

- We call this special vector-vector multiplication the *inner product* (scalar product) of $\mathbf{a}$ and $\mathbf{b}$ (denoted by $\mathbf{a}^{\mathrm{T}}\mathbf{b}$ or $\langle \mathbf{a}, \mathbf{b} \rangle$).
- The matrix-vector multiplication $A\mathbf{u}$ has a row formula as

$$A\mathbf{u} = \begin{bmatrix} \vec{\mathbf{a}}_1\mathbf{u} \\ \vec{\mathbf{a}}_2\mathbf{u} \\ \vdots \\ \vec{\mathbf{a}}_m\mathbf{u} \end{bmatrix} \tag{3.1}$$

$$
\begin{aligned}
A\mathbf{u} &= u_1\mathbf{a}_1 + u_2\mathbf{a}_2 + \cdots + u_n\mathbf{a}_n && \text{(matrix-vector multiplication)}\\
&= u_1\begin{bmatrix} a_{11}\\ a_{21}\\ \vdots\\ a_{m1}\end{bmatrix} + u_2\begin{bmatrix} a_{12}\\ a_{22}\\ \vdots\\ a_{m2}\end{bmatrix} + \cdots + u_n\begin{bmatrix} a_{1n}\\ a_{2n}\\ \vdots\\ a_{mn}\end{bmatrix}\\
&= \begin{bmatrix} u_1a_{11}\\ u_1a_{21}\\ \vdots\\ u_1a_{m1}\end{bmatrix} + \begin{bmatrix} u_2a_{12}\\ u_2a_{22}\\ \vdots\\ u_2a_{m2}\end{bmatrix} + \cdots + \begin{bmatrix} u_na_{1n}\\ u_na_{2n}\\ \vdots\\ u_na_{mn}\end{bmatrix} && \text{(scalar multiplication)}\\
&= \begin{bmatrix} a_{11}u_1 + a_{12}u_2 + a_{13}u_3 + \cdots + a_{1n}u_n\\ a_{21}u_1 + a_{22}u_2 + a_{23}u_3 + \cdots + a_{2n}u_n\\ \vdots\\ a_{m1}u_1 + a_{m2}u_2 + a_{m3}u_3 + \cdots + a_{mn}u_n\end{bmatrix} && \text{(matrix addition)}\\
&= \begin{bmatrix}\vec{\mathbf{a}}_1\mathbf{u} & \vec{\mathbf{a}}_2\mathbf{u} & \cdots & \vec{\mathbf{a}}_m\mathbf{u}\end{bmatrix}^{\mathrm{T}}
\end{aligned}
$$

Consider the linear system of m equations in n unknowns

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n &= b_1\\
a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n &= b_2\\
&\vdots\\
a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n &= b_m.
\end{aligned} \tag{3.2}
$$

The above system can be written as

$$\vec{\mathbf{a}}_i\mathbf{x} = b_i, \quad i = 1, \ldots, m$$

or

$$A\mathbf{x} = \mathbf{b},$$

where $A = (a_{ij})$ is the coefficient matrix,

$$\mathbf{x} = \begin{bmatrix} x_1\\ x_2\\ \vdots\\ x_n\end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1\\ b_2\\ \vdots\\ b_m\end{bmatrix}.$$

The following system

$$
\begin{aligned}
3x_1 + 2x_2 + 3x_3 &= 5\\
x_1 - 2x_2 + 5x_3 &= -2\\
2x_1 + x_2 - x_3 &= 1
\end{aligned}
$$

can be written as

$$\begin{bmatrix} 3 & 2 & 1\\ 1 & -2 & 5\\ 2 & 1 & -3\end{bmatrix}\begin{bmatrix} x_1\\ x_2\\ x_3\end{bmatrix} = \begin{bmatrix} 5\\ -2\\ 1\end{bmatrix}.$$

Complexity of Matrix-Vector Multiplication

- Suppose that A is an $m \times n$ matrix and $\mathbf{x} \in \mathbb{R}^n$.
- Each entry of $A\mathbf{x}$ requires $2n-1$ flops (floating point operations):
  - n multiplications
  - $n-1$ additions
- As $A\mathbf{x}$ has m entries, the complexity of $A\mathbf{x}$ is $m(2n-1) \approx 2mn$ flops.

3.3 Matrix Multiplication

Definition 3.6 If $A = (a_{ij})$ is an $m \times n$ matrix and $B = (b_{ij})$ is an $n \times r$ matrix, then the product AB is an $m \times r$ matrix whose jth column is $A\mathbf{b}_j$, i.e.,

$$AB = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_r \end{bmatrix}.$$

- Applying the row formula of $A\mathbf{b}_j$, we get $(AB)_{ij} = \mathbf{a}_i\mathbf{b}_j$.
- Expending the formula of inner product, we get $(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}$.

■ **Example 3.3** Calculate the product of two matrices

$$\begin{bmatrix} 3 & -2 \\ 2 & 4 \\ 1 & -3 \end{bmatrix} \text{ and } \begin{bmatrix} -2 & 1 & 3 \\ 4 & 1 & 6 \end{bmatrix}.$$

■

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- AB exists if and only if the number of columns of A equals to the number of rows of B.
- AB exists does not imply that BA exists.
- Even if both AB and BA exists, they are in general in different sizes and not equal. (Multiplication of matrices is not commutative.)
- $AB = \mathbf{0}$ does not implies that $A = \mathbf{0}$ or $B = \mathbf{0}$.

Consider

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

We can calculate that $AB = \mathbf{0}$ but $BA \neq \mathbf{0}$.

Properties of Matrix Product

Considers matrices A, B, C such that the indicated operations are defined and real number α:

1. $A(B+C) = AB + AC$,
2. $(A+B)C = AC + BC$,
3. $\alpha(AB) = (\alpha A)B = A(\alpha B)$,
4. $(AB)C = A(BC)$.

Proof of $A(B+C) = AB + AC$. On the one hand, the (i,j) entry of $A(B+C)$ is the multiplication of the ith row of A and the jth column of $B+C$, which is $\mathbf{b}_j + \mathbf{c}_j$, i.e.,

$$\vec{\mathbf{a}}_i(\mathbf{b}_j + \mathbf{c}_j) = \vec{\mathbf{a}}_i\mathbf{b}_j + \vec{\mathbf{a}}_i\mathbf{c}_j. \tag{3.3}$$

On the other hand, the (i,j) entry of $AB + AC$ is the addition of the (i,j) entries of AB and AC, i.e.,

$$\vec{\mathbf{a}}_i\mathbf{b}_j + \vec{\mathbf{a}}_i\mathbf{c}_j.$$

Therefore, $A(B+C) = AB + AC$. ■

The equality in (3.3) needs the following elaboration:

$$\begin{aligned}\vec{\mathbf{a}}_i(\mathbf{b}_j + \mathbf{c}_j) &= \sum_k a_{ik}(b_{kj} + c_{kj}) \\ &= \sum_k a_{ik}b_{kj} + \sum_k a_{ik}c_{kj} \\ &= \vec{\mathbf{a}}_i\mathbf{b}_j + \vec{\mathbf{a}}_i\mathbf{c}_j.\end{aligned}$$

Assume the first three properties are proved.

Proof of associative rule $(AB)C = A(BC)$. We have

$$(AB)C = \begin{bmatrix}(AB)\mathbf{c}_1 & (AB)\mathbf{c}_2 & \cdots & (AB)\mathbf{c}_\ell\end{bmatrix}$$

and because $BC = \begin{bmatrix}B\mathbf{c}_1 & B\mathbf{c}_2 & \cdots & B\mathbf{c}_\ell\end{bmatrix}$, we have

$$A(BC) = \begin{bmatrix}A(B\mathbf{c}_1) & A(B\mathbf{c}_2) & \cdots & A(B\mathbf{c}_\ell)\end{bmatrix}.$$

Therefore, we only need to show $(AB)\mathbf{x} = A(B\mathbf{x})$ for all proper $\mathbf{x}$. ■

The last claim can be shown as follows:

$$\begin{aligned}(AB)\mathbf{x} &= \begin{bmatrix}A\mathbf{b}_1 & \cdots & A\mathbf{b}_r\end{bmatrix}\begin{bmatrix}x_1 \\ \vdots \\ x_r\end{bmatrix} \\ &= x_1(A\mathbf{b}_1) + \cdots + x_r(A\mathbf{b}_r) \\ &= A(x_1\mathbf{b}_1) + \cdots + A(x_r\mathbf{b}_r) \\ &= A(x_1\mathbf{b}_1 + \cdots + x_r\mathbf{b}_r) \\ &= A(B\mathbf{x})\end{aligned}$$

where the first equality is the definition of matrix-matrix multiplication, second and the last equalities follows from the definition of matrix-vector multiplication, the third equality follows the third property and the forth equality follows the first property.

Outer Product

- Given a vector $\mathbf{a} = (a_1, \ldots, a_m)^{\mathrm{T}}$ and a vector $\mathbf{b} = (b_1, \ldots, b_n)^{\mathrm{T}}$, following the rule of matrix-matrix product, we have

$$\mathbf{a}\mathbf{b}^{\mathrm{T}} = (a_i b_j)_{m \times n}.$$

- We call the above special vector-vector multiplication the *outer product* of $\mathbf{a}$ and $\mathbf{b}$ (denoted by $\mathbf{a}\mathbf{b}^{\mathrm{T}}$).
- Consider an $m \times n$ matrix A and an $n \times r$ matrix B.
  - $\mathbf{a}_k \vec{\mathbf{b}}_k = (a_{ik} b_{kj})$
  - $AB = \sum_k \mathbf{a}_k \vec{\mathbf{b}}_k$ (outer product expansion)

To prove the outer product expansion, we write

$$\begin{aligned} AB &= (\sum_k a_{ik} b_{kj}) \\ &= \sum_k (a_{ik} b_{kj}) \\ &= \sum_k \mathbf{a}_k \vec{\mathbf{b}}_k \end{aligned}$$

where the first equality is the definition of matrix-matrix multiplication, the second equality is the definition of matrix addition, and the last equality is the definition of outer product.

Complexity of Matrix Multiplication

- Suppose A and B are $m \times n$ and $n \times r$ matrices, respectively.
- $AB = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_r \end{bmatrix}$
- The computation of $A\mathbf{b}_j$ has $2mn - m$ flops.
- The total computation complexity of AB is $r(2mn - m) \approx 2mnr$ flops.

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Lecture 4

Matrix Algebra

■ **Example 4.1** For

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 & 6 \\ 0 & -4 & 1 & 2 & 3 \\ -5 & 1 & 2 & -3 & 4 \end{bmatrix}, \; B = \begin{bmatrix} 1 & 6 & 2 & 1 \\ -1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 3 \\ 6 & 4 & -1 & 2 \\ 1 & -2 & 3 & 0 \end{bmatrix},$$

what is the $(1, 2)$ entry of AB? ■

Since $(AB)C = A(BC)$, we just write ABC. For a positive integer k and a square matrix A,

$$A^k = \underbrace{AA \dots A}_{k \text{ times}}.$$

■ **Example 4.2** If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, then A^k? ■

What's the computational complexity of A^k?

4.1 Algebraic Rules for Transpose

Recall what is matrix transpose.

For matrix

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} = \begin{bmatrix} \vec{\mathbf{a}}_1 \\ \vec{\mathbf{a}}_2 \\ \vdots \\ \vec{\mathbf{a}}_m \end{bmatrix},$$

we have

$$A^{\mathrm{T}} = \begin{bmatrix} \mathbf{a}_1^{\mathrm{T}} \\ \mathbf{a}_2^{\mathrm{T}} \\ \vdots \\ \mathbf{a}_n^{\mathrm{T}} \end{bmatrix} = \begin{bmatrix} \vec{\mathbf{a}}_1^{\mathrm{T}} & \vec{\mathbf{a}}_2^{\mathrm{T}} & \cdots & \vec{\mathbf{a}}_m^{\mathrm{T}} \end{bmatrix}.$$

Properties of Matrix Transpose

Let $A, B \in \mathbb{R}^{m\times n}$, $\alpha \in \mathbb{R}$, then

1. $(A+B)^{\mathrm{T}} = A^{\mathrm{T}} + B^{\mathrm{T}}$,
2. $(\alpha A)^{\mathrm{T}} = \alpha A^{\mathrm{T}}$,
3. $(A^{\mathrm{T}})^{\mathrm{T}} = A$.

Proof of $(A+B)^{\mathrm{T}} = A^{\mathrm{T}} + B^{\mathrm{T}}$. $((A+B)^{\mathrm{T}})_{ij} = (A+B)_{ji} = a_{ji} + b_{ji} = (A^{\mathrm{T}})_{ij} + (B^{\mathrm{T}})_{ij}$. ■

Transpose of Product

Theorem 4.1 $(AB)^{\mathrm{T}} = B^{\mathrm{T}}A^{\mathrm{T}}$.

Proof. The (i,j) entry of $(AB)^{\mathrm{T}}$ is the (j,i) entry of AB, which is $\sum_{k=1}^{n} a_{jk}b_{ki}$. The (i,j) entry of $B^{\mathrm{T}}A^{\mathrm{T}}$ is the multiplication of the ith row of B^{T} and the jth column of A^{T}. Since the ith row of B^{T} is the transpose of the ith column of B and the jth column of A^{T} is the transpose of the jth row of A, the (i,j) entry of $B^{\mathrm{T}}A^{\mathrm{T}}$ is

$$\mathbf{b}_i^{\mathrm{T}}\vec{\mathbf{a}}_j^{\mathrm{T}} = \sum_{k=1}^{n} a_{jk}b_{ki}.$$

It follows that the (i,j) entries of $(AB)^{\mathrm{T}}$ and $B^{\mathrm{T}}A^{\mathrm{T}}$ are equal. ■

■ **Example 4.3** Consider

$$A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$$

Verify that $(AB)^{\mathrm{T}} = B^{\mathrm{T}}A^{\mathrm{T}}$. ■

Definition 4.1 A matrix A is said to be *symmetric* if $A = A^{\mathrm{T}}$.

- For example,

$$A = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 4 & 5 \\ 4 & 2 & 6 \\ 5 & 6 & 1 \end{bmatrix}$$

  are symmetric.
- A symmetric matrix must be squared.
- The entries of a symmetric matrix are symmetric across the diagonal.

■ **Example 4.4** Show that for any $m \times n$ matrix A, the matrices $A^{\mathrm{T}}A$ and AA^{T} are symmetric. ■

■ **Example 4.5** Show that for any square matrix A, the matrix $A + A^{\mathrm{T}}$ is symmetric.
■

Definition 4.2 A matrix A is called *skew-symmetric* if $A^{\mathrm{T}} = -A$.

- For example,

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 4 & 5 \\ -4 & 0 & 6 \\ -5 & -6 & 0 \end{bmatrix}$$

  are skew-symmetric.
- Show that for any square matrix A, the matrix $A - A^{\mathrm{T}}$ is skew symmetric.
- Any square matrix A can be written as a sum of a symmetric matrix and a skew-symmetric matrix:

$$A = \frac{A + A^{\mathrm{T}}}{2} + \frac{A - A^{\mathrm{T}}}{2}.$$

4.2 Diagonal Matrices

Definition 4.3 A matrix A is called **diagonal** if $a_{ij} = 0$ whenever $i \neq j$.

- Usually "diagonal matrix" refers to a square matrix.
- We will use the following notation for a *square* diagonal matrix:

$$A = \operatorname{diag}(a_{11}, a_{22}, \cdots, a_{nn}).$$

- For example,

$$\operatorname{diag}(1, 2, -5, 3) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -5 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

Let $A = \operatorname{diag}(a_1, a_2, \cdots, a_n)$ and B be an $m \times n$ matrices.

- $BA = \begin{bmatrix} a_1\mathbf{b}_1 & a_2\mathbf{b}_2 & \cdots & a_n\mathbf{b}_n \end{bmatrix}$.
- The multiplication of two diagonal matrices is also diagonal.
- $A^k = \operatorname{diag}(a_1^k, a_2^k, \cdots, a_n^k)$.

Identity Matrix

An important special case of a square diagonal matrix is

$$I = I_n = \operatorname{diag}(1, 1, \cdots, 1),$$

called the **identity matrix** of size n.

Example

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Multiplication with Identity Matrix

- $AI_n = I_m A = A, \quad \forall A \in \mathbb{R}^{m\times n}$.
- For scalar α and vector $\mathbf{x}$, we have $\alpha\mathbf{x} = \alpha I\mathbf{x} = (\alpha I)\mathbf{x}$.

4.3 Matrix Inversion

Recall that we have defined the zero matrix $\mathbf{0}$ which has the property that $A\mathbf{0} = \mathbf{0}A = \mathbf{0}$ for any matrix A. We also defined the identity matrix I which satisfies $AI = IA = A$ for any matrix A.

The additive inversion of a matrix A is $-A = (-1)A$ so that $A + (-A) = \mathbf{0}$. How about the multiplicative inverse?

Definition 4.4 A square matrix A is said to be *invertible* (or *nonsingular*) if there exists a matrix B such that $AB = BA = I$. The matrix B is said to be the (multiplicative) inverse of A.

- If B and C are both multiplicative inverse of A, then

$$B = BI = B(AC) = (BA)C = IC = C.$$

- The above argument implies that if there exist B and C such that $BA = AC = I$, then $B = C$.
- The inverse of a matrix A is unique, and denoted by A^{-1}.

Consider that for two square matrices of the same size, does $BA = I$ imply $AB = I$?

R Suppose that we have an $n \times n$ system of linear equations $A\mathbf{x} = \mathbf{b}$. If A is invertible, then the system has a unique solution given by $\mathbf{x} = A^{-1}\mathbf{b}$.

■ **Example 4.6**

- The inverse of an identity matrix is itself.
- The inverse of a diagonal matrix $A = \text{diag}(a_1, \ldots, a_n)$ exists if and only if $a_1 a_2 \cdots a_n \neq 0$:

$$\text{diag}(a_1, \ldots, a_n)^{-1} = \text{diag}(a_1^{-1}, \ldots, a_n^{-1}).$$

■

■ **Example 4.7** Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. We know that A is invertible if and only if $ad - bc \neq 0$. If $ad - bc \neq 0$,

$$A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

■

In general, to obtain the inverse of an $n \times n$ matrix A, we can solve a system of linear equations $AX = I$ with variables the entries in X, which is equivalent to solving $A\mathbf{x}_j = \mathbf{e}_j$, where $\mathbf{e}_j$ is the jth column of I for all j. Note that it is not necessary to solve these n systems one by one. We will discuss more about matrix inversion in the next lecture.

Theorem 4.2 For square matrices A and B of the same size, if both A and B are invertible, then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

Proof. By assumption, A^{-1} and B^{-1} exist. Thus,

$$(AB)B^{-1}A^{-1} = A(BB^{-1})A^{-1} = AA^{-1} = I.$$

$$B^{-1}A^{-1}(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I.$$

Hence, $(AB)^{-1} = B^{-1}A^{-1}$. ■

In general, if $A_1, \ldots, A_k$ are nonsingular $n \times n$ matrices, then the product $A_1A_2 \cdots A_k$ is nonsingular and

$$(A_1A_2 \cdots A_k)^{-1} = A_k^{-1}A_{k-1}^{-1} \cdots A_2^{-1}A_1^{-1}.$$

As an exercise, prove the above claim by induction.

Another direction of the above theorem is that for square matrices A and B of the same size, if AB is nonsingular, then both A and B are nonsingular.

Theorem 4.3 For square matrices A and B of the same size, if AB is invertible, then both A and B are invertible.

To be proved in the next lecture.

Theorem 4.4 If A and B are square matrices such that $BA = I$, then $AB = I$.

Proof. As I is invertible, by Theorem 4.3, $BA = I$ implies both A and B are invertible. Thus,

$$AB = (B^{-1}B)AB = B^{-1}(BA)B = B^{-1}IB = B^{-1}B = I.$$

■

Suppose A is invertible. Then,

- $(A^{\mathrm{T}})^{-1} = (A^{-1})^{\mathrm{T}}$,
- $(\alpha A)^{-1} = \frac{1}{\alpha}A^{-1}$, for any $\alpha \neq 0$.
- $(A^{-1})^{-1} = A$.

4.4 Triangular Matrices

Definition 4.5 A square matrix is said to be *upper triangular* if $a_{ij} = 0$ for $i > j$ and *lower triangular* if $a_{ij} = 0$ for $i < j$.

- A diagonal matrix is both upper and lower triangular.
- A matrix is triangular if it is either upper or lower triangular.
- A triangular matrix is *strictly triangular* if all the diagonal entries are zero.
- A triangular matrix is said to be *unit* if all the diagonal entries are 1.

Properties of Triangular Matrices

- The inverse of a lower triangular matrix is also lower triangular.
- The product of two upper triangular matrices is also upper triangular.

4.5 Partitioned Matrices

It is often convenient to think about matrices defined in terms of other matrices.

Suppose as an example

$$P_{11} = \begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \end{bmatrix}, \quad P_{12} = \begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix},$$
$$P_{21} = \begin{bmatrix} 1 & 1 & 1 \\ -2 & -2 & -2 \end{bmatrix}, \quad P_{22} = \begin{bmatrix} -2 & -2 \\ -3 & -3 \end{bmatrix}.$$

We can now define

$$P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix}$$

by which we mean

$$P = \left[\begin{array}{ccc|cc} -1 & -1 & -1 & 2 & 2 \\ 2 & 2 & 2 & 3 & 3 \\ \hline 1 & 1 & 1 & -2 & -2 \\ -2 & -2 & -2 & -3 & -3 \end{array}\right].$$

We just have to make sure the blocks are of the right sizes, i.e., blocks in the same (block) row need to have the same number rows, blocks in the same (block) column need to have the same number of columns.

The matrix

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1t} \\ \vdots & \ddots & \vdots \\ A_{s1} & \cdots & A_{st} \end{bmatrix}$$

is a partition of matrix A with $s \times t$ blocks if the matrices A_{ij} satisfies

1. For each fixed i, the number of rows of all A_{ij} are equal.
2. For each fixed j, the number of columns of all A_{ij} are equal.

The matrix A_{ij} is called the (i, j)-block of A.

For

$$A_{11} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad A_{12} = \begin{bmatrix} 5 & -1 & 3 \\ -2 & 1 & 0 \end{bmatrix}, \quad A_{13} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$A_{21} = \begin{bmatrix} 1 & 5 \end{bmatrix}, \quad A_{22} = \begin{bmatrix} 7 & -2 & 3 \end{bmatrix}, \quad A_{23} = \begin{bmatrix} 2 \end{bmatrix}$$

we have

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix} = \left[\begin{array}{cc|ccc|c} 1 & 2 & 5 & -1 & 3 & 4 \\ 3 & 4 & -2 & 1 & 0 & 6 \\ \hline 1 & 5 & 7 & -2 & 3 & 2 \end{array}\right].$$

Partitioned Matrices

- Augmented matrix:

$$\begin{bmatrix} A & \mathbf{b} \end{bmatrix}$$

- Row and column form:

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} = \begin{bmatrix} \vec{\mathbf{a}}_1 \\ \vec{\mathbf{a}}_2 \\ \vdots \\ \vec{\mathbf{a}}_m \end{bmatrix}$$

- Matrix multiplication

$$AB = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_r \end{bmatrix}$$

The matrix algebra rules defined component-wisely can be similarly defined partitioned matrices block-wisely. The block-wise matrix addition and scalar multiplication are straightforward. In general, if the blocks have the proper dimensions, matrix multiplication can be carried out block-wisely.

Block-wise Multiplication

Consider two matrices A and B with AB defined.

- If $A = (A_{ij})$ and $B = (B_{ij})$,
- then $AB = (C_{ij})$ with

$$C_{ij} = \sum_k A_{ik} B_{kj}.$$

■ **Example 4.8**

- $A \begin{bmatrix} B_1 & B_2 \end{bmatrix} = \begin{bmatrix} AB_1 & AB_2 \end{bmatrix}$.
- $\begin{bmatrix} A_1 & A_2 \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = A_1B_1 + A_2B_2$.
- $\begin{bmatrix} A_1 \\ A_2 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \end{bmatrix} = \begin{bmatrix} A_1B_1 & A_1B_2 \\ A_2B_1 & A_2B_2 \end{bmatrix}$.
- $\begin{bmatrix} A_{11} & \mathbf{0} \\ \mathbf{0} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & \mathbf{0} \\ \mathbf{0} & B_{22} \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} & \mathbf{0} \\ \mathbf{0} & A_{22}B_{22} \end{bmatrix}$.

■

Partitioned Matrix View of Matrix Multiplication

- $$AB = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} \vec{\mathbf{b}}_1 \\ \vec{\mathbf{b}}_2 \\ \vdots \\ \vec{\mathbf{b}}_n \end{bmatrix} = \sum_{k=1}^{n} \mathbf{a}_k \vec{\mathbf{b}}_k$$

- $$AB = \begin{bmatrix} \vec{\mathbf{a}}_1 \\ \vec{\mathbf{a}}_1 \\ \vdots \\ \vec{\mathbf{a}}_n \end{bmatrix} \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \end{bmatrix} = (\vec{\mathbf{a}}_i \mathbf{b}_j)$$

The inverse of a general block-diagonal matrix has the similar form. Let A be a matrix of the form

$$\begin{bmatrix} A_{11} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & A_{22} & & \\ \vdots & & \ddots & \\ \mathbf{0} & & & A_{nn} \end{bmatrix}$$

where A_{kk} is an invertible, square matrix for $k = 1, \ldots, n$. Then

$$A^{-1} = \begin{bmatrix} A_{11}^{-1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & A_{22}^{-1} & & \\ \vdots & & \ddots & \\ \mathbf{0} & & & A_{nn}^{-1} \end{bmatrix}.$$

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Lecture 5

Elementary Matrices

5.1 Elementary Matrices

For a system of linear equations, we can apply elementary row operations to transform its augmented matrix to a simpler form (e.g., the (reduced) row-echelon form), while preserving the equivalence of the system. In the matrix form of a system of linear equations, elementary row operations can be performed using matrix multiplication.

Equivalent Systems

- Consider an $m \times n$ system $A\mathbf{x} = \mathbf{b}$.
- For any nonsingular $m \times m$ matrix M,

  $$(MA)\mathbf{x} = M\mathbf{b}$$

  is an equivalent system. (Compare the solution sets.)

  $$(E_k \cdots E_1 A)\mathbf{x} = E_k \cdots E_1 \mathbf{b}$$

  whose augmented matrix is

  $$E_k \cdots E_1 \begin{bmatrix} A & \mathbf{b} \end{bmatrix}.$$

Elementary Matrices

Definition 5.1 Starting with the identity matrix I and then performing exactly one elementary row operation, the resulting matrix is called an *elementary matrix*.

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(a) type I

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

(b) type II

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(c) type III

Properties Type I Elementary Matrices

Let E_1 be the matrix obtained by interchanging the ith and the jth rows of I. For example

$$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- E_1 is symmetric, but not diagonal.
- E_1A is the matrix obtained by interchanging the ith and the jth rows of A.

$$E_1A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \vec{\mathbf{a}}_1 \\ \vec{\mathbf{a}}_2 \\ \vec{\mathbf{a}}_3 \end{bmatrix} = \begin{bmatrix} \vec{\mathbf{a}}_2 \\ \vec{\mathbf{a}}_1 \\ \vec{\mathbf{a}}_3 \end{bmatrix}.$$

- The inverse of E_1 is itself.

Properties of Type II Elementary Matrices

Let E_2 be the matrix obtained by multiplying $\alpha \neq 0$ to the ith row of I. For example

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

- E_2 is diagonal and hence symmetric.
- E_2A is the matrix obtained by multiplying α to the ith row of A.

$$E_2A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} \vec{\mathbf{a}}_1 \\ \vec{\mathbf{a}}_2 \\ \vec{\mathbf{a}}_3 \end{bmatrix} = \begin{bmatrix} \vec{\mathbf{a}}_1 \\ \vec{\mathbf{a}}_2 \\ 3\vec{\mathbf{a}}_3 \end{bmatrix}.$$

- E_2^{-1} is matrix obtained by multiplying $1/\alpha$ to the ith row of I.

Let E be an type I or II elementary matrix. Due to the symmetry of E, we have $AE = (E^{\mathrm{T}}A^{\mathrm{T}})^{\mathrm{T}} = (EA^{\mathrm{T}})^{\mathrm{T}}$. In other words, right multiplication of A by E is equivalent to interchanging columns of A or multiplying a column of A by a constant.

Properties of Type III Elementary Matrices

Let E_3 be the matrix obtained from I by adding $\alpha \neq 0$ times of the jth row to the ith row, $i \neq j$. For example

$$E_3 = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- E_3 is triangular.
- E_3A is the matrix obtained from A by by adding α times of the jth row to the ith row.

$$E_3A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \vec{\mathbf{a}}_1 \\ \vec{\mathbf{a}}_2 \\ \vec{\mathbf{a}}_3 \end{bmatrix} = \begin{bmatrix} \vec{\mathbf{a}}_1 + 3\vec{\mathbf{a}}_3 \\ \vec{\mathbf{a}}_2 \\ \vec{\mathbf{a}}_3 \end{bmatrix}$$

- E_3^{-1} is same as E_3 except that the (i, j) entry is $-\alpha$.

■ **Example 5.1**

$$AE_3^{\mathrm{T}} = \begin{bmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 3 & 0 & 1\end{bmatrix} = \begin{bmatrix}\mathbf{a}_1 + 3\mathbf{a}_3 & \mathbf{a}_2 & \mathbf{a}_3\end{bmatrix}$$

$$AE_3 = \begin{bmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3\end{bmatrix}\begin{bmatrix}1 & 0 & 3\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} = \begin{bmatrix}\mathbf{a}_1 & \mathbf{a}_2 & 3\mathbf{a}_1 + \mathbf{a}_3\end{bmatrix}.$$

■

The three types of elementary matrices are invertible, which can be used to reverse the operation of the original matrix.

Performing Gauss-Jordan elimination on an augmented matrix $\begin{bmatrix}A & \mathbf{b}\end{bmatrix}$ can now be viewed as premultiplying $\begin{bmatrix}A & \mathbf{b}\end{bmatrix}$ by elementary matrices!

$$\begin{aligned}\begin{bmatrix}A & \mathbf{b}\end{bmatrix} &\xrightarrow{op_1} E_1\begin{bmatrix}A & \mathbf{b}\end{bmatrix}\\ &\xrightarrow{op_2} E_2E_1\begin{bmatrix}A & \mathbf{b}\end{bmatrix}\\ &\xrightarrow{op_2} E_3E_2E_1\begin{bmatrix}A & \mathbf{b}\end{bmatrix}\\ &\ \ \vdots\\ &\xrightarrow{op_k} E_k\cdots E_3E_2E_1\begin{bmatrix}A & \mathbf{b}\end{bmatrix}\end{aligned}$$

where op_i, $i = 1, \ldots k$ is the sequence of row operations performed by GJ elimination and $E_iM = op_i(M)$ for a matrix M.

Recall the definition of row equivalence in Definition 1.4.

Theorem 5.1 If A and B are two row equivalent matrices, then there exists a finite sequence $E_1, E_2, \ldots, E_k$ of elementary matrices such that

$$B = E_kE_{k-1}\cdots E_1A.$$

Proof. A and B row equivalent implies the existence of a sequence elementary row operations $\mathrm{op}_1, \mathrm{op}_2, \cdots, \mathrm{op}_k$ that transform A to B, i.e.,

$$B = op_k(op_{k-1}(\cdots op_2(op_1(A))\cdots)).$$

Let E_i be the corresponding elementary matrix for row operation $\mathrm{op_i}$ $(i = 1, \cdots, k)$ such that $op_i(M) = E_iM$ for a matrix M. Then, we have $B = E_kE_{k_1}\cdots E_1A$. ■

Row equivalence is an *equivalence relation*.

1. (Reflexivity) A is row equivalent to itself.
2. (Symmetry) If A is row equivalent to B, then B is row equivalent to A.
3. (Transitivity) If A is row equivalent to B, and B is row equivalent to C, then A is row equivalent to C.

Permutation Matrices

- A *permutation matrix* is a square matrix formed from the identity matrix by reordering its rows (columns).
- A permutation matrix has exactly one 1 in each row/column.
- An elementary matrix of type I is a permutation matrix of interchanging two rows/columns, also called a *elementary permutation matrix*.
- A permutation matrix P can be decomposed into a multiplication of a finite number of elementary matrix of type I, i.e.,

$$P = E_1 E_2 \cdots E_k$$

  where E_i is an elementary matrix of type I, $i = 1, \ldots, k$.
- $P^{-1} = E_k E_{k-1} \cdots E_2 E_1 = P^{\mathrm{T}}$.

Suppose that P is reordering the rows of I in the order $(\pi_1, \ldots, \pi_n)$. Then

$$PA = P \begin{bmatrix} \vec{\mathbf{a}}_1 \\ \vdots \\ \vec{\mathbf{a}}_n \end{bmatrix} = \begin{bmatrix} \vec{\mathbf{a}}_{\pi_1} \\ \vdots \\ \vec{\mathbf{a}}_{\pi_n} \end{bmatrix}$$

and

$$BP^{\mathrm{T}} = \begin{bmatrix} \mathbf{b}_1 & \cdots & \mathbf{b}_n \end{bmatrix} P^{\mathrm{T}} = \begin{bmatrix} \mathbf{b}_{\pi_1} & \cdots & \mathbf{b}_{\pi_n} \end{bmatrix}$$

5.2 Properties of Nonsingular Matrices

Theorem 5.2 Let A be an $n \times n$ matrix. The following are equivalent:

1. A is nonsingular.
2. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{0}$.
3. A is row equivalent to I.
4. A is a product of a sequence of finite elementary matrices.

Proof. 1) $\Rightarrow$ 2): If A is nonsingular, for any solution $\hat{\mathbf{x}}$ of $A\mathbf{x} = \mathbf{0}$, we have

$$\hat{x} = (A^{-1}A)\hat{\mathbf{x}} = A^{-1}(A\hat{\mathbf{x}}) = A^{-1}\mathbf{0} = \mathbf{0}.$$

2) $\Rightarrow$ 3): Suppose $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. By transforming A to the reduced row echelon form U, we get the equivalent system $U\mathbf{x} = \mathbf{0}$. As the system has only one solution, we know from Lecture 2 that U must have n pivotal columns (which implies no free variables), i.e., each column of U has one 1. We can then check inductively that the 1 in the ith column must be the (i, i) entry since otherwise the definition of row echelon form will be violated.

3) $\Rightarrow$ 4): As A is row equivalent to I, by Theorem 5.1, there exists a finite sequence $E_1, E_2, \ldots, E_k$ of elementary matrices such that

$$A = E_k E_{k_1} \cdots E_1 I = E_k E_{k_1} \cdots E_1.$$

4) $\Rightarrow$ 1): As E_i, $i = 1, \ldots, k$, are invertible, we have $A = E_k E_{k_1} \cdots E_1$ is invertible. ■

Corollary 5.3 An $n \times n$ system $A\mathbf{x} = \mathbf{b}$ has a unique solution if and only if A is nonsingular.

Proof. If A is nonsingular, we have $\mathbf{x} = A^{-1}\mathbf{b}$, the unique solution.

If $A\mathbf{x} = \mathbf{b}$ has a unique solution $\hat{\mathbf{x}}$, we show that A is nonsingular by contradiction. Assume A is singular. By Theorem 5.2, the system $A\mathbf{x} = \mathbf{0}$ would have a solution $\mathbf{z} \neq \mathbf{0}$. Then $\hat{\mathbf{x}} + \mathbf{z}$ would be another solution of $A\mathbf{x} = \mathbf{b}$. ■

Proof of Theorem 4.3. The theorem is equivalent to that if either A or B is singular, then AB is singular.

Suppose B is singular, i.e., there exists $\mathbf{z} \neq \mathbf{0}$ such that $B\mathbf{z} = \mathbf{0}$. Hence, $AB\mathbf{z} = A\mathbf{0} = \mathbf{0}$, i.e., AB is singular.

Suppose B is nonsingular, but A is singular. If AB is nonsingular, then there exists C such that $ABC = I$ and hence $A = C^{-1}B^{-1}$, a contradiction to A is singular. ■

Theorem 5.4 Two matrices A and B are row equivalent if and only if there exists an invertible matrix E such that $EA = B$.

Proof. The "only if" part is shown by Theorem 5.1.

To show the "if" part, we know from Theorem 5.2 that the invertible matrix E can be written as the product of a finite sequence of elementary matrices, each of which performs an elementary row operation when premultiplied to a matrix. ■

Equivalent Statements for Square Matrix A

1. A is nonsingular.
2. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{0}$.
3. A is row equivalent to I.
4. A is a product of a sequence of finite elementary matrices.
5. $A\mathbf{x} = \mathbf{b}$ has a unique solution for any $\mathbf{b}$.
6. There exists an invertible matrix E such that $EA = I$.

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Lecture 6

LU Decomposition

LU Decomposition is a matrix representation of the Gaussian elimination process.
Sec 6.1.1, 6.2 and 6.3 are NOT required for the course!

6.1 LU Decomposition without Interchanges

■ **Example 6.1** The following row operations transform matrix A to an upper triangular matrix U:

$$A = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 2 \\ 4 & -1 & 9 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 1 \\ 4 & -1 & 9 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 2 \\ 0 & -9 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & 8 \end{bmatrix} = U.$$

This sequence of transformation can be represented in terms of elementary matrices

$$E_3 E_2 E_1 A = U$$

where

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}.$$

As elementary matrices are nonsingular, we can write

$$A = LU$$

where

$$L = E_1^{-1} E_2^{-1} E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 2 & -3 & 1 \end{bmatrix}.$$

So A can be written as the multiplication of a lower triangular matrix and an upper triangular matrix.

■

Suppose that we can transform a square matrix A to upper triangular form by Gaussian elimination using only elementary row operation III, which constitutes of a sequence of elementary row operations gives by $E_1, E_2, \ldots, E_u$. Therefore,

$$E_u E_{u-1} \cdots E_1 A = U,$$

where U is upper triangular. Let

$$L = (E_u E_{u-1} \cdots E_1)^{-1} = E_1^{-1} E_2^{-1} \cdots E_u^{-1}.$$

We have $A = LU$. We can show that L is unit, lower triangular.

■ **Example 6.2** Check that

$$\begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & b & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ 0 & b & 1 \end{bmatrix}$$

but

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & b & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ ab & b & 1 \end{bmatrix}$$

■

An illustration of 4×4 matrices

$$\begin{aligned} & E_1^{-1} E_2^{-1} \cdots E_6^{-1} \\ &= \begin{bmatrix} 1 & & & \\ a & 1 & & \\ 0 & 0 & 1 & \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ b & 0 & 1 & \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0 & 0 & 1 & \\ c & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0 & d & 1 & \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ & \quad \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0 & 0 & 1 & \\ 0 & e & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ 0 & 1 & & \\ 0 & 0 & 1 & \\ 0 & 0 & f & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & & & \\ a & 1 & & \\ b & d & 1 & \\ c & e & f & 1 \end{bmatrix}. \end{aligned}$$

Gaussian elimination using only row operation III can be stated using the following pseudo-code:

Let U be the upper triangular part of the resulting matrix, and let L be an unit, lower triangular matrix with the (i, j) entry, $i > j$, l_{ij}. Then $A = LU$.

Algorithm 1 Gaussian elimination without row interchanges.

Gaussian Elimination without row Interchanges

1: **for** $j = 1, 2, \ldots, n-1$ **do** ▷ provided that $a_{jj} \neq 0$
2: **for** $i = j+1, \ldots, n$ **do**
3: Set $l_{ij} = \dfrac{a_{ij}}{a_{jj}}$
4: **for** $k = j+1, \ldots, n$ **do**
5: Set $a_{ik} = a_{ik} - l_{ij}a_{jk}$
6: **end for**
7: **end for**
8: **end for**

LU Decomposition

- If a square matrix A can be reduced to upper triangular form using only row operation III, then A has an LU decomposition.
  - U is formed by reducing A to upper triangular form using only row operation III.
  - L is a unit triangular matrix with l_{ij} $(i > j)$ be the multiple of the jth row subtracted from the ith row during the reduction.
- The value of l_{ij} $(i > j)$ can be stored in the (i, j) entry of A.

R LU decomposition of an *invertible matrix* is unique. LU decomposition of a matrix is not unique in general.

By counting the number of iterations in Algorithm 1, we can get the computation cost of Gaussian elimination. The same computation gives us L and U.

Operation Count of Gaussian Elimination

- div: $\sum_{j=1}^{n-1}\sum_{i=j+1}^{n} 1 = n(n-1)/2 \approx n^2/2$
- mul: $\sum_{j=1}^{n-1}\sum_{i=j+1}^{n}\sum_{k=j+1}^{n} 1 = n(2n-1)(n-1)/6 \approx n^3/3$
- add: $\sum_{j=1}^{n-1}\sum_{i=j+1}^{n}\sum_{k=j+1}^{n} 1 = n(2n-1)(n-1)/6 \approx n^3/3$

6.1.1 L is unit, lower triangular*

Consider two unit, lower triangular matrices A and B. Let $A' = A - I$ and $B' = B - I$. Then

$$AB = (I + A')(I + B') = I + A' + B' + A'B'.$$

We know $A'B'$ is the zero matrix if A and B satisfy some constraints. Suppose that the (i_k, j_k) entry is the only nonzero off-diagonal entry of E_k and E_k^{-1}. We know from the GE process that for $k' < k$, $j_{k'} \leq j_k$.

Proposition 6.1 $L = I + \tilde{E}_1 + \tilde{E}_2 + \cdots \tilde{E}_u$, where $\tilde{E}_k = E_k^{-1} - I$.

Proof. We prove by induction. Let $L^{(k)} \triangleq E_1^{-1}E_2^{-1}\cdots E_k^{-1}$. We know that $L^{(1)} = I + \tilde{E}_1$. Assume that $L^{(k)} = I + \tilde{E}^{(k)}$ where $\tilde{E}^{(k)} \triangleq \tilde{E}_1 + \tilde{E}_2 + \cdots \tilde{E}_k$ for certain

$k \geq 1$. We can write

$$\begin{aligned} L^{(k+1)} &= L^{(k)} E_{k+1}^{-1} \\ &= (I + \tilde{E}^{(k)})(I + \tilde{E}_{k+1}) \\ &= I + \tilde{E}^{(k)} + E_{k+1} + \tilde{E}^{(k)} \tilde{E}_{k+1}. \end{aligned}$$

To complete the induction, we only need to show that $\tilde{E}^{(k)} \tilde{E}_{k+1} = \mathbf{0}$. We know that $\tilde{E}_{k+1}$ has only (i_{k+1}, j_{k+1}) entry $l_{i_{k+1}, j_{k+1}}$ nonzero. Therefore, all the (i, j), $j \neq j_{k+1}$ entries of $\tilde{E}^{(k)} \tilde{E}_{k+1}$ are zero. The (i, j_{k+1}) entry of $\tilde{E}^{(k)} \tilde{E}_{k+1}$ is $[\tilde{E}^{(k)}]_{i, i_{k+1}} l_{i_{k+1}, j_{k+1}}$. As $\tilde{E}^{(k)} = \tilde{E}_1 + \tilde{E}_2 + \cdots \tilde{E}_k$, we know that an (i, j) entry of $\tilde{E}^{(k)}$ is zero if $j > j_{k+1}$. As $i_{k+1} > j_{k+1}$, we have $[\tilde{E}^{(k)}]_{i, i_{k+1}} = 0$, and hence the (i, j_{k+1}) entry of $\tilde{E}^{(k)} \tilde{E}_{k+1}$ is zero. ■

6.1.2 Solving Systems of Linear Equations

Solving by LU Decomposition

- Consider a system $A\mathbf{x} = \mathbf{b}$, where A has an LU decomposition $A = LU$.
- The system can be solved in two steps:
  1. Solve $L\mathbf{y} = \mathbf{b}$ and get a solution $\hat{\mathbf{y}}$ of $\mathbf{y}$.
  2. Solve $U\mathbf{x} = \hat{\mathbf{y}}$.
- Computation cost of the above two steps:
  - div: n
  - mul: $n(n-1)$
  - add: $n(n-1)$

■ **Example 6.3** Consider the system $A\mathbf{x} = \mathbf{b}$ with

$$A = \begin{bmatrix} 2 & 4 & 2 \\ 1 & 5 & 2 \\ 4 & -1 & 9 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}.$$

We know that A has an LU decomposition with

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 2 & -3 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & 4 & 2 \\ 0 & 3 & 1 \\ 0 & 0 & 8 \end{bmatrix}.$$

First solve $L\mathbf{y} = \mathbf{b}$ using "forward substitution" and obtain $\mathbf{y} = \begin{bmatrix} 2 & 0 & -4 \end{bmatrix}^{\mathrm{T}}$. Then solve $U\mathbf{x} = \mathbf{y}$ using backward substitution and obtain a solution $\begin{bmatrix} \frac{5}{6} & \frac{1}{3} & -\frac{1}{2} \end{bmatrix}^{\mathrm{T}}$ of $\mathbf{x}$. ■

Solving multiple Systems of the same Coefficient Matrices

- In general, we can solve a system $AX = B$ using LU decomposition by
  - first solving $LY = B$ using forward substitution, and
  - then solving $UX = Y$ using forward substitution.
- Computation cost:
  - The LU decomposition of A is only need to be done once, and
  - The back/forward substitutions need to be done multiple times for each column of B.

Calculating of Matrix Inversion

- Approach 1: Transform $[A\ I]$ to $[I\ A^{-1}]$ by Gaussian-Jordan elimination.
- Approach 2: Solve $AX = I$ by LU decomposition.
- The computation cost:
  - div: $\approx n^2$
  - mul: $\approx n^3$
  - add: $\approx n^3$

The computation cost of Approach 1 can be calculated similar to that of Algorithm 1, except that the loop in Line 4 runs to $n + i$:

- div: $\sum_{i=1}^{n-1}\sum_{k=i+1}^{n} 1 = n(n-1)/2 \approx n^2/2$
- mul: $\sum_{i=1}^{n-1}\sum_{k=i+1}^{n}\sum_{j=i+1}^{n+i} 1 = n^2(n-1)/2 \approx n^3/2$
- add: $\sum_{i=1}^{n-1}\sum_{k=i+1}^{n}\sum_{j=i+1}^{n+i} 1 = n^2(n-1)/2 \approx n^3/2$

together with the back substitution cost:

- div: $n^2/2$
- mul: $n^3/2$
- add: $n^3/2$

Approach 2 has a lower computation cost than using the LU decomposition to solve $AX = B$, where B is a general $n \times n$ matrix. Due to I's special structure, the forward substitution step has only $n^3/6$ mul/add operations instead of $n^3/2$.

Given A^{-1} the computation of $A^{-1}\mathbf{b}$ has the cost

- mul: n^2,
- add: $n(n-1)$.

The number of operations of backward and forward substitutions is almost the same as that of matrix vector multiplication $A^{-1}\mathbf{b}$, provided that A^{-1} is given. The major difference of these two approach lies in the complexity of computing inverse and LU decomposition.

Computation Cost Comparison

Table 6.1: Computation cost comparison

	div	mul	add
$[A\ I] \xrightarrow{G.J.} [I\ A^{-1}]$	$\approx n^2$	$\approx n^3$	$\approx n^3$
$A^{-1}\mathbf{b}$		n^2	$n(n-1)$
$A = LU$	$\approx \frac{n^2}{2}$	$\approx \frac{n^3}{3}$	$\approx \frac{n^3}{3}$
Substitution	n	$n(n-1)$	$n(n-1)$

6.2 LU Decomposition with Interchanges*

■ **Example 6.4** Let

$$A = \begin{bmatrix} 6 & -4 & 2 \\ 4 & 2 & 1 \\ 2 & -1 & 1 \end{bmatrix}.$$

We can reduce A to upper triangular form by row operations I and III:

$$A \to \begin{bmatrix} 2 & -1 & 1 \\ 4 & 2 & 1 \\ 6 & -4 & 2 \end{bmatrix} \to \begin{bmatrix} 2 & -1 & 1 \\ 0 & 4 & -1 \\ 6 & -4 & 2 \end{bmatrix} \to \begin{bmatrix} 2 & -1 & 1 \\ 0 & 4 & -1 \\ 0 & -1 & -1 \end{bmatrix}$$
$$\to \begin{bmatrix} 2 & -1 & 1 \\ 0 & -1 & -1 \\ 0 & 4 & -1 \end{bmatrix} \to \begin{bmatrix} 2 & -1 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & -5 \end{bmatrix} \triangleq U.$$

This sequence of row operations can be represented by elementary matrices as $E_{32}P_{23}E_{31}E_{21}P_{13}A = U$, where

$$P_{13} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix},$$
$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}, \quad E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix}.$$

Note that

$$\begin{aligned} P_{23}E_{31}E_{21} &= P_{23}E_{31}(P_{23}P_{23})E_{21}(P_{23}P_{23}) \\ &= (P_{23}E_{31}P_{23})(P_{23}E_{21}P_{23})P_{23} \\ &= E'_{31}E'_{21}P_{23}, \end{aligned}$$

where

$$E'_{21} = P_{23}E_{31}P_{23}, \quad E'_{31} = P_{23}E_{21}P_{23}.$$

Note that E'_{21} and E'_{31} are elementary matrices. Therefore, we can write the reduction as

$$E_{32}E'_{31}E'_{21}P_{23}P_{13}A = U,$$

i.e., $PA = LU$ with $P = P_{23}P_{13}$ and $L = (E_{32}E'_{31}E'_{21})^{-1}$.

■

In general, for any nonsingular matrix A, there exists a permutation matrix P such that PA has an LU decomposition.

- The multiplier $l_{\pi_k i}$ is stored in the (π_k, i) entry of A being eliminated.
- The vector π can be used to form a permutation matrix P with ith row being the π_ith row of the identity matrix.
- The matrix PA can be factored into a product LU where L is unit lower triangular with the (k, i) entry $l_{\pi_k i}$ for $k > i$ and U is upper triangular with (k, i) entry $a_{\pi_k i}$ for $k \leq i$.

Three steps of solving $A\mathbf{x} = \mathbf{b}$ given that $A = P^{-1}LU$:

1. Reordering: $\mathbf{c} = P\mathbf{b}$. It follows that $LU\mathbf{x} = PA\mathbf{x} = P\mathbf{b} = \mathbf{c}$.
2. Forward substitution: Solve the system $L\mathbf{y} = \mathbf{c}$.
3. Backward substitution: Solve the system $U\mathbf{x} = \mathbf{y}$.

Algorithm 2 Gaussian elimination with interchanges.

Gaussian Elimination with Interchanges

Initialize $\pi = (\pi_1, \ldots, \pi_n)$ with $\pi_i = i$ for $i = 1, \ldots, n$
for $j = 1, 2, \ldots, n-1$ **do**
    Choose a pivot element $a_{\pi_i j}$ from $a_{\pi_j j}, a_{\pi_{j+1} j}, \ldots, a_{\pi_n j}$
    Switch the ith and the jth entries of π
    for $i = j+1, \ldots, n$ **do**
        Set $l_{\pi_i j} = \dfrac{a_{\pi_i j}}{a_{\pi_j j}}$
        for $j' = j+1, \ldots, n$ **do**
            Set $a_{\pi_i j'} = a_{\pi_i j'} - l_{\pi_i j} a_{\pi_j j'}$
        end for
    end for
end for

6.3 LU Decomposition for General Matrices*

LU decomposition for general matrices.

Algorithm 3 Gaussian elimination for a general matrix.

Initialize $\pi = (\pi_1, \ldots, \pi_m)$ with $\pi_i = i$ for $i = 1, \ldots, m$
set $i = 1$
for $j = 1, 2, \ldots, n$ **do**
    if there exists $a_{\pi_i j} \neq 0$ from $a_{\pi_j j}, a_{\pi_{j+1} j}, \ldots, a_{\pi_n j}$ **then**
        Switch the ith and the jth entries of π
        for $i' = i+1, \ldots, m$ **do**
            Set $l_{\pi_{i'} i} = \dfrac{a_{\pi_{i'} j}}{a_{\pi_i j}}$
            for $j' = j+1, \ldots, n$ **do**
                Set $a_{\pi_{i'} j'} = a_{\pi_{i'} j'} - l_{\pi_{i'} i} a_{\pi_i j'}$
            end for
        end for
        Set $i = i+1$. End the loop if $i > m$.
    end if
end for

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Part II

Vectors and Vector Spaces

Draft of Linear Algebra Text in Progress (UNR-22)
Ron Buckmire
Fall 2022 Math 322 Class Notes
Please refer to the most recent version
http://faculty.oxy.edu/ron/math/322/22/

Printed by Shervine Amidi
For linear algebra lecture at CentraleSupélec
Please refer to the original lecture at cs229.stanford.edu
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Lecture 7

Linear Span and Independence

7.1 Linear Combinations

Let's recall linear combinations.

Definition 7.1 Let $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ be column vectors of length m, and $c_1, c_2, \ldots, c_n$ be scalars. The *linear combination* of $a_1, a_2, \ldots, a_n$ with coefficients $c_1, c_2, \ldots, c_n$ is the vector

$$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \ldots + c_n\mathbf{a}_n.$$

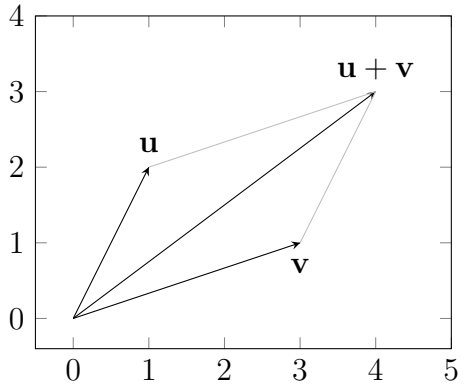
When we say that a vector $\mathbf{b}$ is "**a linear combination of** $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$" it will mean that $\mathbf{b} = c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \ldots + c_n\mathbf{a}_n$ for some choice of $c_1, c_2, \ldots, c_n$.

Let's think about vector addition and scalar multiplication geometrically.

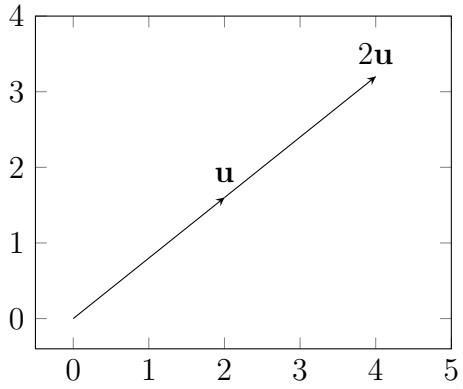
Connection with Geometry

- Consider the coordinate system $\mathbb{R}^2$ — each coordinate specifies the distance from the origin on an axis, where the axes are perpendicular.
- We can associate each geometric point (x, y) in the plane with a column vector $\begin{bmatrix} x \\ y \end{bmatrix}$.

Geometric Illustration of Vector Addition



Geometric Illustration of Scaling a Vector



Given a set of vectors $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n \in \mathbb{R}^m$, a new vector in $\mathbb{R}^m$ can be formed by their linear combination.

- Is $\mathbf{b} \in R^m$ a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$?
- What are the linear combination coefficients?
- Whether $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ have a unique linear combination to be $\mathbf{b}$?

The first question is equivalent to ask whether $A\mathbf{x} = \mathbf{b}$ is consistent, where $A = \begin{bmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \ldots & \mathbf{a}_n\end{bmatrix}$, as the system can be written as $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$.

The second question is answered by Gaussian elimination.

The third question is equivalent to ask whether the system $A\mathbf{x} = \mathbf{b}$ has a unique solution.

7.2 Linear Span

Definition 7.2 If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are vectors in $\mathbb{R}^m$, then the set of all linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, denoted by $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$, is called the subset of $\mathbb{R}^m$ **spanned** (or **generated**) by $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$.

- In other words,

$$\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\} = \{t_1\mathbf{v}_1 + t_2\mathbf{v}_2 + \cdots + t_n\mathbf{v}_n : \forall t_i \in \mathbb{R}\}.$$

- A set $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is called a **spanning set** of V if $V = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$.

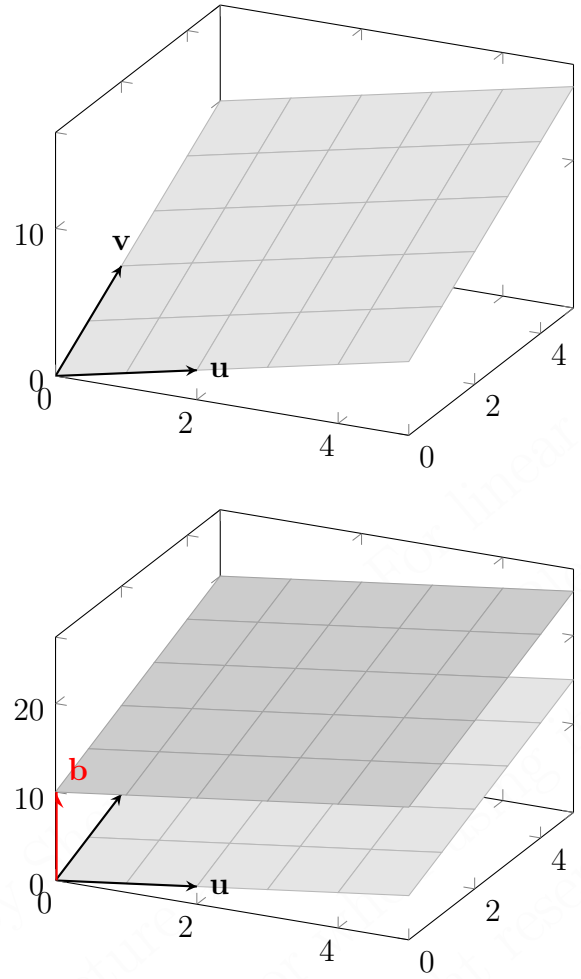
■ **Example 7.1** Illustration of the span of $\mathbf{u} = \begin{bmatrix}2 & 0 & 2\end{bmatrix}^{\mathrm{T}}$, the span of $\mathbf{v} = \begin{bmatrix}0 & 2 & 4\end{bmatrix}^{\mathrm{T}}$, and the span of $\mathbf{u}$ and $\mathbf{v}$. ■

It is not necessary that the linear span of two vectors be a plane. For example,

$$\text{Span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\3\\0\end{bmatrix}\right\} = \text{Span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}\right\}.$$

For a set of vectors V and a vector $\mathbf{b}$, define

$$\mathbf{b} + V = \{\mathbf{b} + \mathbf{v} : \mathbf{v} \in V\}.$$



7.3 Solution Sets Revisited

We will now revisit the solution set of a system of linear equations.

Consider the system of linear equations

$$\begin{aligned} 2x_1 + x_2 + 7x_3 - 7x_4 &= 8 \\ -3x_1 + 4x_2 - 5x_3 - 6x_4 &= -12 \\ x_1 + x_2 + 4x_3 - 5x_4 &= 4 \end{aligned}$$

The augmented matrix can be transformed into reduced row-echelon form as

$$\left[\begin{array}{cccc|c} \boxed{1} & 0 & 3 & -2 & 4 \\ 0 & \boxed{1} & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

We know from the augmented matrix that the system has infinitely many solutions. An equivalent system is

$$\begin{aligned} x_1 &= 4 - 3x_3 + 2x_4 \\ x_2 &= -x_3 + 3x_4, \end{aligned}$$

where x_3 and x_4 are free variables.

Add two equations $x_3 = x_3$ and $x_4 = x_4$, we obtain

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4 - 3x_3 + 2x_4 \\ -x_3 + 3x_4 \\ x_3 \\ x_4 \end{bmatrix},$$

which can be written further as

$$\mathbf{x} = \begin{bmatrix} 4 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ -1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ 3 \\ 0 \\ 1 \end{bmatrix}.$$

All the solutions of the system are linear combinations of the three vectors, where the coefficient of the first vector is 1, and the coefficients of the other two vectors can be any real value. This form of the solution set is called *the parametric vector form*. Using the notations of linear span, the solution set is

$$\begin{bmatrix} 4 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \text{Span} \left\{ \begin{bmatrix} -3 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

■ **Example 7.2** What about the solution set of the homogeneous system with the same coefficient matrix?

$$\left[\begin{array}{cccc|c} \boxed{1} & 0 & 3 & -2 & 0 \\ 0 & \boxed{1} & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

■

In general, we know that for a consistent linear system, the augmented matrix is row equivalent to a matrix of the form $[I\ B \mid b]$ by changing the order of the variables. Write $\mathbf{x} = \begin{bmatrix} \mathbf{x}_a \\ \mathbf{x}_f \end{bmatrix}$, where $\mathbf{x}_f$ includes all the free variables. The equivalent system can written as

$$\begin{bmatrix} I & B \end{bmatrix} \mathbf{x} = \begin{bmatrix} I & B \end{bmatrix} \begin{bmatrix} \mathbf{x}_a \\ \mathbf{x}_f \end{bmatrix} = \mathbf{b},$$

i.e.,

$$\mathbf{x}_a = \mathbf{b} - B\mathbf{x}_f.$$

Hence,

$$\mathbf{x} = \begin{bmatrix} \mathbf{x}_a \\ \mathbf{x}_f \end{bmatrix} = \begin{bmatrix} \mathbf{b} - B\mathbf{x}_f \\ \mathbf{x}_f \end{bmatrix} = \begin{bmatrix} \mathbf{b} - B\mathbf{x}_f \\ \mathbf{0} + \mathbf{x}_f \end{bmatrix} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} -B\mathbf{x}_f \\ \mathbf{x}_f \end{bmatrix} = \begin{bmatrix} \mathbf{b} \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} -B \\ I \end{bmatrix} \mathbf{x}_f.$$

Solution Set of a Homogenerous System

- The solution set S_H of a homogeneous system of linear equations $A\mathbf{x} = 0$ is the set spanned by $n-r$ vectors, where r is the number of dependent variables.
- A spanning set of S_H can be found from the row echelon form of A.

Consistency of Non-Homogenerous Systems

- When $A\mathbf{x} = \mathbf{b}$ has a solution?
- If $\mathbf{b} \in \text{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$, the system $A\mathbf{x} = \mathbf{b}$ has a solution.
- If $\mathbf{b} \notin \text{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$, the system $A\mathbf{x} = \mathbf{b}$ has no solution.

R The question whether a vector $\mathbf{b}$ is in $\text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\}$, is equivalent to the question whether $A\mathbf{x} = \mathbf{b}$ is consistent.

Theorem 7.1 Suppose that $\hat{\mathbf{x}}$ is a solution to $A\mathbf{x} = \mathbf{b}$, then a vector $\mathbf{y}$ is also a solution if and only if $\mathbf{y} = \hat{\mathbf{x}} + \mathbf{z}$ where $A\mathbf{z} = \mathbf{0}$.

Corollary 7.2 A consistent system $A\mathbf{x} = \mathbf{b}$ has a unique solution if and only if $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

- Let S_H be the solution set to the homogeneous system $A\mathbf{x} = \mathbf{0}$.
- The theorem tells that the solution set S to the system $A\mathbf{x} = \mathbf{b}$ is

$$S = \hat{\mathbf{x}} + S_H,$$

where $\hat{\mathbf{x}}$ is any solution of $A\mathbf{x} = \mathbf{b}$.

Proof. "if": Suppose $\mathbf{y} = \hat{\mathbf{x}} + \mathbf{z}$ where $A\mathbf{z} = \mathbf{0}$. Then $A\mathbf{y} = A(\hat{\mathbf{x}} + \mathbf{z}) = A\hat{\mathbf{x}} + A\mathbf{z} = \mathbf{b} + \mathbf{0} = \mathbf{b}$.

"only if": Suppose that $\hat{\mathbf{x}}$ and $\mathbf{y}$ are both solutions of $A\mathbf{x} = \mathbf{b}$. Let $\mathbf{z} = \mathbf{y} - \hat{\mathbf{x}}$. Then $A\mathbf{z} = A(\mathbf{y} - \hat{\mathbf{x}}) = \mathbf{b} - \mathbf{b} = \mathbf{0}$. ■

7.4 Linear Independence

- Whether the homogeneous system $A\mathbf{x} = 0$ has a unique solution or many solutions is an important question.
- The question is equivalent to whether there exists $x_1, \ldots, x_n$, not all zero, such that

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{0}.$$

Definition 7.3 The vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in $\mathbb{R}^m$ are said to be *linearly independent* if

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_n\mathbf{v}_n = \mathbf{0}$$

implies that all the scalars $c_1, \ldots, c_n$ are 0.

Definition 7.4 The vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in $\mathbb{R}^m$ are said to be *linearly dependent* if there exists scalars $c_1, c_2, \ldots, c_n$, not all zero, such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_n\mathbf{v}_n = \mathbf{0}.$$

■ **Example 7.3** Determine whether the following sets of vectors are linearly dependent or not.

1. $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \right\}$
2. $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 7 \\ 0 \end{bmatrix} \right\}$
3. $\{\mathbf{0}\}$
4. $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{0}\}$

■

- To determine whether vectors $\mathbf{a}_1, \ldots \mathbf{a}_n$ are linearly dependent or not, we can check whether the system $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots x_n\mathbf{a}_n = A\mathbf{x} = \mathbf{0}$ has a non-trivial solution or not.
- In other words, if the columns of A are linearly independent, the system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. If the columns of A are linearly dependent, the system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.
  - For a square matrix A, its columns are linearly dependent if and only if A is singular.
  - For an $m \times n$ matrix A with $m < n$, its columns are linearly dependent.

Equivalent Statements for Square Matrix A

1. A is nonsingular.
2. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{0}$.
3. A is row equivalent to I.
4. A is a product of a sequence of finite elementary matrices.
5. $A\mathbf{x} = \mathbf{b}$ has a unique solution for any $\mathbf{b}$.
6. There exists an invertible matrix E such that $EA = I$.
7. The columns of A are linearly independent.

Lecture 8

Vector Spaces

We have studied extensively the operations of addition and scalar multiplication of vectors of real numbers, in terms of linear combination, span and independence. We will keep on developing more theory about vectors. But this theory about the operations of addition and scalar multiplication are used in many contexts beyond the vectors of real numbers.

8.1 Vector Space Axioms

Let $\mathcal{V}$ be a set with two operations *addition* and *scalar multiplication* defined:

- addition: for any $\mathbf{x}, \mathbf{y} \in \mathcal{V}$, associate a unique element $\mathbf{x}+\mathbf{y}$ in $\mathcal{V}$,
- scalar multiplication: for any $\mathbf{x} \in \mathcal{V}$ and scalar α, associate a unique element $\alpha\mathbf{x}$ in $\mathcal{V}$.

In other words, $\mathcal{V}$ is *closed* under the operations of addition and scalar multiplication.

We can verify that the addition and scalar multiplication of matrices are defined for both $\mathbb{R}^n$ and $\mathbb{R}^{m\times n}$. Actually, $\mathbb{R}^{m\times n}$ is similar to $\mathbb{R}^{mn}$ for these two operations. Consider the set $\mathcal{W} = \{(a, 1) : a \in \mathbb{R}\}$. Is the addition and scalar multiplication of matrices defined for $\mathcal{W}$?

Let $C[a, b]$ denote the set of all real-valued functions that are defined and continuous on $[a, b]$.

- The *sum* $f + g$ of two functions $f, g \in C[a, b]$ is defined by $(f + g)(x) = f(x) + g(x)$. The new function $f + g$ is in $C[a, b]$ as the sum of two continuous functions are continuous.
- The *scalar multiplication* αf of $f \in C[a, b]$ and scalar α is defined by $(\alpha f)(x) = \alpha f(x)$.

Let P_n denote the set of all polynomials of degree less than n, i.e., $p \in P_n$ can be written as $p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_{n-1}x^{n-1}$.

- Define $p + q$ as $(p + q)(x) = p(x) + q(x)$.
- Define αp as $(\alpha p)(x) = \alpha p(x)$.

Definition 8.1 A set $\mathcal{V}$, on which two operations *addition* and *scalar multiplication* are defined, is said to form a *vector space* if the following axioms are satisfied:

A1. $\mathbf{x}+\mathbf{y} = \mathbf{y}+\mathbf{x}$ for any $\mathbf{x}, \mathbf{y} \in \mathcal{V}$.

A2. $(\mathbf{x}+\mathbf{y})+\mathbf{z}=\mathbf{x}+(\mathbf{y}+\mathbf{z})$ for any $\mathbf{x},\mathbf{y},\mathbf{z}\in\mathcal{V}$.
A3. There exists $\mathbf{0}\in\mathcal{V}$ such that $\mathbf{x}+\mathbf{0}=\mathbf{x}$ for each $\mathbf{x}\in\mathcal{V}$.
A4. For each $\mathbf{x}\in\mathcal{V}$, there exists $\mathbf{x}'\in\mathcal{V}$ such that $\mathbf{x}+\mathbf{x}'=\mathbf{0}$, where $\mathbf{x}'$ is usually denoted as $-\mathbf{x}$.
A5. $\alpha(\mathbf{x}+\mathbf{y})=\alpha\mathbf{x}+\alpha\mathbf{y}$ for each scalar α and any $x,y\in\mathcal{V}$.
A6. $(\alpha+\beta)\mathbf{x}=\alpha\mathbf{x}+\beta\mathbf{x}$ for any scalars α and β and any $\mathbf{x}\in\mathcal{V}$.
A7. $(\alpha\beta)\mathbf{x}=\alpha(\beta\mathbf{x})$ for any scalars α and β and any $\mathbf{x}\in\mathcal{V}$.
A8. $1\mathbf{x}=\mathbf{x}$ for all $\mathbf{x}\in\mathcal{V}$.

A vector space is also called a linear space. The elements in a vector space are also called *vectors*.

■ **Example 8.1**
- $\mathbb{R}^n$, $n\geq 1$ is a vector space.
- $\mathbb{R}^{m\times n}$ is a vector space.
- check that P_n is a vector space.
- check that $C[a,b]$ is a vector space.

■

What is the zero polynomial (the zero vector) in P_n? What is the zero function (the zero vector) in $C[a,b]$?

Theorem 8.1 Let $\mathcal{V}$ be a vector space.
1. The zero vector $\mathbf{0}\in\mathcal{V}$ is unique.
2. $0\mathbf{x}=\mathbf{0}$ for each $\mathbf{x}\in\mathcal{V}$.
3. $c\mathbf{0}=\mathbf{0}$ for any scalar c.
4. For any $\mathbf{x},\mathbf{y}\in\mathcal{V}$, if $\mathbf{y}+\mathbf{x}=\mathbf{0}$ then $\mathbf{y}=-\mathbf{x}$.
5. $(-1)\mathbf{x}=-\mathbf{x}$ for each $\mathbf{x}\in\mathcal{V}$.

Proof.
1. Suppose that $\mathbf{0}$ and $\mathbf{0}'$ are both zero vectors. Then, it follows from axioms A1 and A3 that $\mathbf{0}=\mathbf{0}+\mathbf{0}'=\mathbf{0}'+\mathbf{0}=\mathbf{0}'$.
2. First, $\mathbf{x}\overset{A8}{=}1\mathbf{x}=(1+0)\mathbf{x}\overset{A6}{=}\mathbf{x}+0\mathbf{x}$. Hence, $\mathbf{0}\overset{A4}{=}-\mathbf{x}+\mathbf{x}=-\mathbf{x}+(\mathbf{x}+0\mathbf{x})\overset{A2}{=}(-\mathbf{x}+\mathbf{x})+0\mathbf{x}\overset{A4}{=}\mathbf{0}+0\mathbf{x}\overset{A1,A3}{=}0\mathbf{x}$.
3. $c\mathbf{0}=c\mathbf{0}+\mathbf{0}=c\mathbf{0}+(c\mathbf{0}+(-c\mathbf{0}))=(c\mathbf{0}+c\mathbf{0})+(-c\mathbf{0})=c(\mathbf{0}+\mathbf{0})+(-c\mathbf{0})=c\mathbf{0}+(-c\mathbf{0})=\mathbf{0}$, where we apply A3, A4, A2, A5, A3 and A4, respectively.
4. $\mathbf{y}\overset{A3}{=}\mathbf{y}+\mathbf{0}\overset{A4}{=}\mathbf{y}+(\mathbf{x}+(-\mathbf{x}))\overset{A2}{=}(\mathbf{y}+\mathbf{x})+(-\mathbf{x})\overset{A4}{=}\mathbf{0}+(-\mathbf{x})\overset{A1,A3}{=}-\mathbf{x}$.
5. As $\mathbf{x}+(-1)\mathbf{x}=1\mathbf{x}+(-1)\mathbf{x}=(1+(-1))\mathbf{x}=0\mathbf{x}=\mathbf{0}$ (where we apply A8, A6 and 2), respectively), we have $(-1)\mathbf{x}=-\mathbf{x}$ by 4).

■

Our previous definitions on $\mathbb{R}^n$, e.g., linear combination, span, independence, can be defined on any vector space.

Consider a vector space $\mathcal{V}$ and $\mathbf{v}_1,\ldots,\mathbf{v}_n\in\mathcal{V}$.
- Linear combination: $c_1\mathbf{v}_1+c_2\mathbf{v}_2+\cdots+c_n\mathbf{v}_n\in\mathcal{V}$.
- Linear span: $\mathrm{Span}\{\mathbf{v}_1,\ldots,\mathbf{v}_n\}\subset V$.
- Linear independence: $c_1\mathbf{v}_1+c_2\mathbf{v}_2+\cdots+c_n\mathbf{v}_n=\mathbf{0}$ implies $c_1,\ldots,c_n$ are all zero.

The linear independence of matrices $M_1,\ldots,M_k\in\mathbb{R}^{m\times n}$ is equivalent to that the system

$$(M_1)_{ij}x_1+\cdots+(M_k)_{ij}x_k=0,\quad 1\leq i\leq m, 1\leq j\leq n$$

has only the trivial solution.

How to test whether vectors (polynomials) $p_1, p_2, \ldots, p_k$ are linearly independent in P_n?

■ **Example 8.2** Test whether the vectors

$$p_1(x) = x^2 - 2x + 3, \quad p_2(x) = 2x^2 + x + 8, \quad p_3(x) = x^2 + 8x + 7$$

in P_3 are linearly independent. ■

How to test whether the functions $x, x^2, \sin(x) \in C[-1, 1]$ are linearly independent? Consider the system

$$c_1 x + c_2 x^2 + c_3 \sin(x) = 0, \quad \forall x \in [-1, 1].$$

This is a system of infinitely many linear equations. These three functions are linearly independent if and only if the system has only the trivial solution.

8.2 Subspaces

A subset of a vector space $\mathcal{V}$ can also be a vector space using the operations of $\mathcal{V}$.

Definition 8.2 Let $\mathcal{V}$ be a vector space. A nonempty subset $\mathcal{S}$ of $\mathcal{V}$ is called a *subspace* of $\mathcal{V}$ if $\mathcal{S}$ satisfies the conditions

1. $\alpha\mathbf{x} \in \mathcal{S}$ whenever $\mathbf{x} \in \mathcal{S}$ for any scalar α, and
2. $\mathbf{x} + \mathbf{y} \in \mathcal{S}$ whenever $\mathbf{x}, \mathbf{y} \in \mathcal{S}$.

A subspace is a vector space in its own right:

- A1-2,5-8 are valid for any vectors in $\mathcal{V}$, and hence for vectors in $\mathcal{S}$.
- A3 is valid by Theorem 8.1-2).
- A4 is valid by Theorem 8.1-5).

How to Check a Subset is a Subspace?

- Let $\mathcal{S}$ be a subset of a vector space $\mathcal{V}$.
- Verify that $\mathcal{S}$ is a subspace using the following steps:
  0. if it is not clear whether $\mathcal{S}$ is nonempty, check $\mathbf{0} \in \mathcal{S}$, which implies that $\mathcal{S}$ is nonempty.
  1. check $\alpha\mathbf{x} \in \mathcal{S}$.
  2. check $\mathbf{x} + \mathbf{y} \in \mathcal{S}$.
- $\{\mathbf{0}\}$ and $\mathcal{V}$ are two subspaces of a vector space $\mathcal{V}$.

■ **Example 8.3**

- $\mathbb{R}^2$ is **not** a subspace of $\mathbb{R}^3$.
- $\{(x_1, x_2, 0) : x_1, x_2 \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3$.

■

■ **Example 8.4**

- Let $\mathcal{W} = \{\mathbf{x} \in \mathbb{R}^n : x_1 \geq 0\}$. Is $\mathcal{W}$ a subspace of $\mathbb{R}^n$?
- Let $\mathcal{T} = \{\mathbf{x} \in \mathbb{R}^n : x_1 + x_2 + \cdots + x_n = 1\}$. Is $\mathcal{T}$ a subspace of $\mathbb{R}^n$?

■

■ **Example 8.5** Let

$$\mathcal{W} = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x + y - 3z = 0 \right\}.$$

Is $\mathcal{W}$ a subspace of $\mathbb{R}^3$? ■

Recall that the solution set of a homogeneous system can be regarded as the collection of all the linear combinations of certain vectors, i.e., a span.

Theorem 8.2 If $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are elements of a vector space $\mathcal{V}$, then $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a subspace of $\mathcal{V}$.

Corollary 8.3 The solution set of a homogeneous linear system is a subspace.

Proof. Let $\mathcal{S} = \text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$. First, $\mathcal{S}$ is non-empty. The theorem can then be proved by verifying the two conditions of a subspace (exercise). ■

■ **Example 8.6** Let $\mathcal{S} = \{A \in \mathbb{R}^{2\times 2} : a_{12} = -a_{21}\}$. Is $\mathcal{S}$ a subspace of $\mathbb{R}^{2\times 2}$?
The set of (anti)symmetric $n \times n$ matrices is a subspace of $\mathbb{R}^{n\times n}$. ■

■ **Example 8.7** Let $\mathcal{S}$ be the set of all polynomials of degree less than n with the property that $p(0) = 0$. Is $\mathcal{S}$ a subspace of P_n? ■

■ **Example 8.8**

- Let $C^n[a, b]$ be the set of all functions f that have a continuous nth derivative on $[a, b]$.
  - $C^n[a, b]$ is a (proper) subset of $C^{n-1}[a, b]$.
  - $C^n[a, b]$ is a subspace of $C^{n-1}[a, b]$.
- Let $\mathcal{S}$ be the set of all f in $C^2[a, b]$ such that $f''(x) + f(x) = 0$.
  - $\mathcal{S}$ is a subspace of $C^2[a, b]$.
  - Both $\sin(x)$ and $\cos(x)$ are in $\mathcal{S}$, and hence their linear combinations are in $\mathcal{S}$.

■

R $|x|^{k+1}$ for k even and $x^{k+1}\sin(1/x)$ do not have a $(k+1)$th derivative at 0.

How to test whether vectors (functions) $f_1, f_2, \ldots, f_k$ are linearly independent in $C^n[a, b]$?

Lecture 9

Basis and Coordinates

9.1 More about Independence

Theorem 9.1 The vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in a vector space $\mathcal{V}$ are linearly dependent if and only if for certain $k \in \{1, 2, \ldots, n\}$, $\mathbf{v}_k$ is a linear combination of the other vectors.

Proof. To prove the necessary condition, consider $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly dependent, i.e., there exist scalars $c_1, c_2, \ldots, c_n$, not all zero, such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_n\mathbf{v}_n = \mathbf{0}.$$

Without Loss of Generality (WLOG) assume $c_1 \neq 0$. Then,

$$\mathbf{v}_1 = -\frac{1}{c_1}\sum_{i=2}^{n} c_i\mathbf{v}_i,$$

i.e., $\mathbf{v}_1$ is a linear combination of the other vectors.

To prove the sufficient condition, WLOG consider that $\mathbf{v}_1$ is a linear combination of the other vectors, i.e., there exist scalars $c_2', \ldots, c_2', \ldots, c_n'$ such that

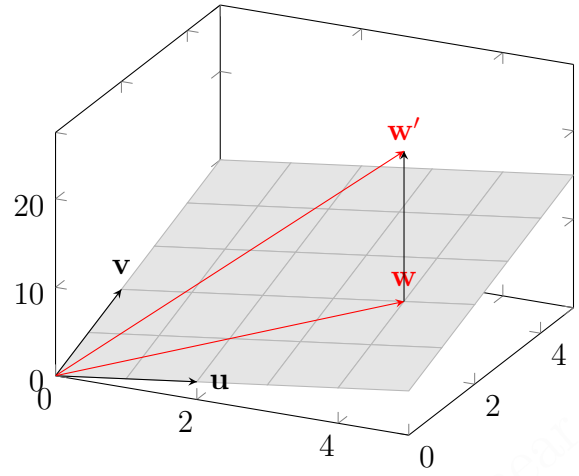
$$\mathbf{v}_1 = \sum_{i=2}^{n} c_i'\mathbf{v}_i.$$

Rewriting the above equality as

$$\mathbf{v}_1 - \sum_{i=2}^{n} c_i'\mathbf{v}_k = \mathbf{0},$$

we see $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly dependent. ■

- Suppose $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\} = \mathcal{V}$.
- When $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linear dependent, one of them, e.g., $\mathbf{v}_n$ is the linear combination of the others.
- Then, $\{\mathbf{v}_1, \ldots, \mathbf{v}_{n-1}\}$ is a *smaller* spanning set of $\mathcal{V}$. (justify the spanning set)
- We can continue the above process to obtain a *minimal* spanning set of $\mathcal{V}$.



Minimal Spanning Set

- Suppose that $\mathcal{V}$ is generated by $\mathcal{S} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$, i.e., $\mathcal{V} = \text{Span}(\mathcal{S})$.
- We say $\mathcal{S}$ is a *minimal spanning set* if $\mathcal{V}$ cannot be generated by any proper subset of $\mathcal{S}$.

Theorem 9.2 Suppose $\mathcal{S} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a minimal spanning set of $\mathcal{V}$. Then the elements of $\mathcal{S}$ are linearly independent.

Proof. Assume that the elements in $\mathcal{S}$ are not linearly independent. By Theorem 9.1, there exist scalars $c'_1, \ldots, c'_{k-1}, c'_{k+1}, \ldots, c'_n$ such that

$$\mathbf{v}_k = \sum_{i \neq k} c'_i \mathbf{v}_i. \tag{9.1}$$

For any vector $\mathbf{v} \in V$, as $\mathcal{S}$ is a spanning set of $\mathcal{V}$, there exist scalars $c_1, c_2, \ldots, c_n$ such that

$$\mathbf{v} = \sum_i c_i \mathbf{v}_i.$$

Substituting (9.1), we obtain

$$\mathbf{v} = \sum_{i \neq k} c_i \mathbf{v}_i + c_k \mathbf{v}_k = \sum_{i \neq k} c_i \mathbf{v}_i + c_k \sum_{i \neq k} c'_i \mathbf{v}_i = \sum_{i \neq k} (c_i + c_k c'_i) \mathbf{v}_i,$$

i.e., $\mathbf{v}$ is a linear combination of the elements in $\mathcal{S} \setminus \{\mathbf{v}_k\}$. Hence, $\mathcal{S} \setminus \{\mathbf{v}_k\}$ is a spanning set of $\mathcal{V}$, which is a contradiction to the condition that $\mathcal{S}$ is a minimal spanning set of $\mathcal{V}$. Therefore, the assumption that the elements in $\mathcal{S}$ are not linearly independent does not hold. ■

9.2 Basis

Any vector space $\mathcal{V}$ has a spanning set. Given a subspace $V = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$, any vector $\mathbf{v}$ in $\mathcal{V}$ can be represented by the linear combination coefficients of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ that generate $\mathbf{v}$. Is this representation unique and simple?

Theorem 9.3 For vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in a vector space $\mathcal{V}$, a vector $\mathbf{u} \in \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ can be written uniquely as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ if and only if $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly independent.

Proof. Consider $\mathbf{u} \in \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$, i.e.,

$$\mathbf{u} = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n. \tag{9.2}$$

To prove the sufficiency, assume that $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly independent. Suppose that $\mathbf{u}$ can be expressed as

$$\mathbf{u} = \beta_1\mathbf{v}_1 + \beta_2\mathbf{v}_2 + \cdots + \beta_n\mathbf{v}_n. \tag{9.3}$$

By (9.2) and (9.3),

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n = \beta_1\mathbf{v}_1 + \beta_2\mathbf{v}_2 + \cdots + \beta_n\mathbf{v}_n$$

and

$$(\alpha_1 - \beta_1)\mathbf{v}_1 + \cdots (\alpha_n - \beta_n)\mathbf{v}_n = \mathbf{0}.$$

As $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly independent, we have

$$\alpha_1 - \beta_1 = \cdots = \alpha_n - \beta_n = 0,$$

which implies that (9.3) is same as (9.2).

To prove the necessary condition, assume that $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly dependent, i.e., there exist $c_1, c_2, \ldots, c_n$, not all zero, such that

$$c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n = \mathbf{0}.$$

We can then check that

$$\mathbf{u} = (\beta_1 + c_1)\mathbf{v}_1 + (\beta_2 + c_2)\mathbf{v}_2 + \cdots + (\beta_n + c_n)\mathbf{v}_n.$$

As $c_1, c_2, \ldots, c_n$ are not all zero, there exists certain k such that $\beta_k \neq \beta_k + c_k$. So we get another linear combination forming $\mathbf{u}$ that is different from (9.2). ■

The previous two theorems say that a linearly independent spanning set only minimal, but also has unique representation.

Definition 9.1 The vectors $v_1, v_2, \ldots, v_n$ form a *basis* for a vector space $\mathcal{V}$ if
1. $v_1, v_2, \ldots, v_n$ are linearly independent, and
2. $v_1, v_2, \ldots, v_n$ span $\mathcal{V}$.

How to determine whether a set $\mathcal{B}$ of vectors form a basis of a vector space $\mathcal{V}$?
- First, check that $\mathcal{B}$ is a subset of $\mathcal{V}$.
- Second, verify that $\mathcal{B}$ is linearly independent.
- Third, verify that for any $\mathbf{v} \in \mathcal{V}$, $\mathbf{v} \in \text{Span}\{\mathcal{B}\}$.

■ **Example 9.1** Are the following sets a basis for $\mathbb{R}^2$ or not?

- $\mathcal{B}_1 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$

- $\mathcal{B}_2 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\}$
- $\mathcal{B}_3 = \left\{ \begin{bmatrix} 3 \\ 6 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$
- $\mathcal{B}_4 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 5 \end{bmatrix} \right\}$
- $\mathcal{B}_5 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$

■

■ **Example 9.2** Let

$$\mathbf{e}_i = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (i\text{th row})$$

Then $\mathcal{B} = \{\mathbf{e}_1, \cdots, \mathbf{e}_n\}$ is a basis for $\mathbb{R}^n$. In particular, B is called the standard basis. ■

■ **Example 9.3** Given a vector space $\mathbb{R}^{2\times 2}$, the set $\mathcal{B}$ consisting of

$$B_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \; B_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \; B_{21} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \; B_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

is a basis of $\mathbb{R}^{2\times 2}$. Notice that $\forall A \in \mathbb{R}^{2\times 2}$,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = aB_{11} + bB_{12} + cB_{21} + dB_{22}.$$

■

■ **Example 9.4** Consider P_2, the set of polynomials of degree at most 1, and $p_1(x) = 1$, $p_2(x) = x$, $p_3(x) = 2 - 3x$. Is $\mathcal{B} = \{p_1, p_2, p_3\}$ a basis for P_2? ■

9.3 Coordinates

Suppose $v_1, \ldots, v_n \in V$, a vector space. We know that $\mathcal{S} \triangleq \text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a subspace of $\mathcal{V}$. If every vector $\mathbf{u}$ in $\mathcal{S}$ can be written uniquely as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ as $c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, then $\mathbf{u}$ can be represented by the coefficients of the linear combination uniquely. In this sense, $v_1, v_2, \ldots, v_n$ form a basis of $\mathcal{S}$ and $(c_1, c_2, \ldots, c_n)$ is the *coordinate vector* of u with respect to the basis. A vector in an abstract vector space can be represented by a coordinate of real numbers!

Definition 9.2 Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a basis for a vector space $\mathcal{V}$. The *coordinates* of $\mathbf{x} \in \mathcal{V}$ with respect to $\mathcal{B}$ are the unique scalars $c_1, c_2, \ldots, c_n$ such that

$$\mathbf{x} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots c_n\mathbf{v}_n.$$

We call $[\mathbf{x}]_{\mathcal{B}} \triangleq (c_1, c_2, \ldots, c_n)^{\mathrm{T}}$ the *coordinate vector* of $\mathbf{x}$ w.r.t. the basis $\mathcal{B}$.

■ **Example 9.5** Consider a basis $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ of $\mathbb{R}^3$. What are the coordinates of $\mathbf{x} = \begin{bmatrix} 5 \\ 39 \\ 17 \end{bmatrix}$ w.r.t. the basis $\mathcal{B}$? ■

■ **Example 9.6** Consider a basis $\mathcal{B} = \left\{ \begin{bmatrix} 5 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$ of $\mathbb{R}^3$. What are the coordinates of $\mathbf{x} = \begin{bmatrix} 5 \\ 16 \\ 2 \end{bmatrix}$ w.r.t. the basis $\mathcal{B}$? ■

■ **Example 9.7** Consider a basis $\mathcal{B} = \{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$ of P_2 where $\mathbf{p}_1(x) = 5 - x$, $\mathbf{p}_2(x) = x$, $\mathbf{p}_3(x) = x + 2x^2$. What are the coordinates of $\mathbf{q}(x) = 5 + 16x + 2x^2$ w.r.t. the basis $\mathcal{B}$? ■

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Lecture 10

Dimension and Change of Basis

10.1 Dimension

Theorem 10.1 If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a spanning set for a vector space $\mathcal{V}$, then any collection of m vectors in $\mathcal{V}$, where $m > n$, is linearly dependent.

Proof. Let $\mathbf{u}_i$, $i = 1, \ldots, m$ be vectors in $\mathcal{V}$. Since $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ spans $\mathcal{V}$, we have $\mathbf{u}_i = \sum_{j=1}^{n} a_{ij}\mathbf{v}_j$. A linear combination of $\mathbf{u}_i$, $i = 1, \ldots, m$ can be written as

$$\begin{aligned}\sum_{i=1}^{m} x_i \mathbf{u}_i &= \sum_{i=1}^{m} x_i \left(\sum_{j=1}^{n} a_{ij}\mathbf{v}_j \right) \\ &= \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij} x_i \right) \mathbf{v}_j.\end{aligned}$$

As $m > n$, the system of linear equations

$$\sum_{i=1}^{m} a_{ij} x_i = 0, \quad j = 1, \ldots, n$$

with $x_1, \ldots, x_m$ as the variables has a non-trivial solution $\hat{c}_1, \ldots, \hat{c}_m$ (check the number of free variables). Hence

$$\sum_{i=1}^{m} \hat{c}_i \mathbf{u}_i = \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij}\hat{c}_i \right) \mathbf{v}_j = \sum_{j=1}^{n} 0\mathbf{v}_j = \mathbf{0}.$$

Therefore, $\mathbf{u}_i$, $i = 1, \ldots, m$ are linearly dependent. ■

Corollary 10.2 If both $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ and $\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ are bases for a vector space $\mathcal{V}$, then $n = m$.

Definition 10.1 If a vector space $\mathcal{V}$ has a basis consisting of n vectors, we say that $\mathcal{V}$ has *dimension* n. The subspace $\{\mathbf{0}\}$ of $\mathcal{V}$ is said to have dimension 0. $\mathcal{V}$ is said to be *finite dimensional* if there is a finite set of vectors that spans $\mathcal{V}$; otherwise,

we say that $\mathcal{V}$ is *infinite dimensional*.

■ **Example 10.1**
- What is the dimension of $\mathbb{R}^n$?
- What is the dimension of $\mathbb{R}^{m\times n}$?
- Let P be the vector space of all polynomials. P is infinite dimensional.
- $C[a,b]$ is infinite dimensional.

Suppose P has a finite dimension n, then any $n+1$ polynomials would be linearly dependent. Find a contradiction. A similar argument shows that $C[a,b]$ is infinite dimensional. ■

■ **Example 10.2**
- A non-zero vector in $\mathbb{R}^3$ spans a 1-dimensional subspace.
- Two linearly independent vectors in $\mathbb{R}^3$ span a 2-dimensional subspace.
- Any three linearly independent vectors in $\mathbb{R}^3$ span a 3-dimensional subspace, which must be $\mathbb{R}^3$.

■

Theorem 10.3 If $\mathcal{V}$ is a vector space of dimension $n > 0$, then
1. any set of n linearly independent vectors spans $\mathcal{V}$, and
2. any n vectors that span $\mathcal{V}$ are linearly independent.

R In other words, a basis is a minimal spanning set of a vector space, and a maximal independent set of a space.

Theorem 10.4 If $\mathcal{V}$ is a vector space of dimension $n > 0$, then
1. no set of fewer than n vectors can span $\mathcal{V}$,
2. any subset of fewer than n linearly independent vectors can be extended to form a basis for $\mathcal{V}$,
3. any spanning set of $\mathcal{V}$ includes a basis for $\mathcal{V}$.

10.2 Change of Basis

- Consider two bases $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and $\mathcal{W} = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_n\}$ for $\mathcal{V}$.
- For a vector $\mathbf{x} \in \mathcal{V}$, what is the relationship between $[\mathbf{x}]_\mathcal{B}$ and $[\mathbf{x}]_\mathcal{W}$?
- How do we determine $[\mathbf{x}]_\mathcal{B}$ if we are given $[\mathbf{x}]_\mathcal{W}$?

Lemma 10.5 Let $\mathcal{V}$ be a vector space with a basis $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$, and let $\mathbf{x}, \mathbf{y} \in \mathcal{V}$. For any scalars α, β,

$$[\alpha\mathbf{x} + \beta\mathbf{y}]_\mathcal{B} = \alpha[\mathbf{x}]_\mathcal{B} + \beta[\mathbf{y}]_\mathcal{B}.$$

This lemma says that the coordinate mapping is *linear*, which will be discussed in Part IV.

Proof. Denote $[\mathbf{x}]_{\mathcal{B}} = (c_1, \ldots, c_n)^{\mathrm{T}}$ and $[\mathbf{y}]_{\mathcal{B}} = (d_1, \ldots, d_n)^{\mathrm{T}}$. We have

$$\mathbf{x} = \sum_{i=1}^{n} c_i \mathbf{b}_i$$
$$\mathbf{y} = \sum_{i=1}^{n} d_i \mathbf{b}_i.$$

Therefore,

$$\alpha\mathbf{x} + \beta\mathbf{y} = \alpha \sum_{i=1}^{n} c_i \mathbf{b}_i + \beta \sum_{i=1}^{n} d_i \mathbf{b}_i = \sum_{i=1}^{n} \alpha c_i \mathbf{b}_i + \sum_{i=1}^{n} \beta d_i \mathbf{b}_i = \sum_{i=1}^{n} (\alpha c_i + \beta d_i)\mathbf{b}_i,$$

i.e.,

$$[\alpha\mathbf{x} + \beta\mathbf{y}]_{\mathcal{B}} = \begin{bmatrix} \alpha c_1 + \beta d_1 \\ \alpha c_2 + \beta d_2 \\ \cdots \\ \alpha c_n + \beta d_n \end{bmatrix} = \alpha[\mathbf{x}]_{\mathcal{B}} + \beta[\mathbf{y}]_{\mathcal{B}}.$$

■

Theorem 10.6 Consider a vector space $\mathcal{V}$ with two bases $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and $\mathcal{W} = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_n\}$. Then,

$$[\mathbf{x}]_{\mathcal{B}} = U[\mathbf{x}]_{\mathcal{W}},$$

where the jth column of U is $[\mathbf{w}_j]_{\mathcal{B}}$.

Proof. Let $[\mathbf{x}]_{\mathcal{W}} = (d_1, d_2, \ldots, d_n)^{\mathrm{T}}$. We can write

$$\mathbf{x} = d_1\mathbf{w}_1 + d_2\mathbf{w}_2 + \cdots d_n\mathbf{w}_n.$$

Suppose we know $[\mathbf{w}_j]_{\mathcal{B}} = \mathbf{u}_j$. According to Lemma 10.5,

$$\begin{aligned} [\mathbf{x}]_{\mathcal{B}} &= d_1[\mathbf{w}_1]_{\mathcal{B}} + d_2[\mathbf{w}_2]_{\mathcal{B}} + \cdots d_n[\mathbf{w}_n]_{\mathcal{B}} \\ &= d_1\mathbf{u}_1 + d_2\mathbf{u}_2 + \cdots d_n\mathbf{u}_n \\ &= \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n \end{bmatrix} [\mathbf{x}]_{\mathcal{W}}. \end{aligned}$$

The proof is completed by letting $U = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n \end{bmatrix}$. ■

We call U the *transition matrix* from basis $\mathcal{W}$ to basis $\mathcal{B}$. We can verify that the columns of U are linearly independent, i.e., U is invertible (nonsingular). Therefore, $[\mathbf{x}]_{\mathcal{W}} = U^{-1}[\mathbf{x}]_{\mathcal{B}}$. Assume that the columns of U are linearly dependent, i.e., there exists $c_1, \ldots, c_n$, not all zero, such that

$$\sum_{j=1}^{n} c_j \mathbf{u}_j = \mathbf{0} \quad \text{or} \quad \sum_{j=1}^{n} c_j u_{ij} = 0, \quad i = 1, \ldots, n.$$

As $\mathbf{u}_j$ is $[\mathbf{w}_j]_{\mathcal{B}}$, we have $\mathbf{w}_j = \sum_i u_{ij}\mathbf{v}_i$. Then

$$\sum_{j=1}^{n} c_j \mathbf{w}_j = \sum_j c_j \sum_i u_{ij}\mathbf{v}_i = \sum_i \left(\sum_j c_j u_{ij} \right) \mathbf{v}_i = \sum_i 0\mathbf{v}_i = \mathbf{0},$$

i.e., $\mathbf{w}_1, \ldots, \mathbf{w}_n$ are linearly dependent, a contradiction.

■ **Example 10.3** Suppose that in P_3 we want to change from the basis $\{1, x, x^2\}$ to the basis $\{1, 2x, 4x^2 - 2\}$.

- What is the transition matrix from $\{1, x, x^2\}$ to $\{1, 2x, 4x^2 - 2\}$?
- Given $p(x) = a + bx + cx^2$, find the coordinates of $p(x)$ w.r.t. $\{1, 2x, 4x^2 - 2\}$.

The transition matrix is a 3×3 matrix $U = (u_{ij})$ satisfying the following system of polynomial equations:

$$\begin{aligned} 1 &= u_{11} \cdot 1 + u_{21}(2x) + u_{31}(4x^2 - 2) \\ x &= u_{12} \cdot 1 + u_{22}(2x) + u_{32}(4x^2 - 2) \\ x^2 &= u_{13} \cdot 1 + u_{23}(2x) + u_{33}(4x^2 - 2). \end{aligned}$$

Note that each of the polynomial equation is equivalent to a 3×3 system of linear equation, all of the same coefficient matrix:

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

The system of polynomial equations is equivalent to

$$I = AU.$$

So U can be solved by backsubstitution

$$U = \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1/4 \end{bmatrix}.$$

As $[p(x)]_{\{1,x,x^2\}} = (a, b, c)^{\mathrm{T}}$, we have

$$[p(x)]_{\{1,2x,4x^2-2\}} = U(a, b, c)^{\mathrm{T}}.$$

■

10.3 Change of Basis for $\mathbb{R}^n$

- For a basis $\mathcal{V} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ for $\mathbb{R}^n$, we write $V = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \end{bmatrix}$.
- When $[\mathbf{x}]_{\mathcal{V}} = \mathbf{c} = (c_1, c_2, \ldots, c_n)^{\mathrm{T}}$, we can write the equality $\mathbf{x} = V\mathbf{c}$ using matrix-vector multiplication.
- For the standard basis $\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ for $\mathbb{R}^n$, we have $[\mathbf{x}]_{\mathcal{E}} = \mathbf{x}$.
- To change the basis from $\mathcal{V}$ to $\mathcal{E}$, we can write

$$[\mathbf{x}]_{\mathcal{E}} = \begin{bmatrix} [\mathbf{v}_1]_{\mathcal{E}} & [\mathbf{v}_2]_{\mathcal{E}} & \cdots & [\mathbf{v}_n]_{\mathcal{E}} \end{bmatrix} [\mathbf{x}]_{\mathcal{V}} = V[\mathbf{x}]_{\mathcal{V}}.$$

Theorem 10.7 Consider $\mathbb{R}^n$ with two bases

$$\mathcal{V} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\} \quad \text{and} \quad \mathcal{W} = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_n\}.$$

Then,

$$[\mathbf{x}]_{\mathcal{V}} = V^{-1}W[\mathbf{x}]_{\mathcal{W}}.$$

Proof. Let $\mathcal{E}$ be the standard basis of $\mathbb{R}^n$. By Theorem 10.6,

$$\begin{aligned}[\mathbf{x}]_{\mathcal{E}} &= V[\mathbf{x}]_{\mathcal{V}} \\ [\mathbf{x}]_{\mathcal{E}} &= W[\mathbf{x}]_{\mathcal{W}},\end{aligned}$$

and hence $V[\mathbf{x}]_{\mathcal{V}} = W[\mathbf{x}]_{\mathcal{W}}$. The proof is completed by V is invertible. ■

■ **Example 10.4** Consider the bases of $\mathbb{R}^2$:

$$\mathcal{E} = \left\{ \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix} \right\}, \quad \mathcal{V} = \left\{ \begin{bmatrix}5\\2\end{bmatrix}, \begin{bmatrix}7\\3\end{bmatrix} \right\}, \quad \mathcal{U} = \left\{ \begin{bmatrix}3\\2\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix} \right\}.$$

- Find the transition matrices from $\mathcal{E}$ to $\mathcal{V}$ and from $\mathcal{U}$ to $\mathcal{E}$, respectively.
- Given a vector $\mathbf{x}$ with the coordinate vector $(1, 2)^{\mathrm{T}}$ w.r.t. $\mathcal{V}$, find its coordinate vectors w.r.t. $\mathcal{E}$ and $\mathcal{U}$, respectively.

■

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Lecture 11

Subspaces related to Matrices

11.1 Null Space of a Matrix

Definition 11.1 The *null space* of an $m \times n$ matrix A, denoted by $\text{Null}(A)$ is the solution set of $A\mathbf{x} = \mathbf{0}$, i.e.,

$$\text{Null}(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}.$$

The null space of a matrix is just a new name of the solution set of the homogeneous system with the matrix as the coefficient matrix. Recall that in Section 7.3, we can write the solution set of $A\mathbf{x} = \mathbf{0}$ as the span of a set of vectors. Therefore, we have the following fact about null spaces:

Theorem 11.1 For an $m \times n$ matrix A, $\text{Null}(A)$ is a subspace of $\mathbb{R}^n$.

Proof. Verify the two conditions in the subspace definition for the solutions of a homogeneous system. ■

A set of vectors that span Null$A$ can be derived using Gauss-Jordan reduction.

■ **Example 11.1** Find $\text{Null}(A)$ for $A = \begin{bmatrix} 2 & -1 & 7 & -3 & -8 \\ 1 & 0 & 2 & 4 & 9 \\ 2 & 2 & -2 & -1 & 8 \end{bmatrix}$, which is row equivalent to $\begin{bmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -3 & 0 & 4 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$. ■

Theorem 11.2 The dimension of $\text{Null}(A)$ is equal to the number of free variables of $A\mathbf{x} = \mathbf{0}$.

Proof. Verify that the vectors spanning $\text{Null}(A)$ found in the parametric vector form are linearly independent. ■

- A square matrix A is invertible if and only if $\text{Null}(A) = \{\mathbf{0}\}$.
- $\text{Null}(A) = \{\mathbf{0}\}$ if and only the columns of A are linearly independent.
- If A and B are two row-equivalent matrices, then $\text{Null}(A) = \text{Null}(B)$.

In other words, row operations do not change the null space. This is an old result presented using new terms. It is equivalent to say that if A and B are row equivalent, then $A\mathbf{x} = \mathbf{0}$ and $B\mathbf{x} = \mathbf{0}$ have the same solution set.

11.2 Row and Column Spaces

Definition 11.2 For an $m \times n$ matrix A over $\mathbb{R}$, the subspace spanned by the column vectors of A is called the *column space* of A, denoted by $\mathrm{Col}(A)$. The subspace spanned by (the transpose) of the row vectors of A is called the *row space* of A, denoted by $\mathrm{Row}(A)$.

- The column space of A is a subspace of $\mathbb{R}^m$.
- The row space of A is a subspace of $\mathbb{R}^n$.
- $\mathrm{Row}(A) = \mathrm{Col}(A^{\mathrm{T}})$.

Theorem 11.3 If A and B are two row-equivalent matrices, then $\mathrm{Row}(A) = \mathrm{Row}(B)$.

Proof. If B is row equivalent to A, then B can be obtained from A by applying row operations. Thus, the rows of B are linear combinations of the rows of A, i.e., $\mathrm{Row}(B) \subset \mathrm{Row}(A)$. Similarly, we have $\mathrm{Row}(A) \subset \mathrm{Row}(B)$. ■

If the columns of A are linearly independent, they form a basis of the column space of A, and the dimension of the column space of A is n.

Theorem 11.4 Suppose a matrix A has (reduced) row-echelon form B. Let $\mathcal{W}$ be the set of columns of A which correspond to the pivot columns of B. Then $\mathcal{W}$ is a basis of the column space of A.

Proof. As B is row equivalent to A, there exists a nonsingular matrix E such that $B = EA$. Then we can show that any sets of columns of A and B with the same indices are both linearly dependent or independent simultaneously: Consider the matrices A_1 and B_1 formed by the columns of A and B, respectively, of the same indices. We know that $B_1 = EA_1$ and hence A_1 and B_1 are row equivalent. As $A_1\mathbf{x} = \mathbf{0}$ and $B_1\mathbf{x} = \mathbf{0}$ have the same solution set, the columns of A_1 are linearly independent if and only if the columns of B_1 are linearly independent.

As the pivotal columns of B form a basis of the column space of B (how to prove?), we know that the vectors in $\mathcal{W}$ are linear independent and all the other columns of A can be generated by the vectors in $\mathcal{W}$. Hence $\mathcal{W}$ is a basis of $\mathrm{Col}(A)$. ■

R The column spaces of two row-equivalent matrices A and B may not be the same.

How to find the column space of A?

■ **Example 11.2** Let

$$A = \begin{bmatrix} 1 & 2 & 7 & 1 & -1 \\ 1 & 1 & 3 & 1 & 0 \\ 3 & 2 & 5 & -1 & 9 \\ 1 & -1 & -5 & 2 & 0 \end{bmatrix}.$$

Find the columns space of A. The reduced row-echelon form of A is

$$\begin{bmatrix} \boxed{1} & 0 & -1 & 0 & 3 \\ 0 & \boxed{1} & 4 & 0 & -1 \\ 0 & 0 & 0 & \boxed{1} & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

So the column space of A is spanned by its 1st, 2nd and 4th columns. ■

Theorem 11.5 A linear system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ is in the column space of A.

The theorem gives us another approach to find the column space. Form an augmented matrix

$$\left[\begin{array}{ccccc|c} 1 & 2 & 7 & 1 & -1 & b_1 \\ 1 & 1 & 3 & 1 & 0 & b_2 \\ 3 & 2 & 5 & -1 & 9 & b_3 \\ 1 & -1 & -5 & 2 & 0 & b_4 \end{array}\right]$$

which can be reduced to the form

$$\left[\begin{array}{ccccc|c} \boxed{1} & 0 & -1 & 0 & 3 & -3b_1 + 5b_2 - b_4 \\ 0 & \boxed{1} & 4 & 0 & -1 & b_1 - b_2 \\ 0 & 0 & 0 & \boxed{1} & -2 & 2b_1 - 3b_2 + b_4 \\ 0 & 0 & 0 & 0 & 0 & 9b_1 - 16b_2 + b_3 + 4b_4 \end{array}\right]$$

The system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $9b_1 - 16b_2 + b_3 + 4b_4 = 0$. Therefore,

$$\text{Col}(A) = \{\mathbf{b} : 9b_1 - 16b_2 + b_3 + 4b_4 = 0\} = \text{Null}\{(9, -16, 1, 4)\}.$$

11.3 Rank

Theorem 11.6 For an $m \times n$ matrix A, the dimension of $\text{Row}(A)$ equals the dimension of $\text{Col}(A)$.

Proof. Suppose B is the reduced row-echelon form of A. Theorem 11.4 implies that the dimension of $\text{Col}(A)$ is equal to the number of pivots in B, and the dimension of $\text{Row}(A)$ is equal to that of $\text{Row}(B)$, where the latter is also the same as the number of pivots in B. ■

Definition 11.3 The rank of a matrix A, denoted by $\text{rank}(A)$, is the dimension of the row space of A.

- $\text{rank}(A) = \dim(\text{Row}(A)) = \dim(\text{Col}(A))$ = number of pivots.

Theorem 11.4 implies that the rank of A is equal to the number of pivots. Recall that the dimension of the null space of A is the number of free variables of $A\mathbf{x} = \mathbf{0}$. We have the *rank-nullity theorem*:

Theorem 11.7 If A is an $m \times n$ matrix, then

$$\operatorname{rank}(A) + \dim(\operatorname{Null}(A)) = n.$$

In other words, the solution set of $A\mathbf{x} = \mathbf{0}$ has dimension $n - \operatorname{rank}(A)$.

■ **Example 11.3** Let

$$A = \begin{bmatrix} 1 & 2 & -1 & 1 \\ 2 & 4 & -3 & 0 \\ 1 & 2 & 1 & 5 \end{bmatrix}.$$

Find a basis for the row space of A and a basis for $\operatorname{Null}(A)$. The reduced row-echelon form of A is

$$U = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

■

Properties

- $\operatorname{rank}(A) = \operatorname{rank}(A^\top)$.
- $\operatorname{rank}(A) \leq m$.
- $\operatorname{rank}(A) \leq n$.

Definition 11.4 An $m \times n$ matrix is called *full rank* if $\operatorname{rank}(A) = \min\{m, n\}$.

Equivalent Statements for Square matrix A

1. A is nonsingular (invertible).
2. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{0}$.
3. $A\mathbf{x} = \mathbf{b}$ has a unique solution for any $\mathbf{b}$.
4. A is row equivalent to I.
5. A is a product of a sequence of finite elementary matrices.
6. The columns/rows of A are linearly independent.
7. The columns/rows of A span $\mathbb{R}^n$, i.e., $\operatorname{Col}(A) = \operatorname{Row}(A) = \mathbb{R}^n$.
8. The columns/rows of A form a basis of $\mathbb{R}^n$.
9. A is full rank, i.e., $\operatorname{rank}(A) = n$.
10. $\dim(\operatorname{Null}(A)) = 0$.

We first know that 6), 7), 8) are equivalent (see Theorem 10.3). We can then check that 2) $\Rightarrow$ 6) $\Rightarrow$ 9) $\Rightarrow$ 10) $\Rightarrow$ 2).

11.4 More about Ranks

Theorem 11.8 For any matrix A and B with AB defined,

$$\operatorname{rank}(AB) \leq \operatorname{rank}(A) \quad \text{and} \quad \operatorname{rank}(AB \leq \operatorname{rank}(B).$$

Proof. As each column of AB is a linear combination of the columns of B, we have $\mathrm{Col}(AB) \subset \mathrm{Col}(A)$, which implies $\mathrm{rank}(AB) \leq \mathrm{rank}(A)$.

As each row of AB is a linear combination of the rows of A, we have $\mathrm{Row}(AB) \subset \mathrm{Row}(B)$, which implies $\mathrm{rank}(AB) \leq \mathrm{rank}(B)$. ■

Theorem 11.9 $\mathrm{rank}(A) = \mathrm{rank}(A^{\mathrm{T}}A)$

Proof. Prove $\mathrm{Null}(A) = \mathrm{Null}(A^{\mathrm{T}}A)$ and apply the rank-nullity theorem. ■

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Part III

Determinants

Draft of Linear algebra text in Telugu
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Lecture 12

Properties of Determinants

12.1 Motivation of Determinants

What is Determinant?

- The determinant of a squared matrix A is a value that is computed from the entries of A, denoted by $\det(A)$ or $|A|$.
- If $\det(A) = 0$ then A is singular, otherwise, nonsingular.
- A^{-1} can be explicitly solved in terms of $\det(A)$, by giving each entry of A^{-1}.
- Many more applications and very interesting properties.

The determinant has a *complicated* formula. Before giving the definition, let us consider the properties that determinants should satisfy.

Desired Properties of Determinants

- $\det(A) = 0$ if and only if A is singular. To have this property:
  - $\det(A) = 0$ if one row/column of A is all zero.
  - $\det(A) = 0$ if two rows/columns of A are the same.
  - $\det(A) = 0$ if the rows/columns of A are linearly dependent.
- It would also make sense if

$$\det(A) = \det(A^{\mathrm{T}}).$$

Diagonal Matrices

- For $A = [a]$, a 1×1 matrix, A has an inverse if and only if $a \neq 0$. Thus, a reasonable definition of determinant is

$$\det(A) = a.$$

- For an $n \times n$ diagonal matrix A, A is invertible if and only if $a_{ii} \neq 0$ for all $i = 1, 2, \ldots, n$. Therefore, we can define

$$\det(A) = a_{11}a_{22}\cdots a_{nn}.$$

12.2 Properties of Determinants

In this section, we give three basic properties that (we hope) determinants should satisfies, and derive some other properties based on these three basic properties.

From our previous study, we know that

- AB is invertible if and only if both A and B are invertible (see Theorem 4.2 and 4.3).
- When A is an $n \times n$ triangular matrix, A is invertible if and only if $a_{ii} \neq 0$ for all $i = 1, 2, \ldots, n$.

The above results motivate us to define determinants of square matrices satisfying the following two properties:

First two Basic Properties

- For two $n \times n$ matrices A and B

$$\det(AB) = \det(A)\det(B)$$

- For an $n \times n$ triangular matrix A:

$$\det(A) = a_{11}a_{22}\cdots a_{nn}$$

We will later show that these two properties give us a complete and unique definition of determinant.

Direct Results of Basic Properties

- $\det(I) = 1$
- If A is invertible, then $\det(A^{-1}) = \frac{1}{\det(A)}$.
- Type II and III elementary matrices are triangular, and hence:
  - If E is formed by adding multiples of one row to another row: $\det(E) = 1$.
  - If E is formed by multiplying a constant α to a row: $\det(E) = \alpha$.

Property 12.1 Let A be a square matrix and B be a matrix obtained by multiplying a constant α to a row/column of A. Then $\det(B) = \alpha\det(A)$.

Proof of the row case. We have $B = EA$ for certain type II elementary matrix E with $\det(E) = \alpha$. Then $\det(B) = \det(E)\det(A) = \alpha\det(A)$. ■

Note that $\det(\alpha A) = \alpha^n \det(A)$.

Property 12.2 Let A be a square matrix and B be a matrix obtained by adding a constant times of one row/column to another row/column of A. Then $\det(B) = \det(A)$.

Proof of the row case. We have $B = EA$ for certain type III elementary matrix E. As $\det(E) = 1$, we have $\det(B) = \det(E)\det(A) = \det(A)$. ■

That is the determinant of a type I elementary matrix? Recall that a type I row operation can be achieved using type II and III row operations. Consider the

determinant of $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. We can transform the matrix to a triangular one using a type I row operation, or using only type II and III row operations:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \xrightarrow{R_2+R_1} \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \xrightarrow{R_1-R_2} \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}$$

As type III row operations do not change the determinant, we have

$$\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = \begin{vmatrix} -1 & 0 \\ 1 & 1 \end{vmatrix} = -1.$$

Determinant of Type I Elementary Matrices

- Suppose that E is formed by interchanging row i and row j, $i < j$.
- We have
$$E = E_4E_3E_2E_1$$
where
  - E_1 is formed by adding row i to row j.
  - E_2 is formed by adding -1 times row j to row i.
  - E_3 is formed by adding row i to row j.
  - E_4 is formed by multiplying -1 to row i.
- E_1, E_2, E_3 are type III, while E_4 is type II.
- By the two basic properties, $\det(E) = -1$!

We may regard this as the third basic property of determinants, i.e., the determinant of a type I elementary matrix is -1.

Property 12.3 Let A be a square matrix and B be a matrix obtained by interchanging two rows/columns of A. Then $\det(B) = -\det(A)$.

Proof of the row case. We have $B = EA$ for certain type I elementary matrix E. As $\det(E) = -1$, $\det(B) = \det(E)\det(A) = -\det(A)$. ■

Determinant of a Matrix with LU Decomposition

- Suppose that A has an LU decomposition as $A = LU$, where L is unit, lower triangular and U is upper triangular.
- we have
$$\det(A) = \det(L)\det(U) = \det(U),$$
where the second equality follows that L is a unit triangular matrix.

Determinant of a General Matrix

- In general, permutation of the rows of A is required to obtain an LU decomposition, i.e., $A = PLU$, where P is a permutation matrix.
- $\det(A) = \det(P)\det(L)\det(U)$.

Recall permutation matrices defined in Sec 5.1. We know that P can be written as the product of a sequence type I elementary matrices: $P = E_k E_{k-1} \cdots E_1$. Hence, $\det(P) = (-1)^k$. Though such expressions as type I elementary matrices are not unique, either they all contain an even number of type I elementary matrices, or odd.

When evaluating the determinant, we do not need to calculate P explicitly as we only need to count how many times of row interchanging are applied when performing Gaussian elimination.

Property 12.4 For a square matrix A, $\det(A) = 0$ if and only if A is singular.

Proof. If A is singular, its PLU decomposition must has U with certain diagonal entry 0. Therefore, $\det(P)\det(L)\det(U) = 0$ for any PLU decomposition of A. Hence, $\det(A) = 0$.

If A is nonsingular, all the diagonal entry of U must be nonzero and hence $\det(P)\det(L)\det(U) \neq 0$. ■

Property 12.5 $\det(A^{\mathrm{T}}) = \det(A)$.

Proof. Suppose $A = PLU$, a PLU decomposition. As $A^{\mathrm{T}} = U^{\mathrm{T}} L^{\mathrm{T}} P^{\mathrm{T}}$, we have

$$\det(A^{\mathrm{T}}) = \det(U^{\mathrm{T}})\det(L^{\mathrm{T}})\det(P^{\mathrm{T}}) = \det(U)\det(L)\det(P),$$

where $\det(P^{\mathrm{T}}) = \det(P^{-1}) = 1/\det(P) = \det(P)$. ■

Note that in general $\det(A + B) \neq \det(A) + \det(B)$.

Property 12.6 Consider $n \times n$ matrices A, B and C defined as follows:

$$A = \begin{bmatrix} \vec{\mathbf{a}} \\ \vec{\mathbf{c}}_2 \\ \vdots \\ \vec{\mathbf{c}}_n \end{bmatrix}, \quad B = \begin{bmatrix} \vec{\mathbf{b}} \\ \vec{\mathbf{c}}_2 \\ \vdots \\ \vec{\mathbf{c}}_n \end{bmatrix}, \quad C = \begin{bmatrix} \vec{\mathbf{a}} + \vec{\mathbf{b}} \\ \vec{\mathbf{c}}_2 \\ \vdots \\ \vec{\mathbf{c}}_n \end{bmatrix}.$$

We have $\det(C) = \det(A) + \det(B)$.

Proof. We consider three cases: In the first case: $\vec{\mathbf{a}} \in \mathrm{Span}\{\vec{\mathbf{b}}, \vec{\mathbf{c}}_2, \ldots \vec{\mathbf{c}}_n\}$, i.e.,

$$\vec{\mathbf{a}} = \alpha\vec{\mathbf{b}} + \sum_{i=2}^{n} \alpha_i \vec{\mathbf{c}}_i.$$

By a sequence of type III row operations, we can transform:

$$A = \begin{bmatrix} \alpha\vec{\mathbf{b}} + \sum_{i=2}^{n} \alpha_i \vec{\mathbf{c}}_i \\ \vec{\mathbf{c}}_2 \\ \vdots \\ \vec{\mathbf{c}}_n \end{bmatrix} \xrightarrow{\text{type III}} \begin{bmatrix} \alpha\vec{\mathbf{b}} \\ \vec{\mathbf{c}}_2 \\ \vdots \\ \vec{\mathbf{c}}_n \end{bmatrix}.$$

As type III row operations do not change determinants (Property 12.2), we have $\det(A) = \alpha \det(B)$. Similarly, we can transform

$$C = \begin{bmatrix} \vec{\mathbf{b}} + \alpha\vec{\mathbf{b}} + \sum_{i=2}^{n} \alpha_i \vec{\mathbf{c}}_i \\ \vec{\mathbf{c}}_2 \\ \vdots \\ \vec{\mathbf{c}}_n \end{bmatrix} \xrightarrow{\text{type III}} \begin{bmatrix} (1+\alpha)\vec{\mathbf{b}} \\ \vec{\mathbf{c}}_2 \\ \vdots \\ \vec{\mathbf{c}}_n \end{bmatrix},$$

and get $\det(C) = (1+\alpha)\det(B) = \det(B) + \det(A)$.

The second case $\vec{\mathbf{b}} \in \text{Span}\{\vec{\mathbf{a}}, \vec{\mathbf{c}}_2, \ldots \vec{\mathbf{c}}_n\}$ can be proved similar to the above case.

In the third case, $\vec{\mathbf{a}} \notin \text{Span}\{\vec{\mathbf{b}}, \vec{\mathbf{c}}_2, \ldots \vec{\mathbf{c}}_n\}$ and $\vec{\mathbf{b}} \notin \text{Span}\{\vec{\mathbf{a}}, \vec{\mathbf{c}}_2, \ldots \vec{\mathbf{c}}_n\}$. Assume that $\vec{\mathbf{c}}_2, \ldots, \vec{\mathbf{c}}_n$ are linearly independent. By $\vec{\mathbf{b}} \notin \text{Span}\{\vec{\mathbf{a}}, \vec{\mathbf{c}}_2, \ldots \vec{\mathbf{c}}_n\}$, we have $\vec{\mathbf{b}}$ is not a linear combination of $\vec{\mathbf{c}}_2, \ldots, \vec{\mathbf{c}}_n$. Hence $\{\vec{\mathbf{b}}, \vec{\mathbf{c}}_2, \ldots \vec{\mathbf{c}}_n\}$ is linearly independent and forms a basis of $\mathbb{R}^n$ (of row vectors), which is a contradiction that $\vec{\mathbf{a}} \notin \text{Span}\{\vec{\mathbf{b}}, \vec{\mathbf{c}}_2, \ldots \vec{\mathbf{c}}_n\}$. Therefore, $\vec{\mathbf{c}}_2, \ldots, \vec{\mathbf{c}}_n$ are linearly dependent, and hence A, B and C are singular. The equality holds as all the determinants are zero (by Property 12.4). ■

Similarly, let $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}_i$, $i = 2, \ldots, n$ be (column) vectors in $\mathbb{R}^n$. Then,

$$\det([\mathbf{a} + \mathbf{b}, \mathbf{c}_2, \ldots \mathbf{c}_n]) = \det([\mathbf{a}, \mathbf{c}_2, \ldots \mathbf{c}_n]) + \det([\mathbf{b}, \mathbf{c}_2, \ldots \mathbf{c}_n]).$$

Property 12.1 and 12.6 together give certain linear property of determinants.

Among these properties we have defined or proved for determinants, a set of them can be used to prove the others, e.g., the three basic properties we have mentioned. Another example of such a set of basic properties is formed by

- $\det(I) = 1$,
- $\det(E) = -1$ for a type I elementary matrix, and
- Property 12.6.

As a summary of this lecture, we define three basic properties of determinants. We use these basic properties to further derives other properties of determinants and a Gaussian elimination approach for calculating determinants. In the next lecture, we will give a formula of determinant that satisfies all these three basic properties.

12.3 More about Permutation

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Lecture 13

Formulae of Determinants

13.1 Cofactor Expansion

In this section, we first derive an iterative formula of determinants from the basic properties:

P1. $\det(AB) = \det(A)\det(B)$ for any $n \times n$ matrices A and B.
P2. If A is triangular, $\det(A) = a_{11}a_{22}\cdots a_{nn}$.
P3. If A is a type I elementary matrix, $\det(A) = -1$.

Definition 13.1 Let $A = (a_{ij})$ be an $n \times n$ matrix and let M_{ij} denotes the $(n-1) \times (n-1)$ matrix obtained by deleting the ith row and the jth column.

A Further Property

Property 13.1 Suppose that A is a square matrix with the entries in the first column are all zero except that a_{11} may be nonzero, i.e.,

$$A = \begin{bmatrix} a_{11} & \cdots \\ \mathbf{0} & M_{11} \end{bmatrix}$$

We have $\det(A) = a_{11}\det(M_{11})$.

Proof. When $a_{11} = 0$, $\det(A) = 0$ and hence the property holds. When $a_{11} \neq 0$, suppose $M_{11} = P_1L_1U_1$, a PLU decomposition. We have a PLU decomposition of A as

$$A = \begin{bmatrix} 1 & \vec{\mathbf{0}} \\ \mathbf{0} & P_1 \end{bmatrix} \begin{bmatrix} 1 & \vec{\mathbf{0}} \\ \mathbf{0} & L_1 \end{bmatrix} \begin{bmatrix} a_{11} & \cdots \\ \mathbf{0} & U_1 \end{bmatrix},$$

where the first row of the last matrix is the same as the first row of A. Then

$$\det(A) = \begin{vmatrix} 1 & \vec{\mathbf{0}} \\ \mathbf{0} & P_1 \end{vmatrix} \begin{vmatrix} 1 & \vec{\mathbf{0}} \\ \mathbf{0} & L_1 \end{vmatrix} \begin{vmatrix} a_{11} & \cdots \\ \mathbf{0} & U_1 \end{vmatrix} = a_{11}|P_1||L_1||U_1| = a_{11}\det(M_{11}),$$

where $\begin{vmatrix} 1 & \vec{\mathbf{0}} \\ \mathbf{0} & P_1 \end{vmatrix} = |P_1|$ is verified as follows.

Suppose $P_1 = E_k \cdots E_1$, where $E_1, \ldots, E_k$ is a sequence of type I elementary matrices. Then $\begin{vmatrix} 1 & \vec{\mathbf{0}} \\ \mathbf{0} & P_1 \end{vmatrix} = \begin{bmatrix} 1 & \\ & E_k \end{bmatrix} \cdots \begin{bmatrix} 1 & \\ & E_1 \end{bmatrix}$. Hence, $\begin{vmatrix} 1 & \vec{\mathbf{0}} \\ \mathbf{0} & P_1 \end{vmatrix} = (-1)^k = \det(P_1)$. ■

Definition 13.2 For matrix (a_{ij}), we call $\det(M_{ij})$ the *minor* of a_{ij}, and $C_{ij} = (-1)^{i+j} \det(M_{ij})$ the *cofactor* of a_{ij}.

Property 13.2 If A is an $n \times n$ matrix with $n \geq 2$, then

$$\begin{aligned} \det(A) &= a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in} \\ &= a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj} \end{aligned}$$

for $i, j = 1, 2, \ldots, n$.

This is the cofactor expansion formula or the **Laplace formula** for determinants.

Proof. We prove the case $\det(A) = a_{11}C_{11} + a_{21}C_{21} + \cdots + a_{n1}C_{n1}$. Write the first column of A as

$$a_{11}\mathbf{e}_1 + a_{21}\mathbf{e}_2 + \cdots + a_{n1}\mathbf{e}_n$$

where $\mathbf{e}_k$ is the k column of I_n. By the linear properties of determinants for a column (see Property 12.1 and 12.6), we have

$$\det(A) = \sum_{i=1}^{n} a_{i1} \det(B_i),$$

where B_j is obtained by replace the first column of A by $\mathbf{e}_j$, i.e.,

$$B_j = \begin{bmatrix} \mathbf{e}_j & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}.$$

Performing a permutation of the first j rows of B_j, we obtain

$$\begin{bmatrix} 1 & \cdots \\ \mathbf{0} & M_{j1} \end{bmatrix}.$$

Compared with B_j, the ith row, $i = 1, \ldots, j-1$ is moved to the $(i+1)$th row, and the jth row is moved to the first row. This permutation can be done by $j-1$ row interchanging. By Property 13.1,

$$\det(B_j) = (-1)^{j-1} \det(M_{j1}) = C_{ij},$$

where the term $(-1)^{j-1}$ comes from the permutation of the first j rows of B_j. ■

This formula can be used to calculate the determinant of small matrices. For a large matrix, it is much more efficient to use LU decomposition.

■ **Example 13.1** Evaluate

$$\begin{vmatrix} 0 & 2 & 3 & 0 \\ 0 & 4 & 5 & 0 \\ 0 & 1 & 0 & 3 \\ 2 & 0 & 1 & 3 \end{vmatrix}$$

$$(-1)^5 2\begin{vmatrix}2 & 3 & 0\\ 4 & 5 & 0\\ 1 & 0 & 3\end{vmatrix} = -2\cdot(-1)^6 3\begin{vmatrix}2 & 3\\ 4 & 5\end{vmatrix} = -2\cdot 3\cdot(10-12) = 12.$$

■

Adjoint Matrix

- Define the adjoint of matrix A by

$$\operatorname{adj}(A) = \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1}\\ C_{12} & C_{22} & \cdots & C_{n2}\\ \vdots & & & \\ C_{1n} & C_{2n} & \cdots & C_{nn}\end{bmatrix}.$$

- Note that

$$\begin{vmatrix}\mathbf{a}_1 & \cdots & \mathbf{a}_{j-1} & \mathbf{a}_i & \mathbf{a}_{j+1} & \cdots & \mathbf{a}_n\end{vmatrix} = \sum_{k=1}^{n} a_{ki}C_{kj} = \begin{cases}\det(A) & \text{if } i=j\\ 0 & \text{if } i\neq j\end{cases}.$$

■ **Example 13.2** The adjoint matrix of $A = \begin{bmatrix}a_{11} & a_{12}\\ a_{21} & a_{22}\end{bmatrix}$, is

$$\operatorname{adj}(A) = \begin{bmatrix} a_{22} & -a_{12}\\ -a_{21} & a_{11}\end{bmatrix}.$$

Check $\operatorname{adj}(A)A$. ■

A Formula of Matrix Inverse

- The (i,j) entry of $\operatorname{adj}(A)A$ is

$$\sum_{k=1}^{n} a_{kj}C_{ki} = \begin{cases}\det(A) & \text{if } i=j\\ 0 & \text{if } i\neq j\end{cases}.$$

- So

$$\operatorname{adj}(A)A = \det(A)I, \quad \text{i.e.,} \quad A^{-1} = \frac{1}{\det(A)}\operatorname{adj}(A),$$

and

$$(A^{-1})_{ij} = \frac{C_{ji}}{\det(A)} = \frac{(-1)^{i+j}\det(M_{ji})}{\det(A)}.$$

Cramer's Rule

- Consider a system of linear equations $A\mathbf{x} = \mathbf{b}$, where A is an invertible $n\times n$ matrix.
- Using adjoint matrix, we have

$$\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{\det(A)}\operatorname{adj}(A)\mathbf{b}.$$

- Then $x_i = \frac{1}{\det(A)}\sum_{k=1}^{n} b_k C_{ki} = \frac{1}{\det(A)}\det(B_i)$, where

$$B_i = \begin{bmatrix}\mathbf{a}_1 & \cdots & \mathbf{a}_{i-1} & \mathbf{b} & \mathbf{a}_{i+1} & \cdots & \mathbf{a}_n\end{bmatrix}.$$

13.2 A close-form Formula

- Let's check a 2×2 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

According to the cofactor exapansion

$$\det(A) = a_{11}a_{22} - a_{12}a_{21}.$$

- Let's check a 3×3 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

According to the cofactor exapansion

$$\begin{aligned} \det(A) = {} & a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ & -a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}. \end{aligned}$$

Leibniz Formula

- Observe that the determinant of an $n \times n$ ($n = 2$ or 3) matrix can be written as the summation of $n!$ terms, each of the multiplication of n entries in A, which are not on the same row nor column.
- We can derive that

$$\det(A) = \sum_{\tau \in S_n} \operatorname{sgn}(\tau) a_{1\tau(1)} a_{2\tau(2)} \cdots a_{n\tau(n)}, \tag{13.1}$$

where S_n is the set of all permutations of n elements, and $\operatorname{sgn}(\tau) \in \{-1, 1\}$ is the *sign function* of the permutation τ.

If a permutation τ is odd, $\operatorname{sgn}(\tau) = -1$, otherwise 1.

To derive the Leibniz formula, follow the proof of Property 13.2,

$$\det(A) = \sum_{i=1}^{n} a_{i1} \begin{vmatrix} \mathbf{e}_i & \mathbf{a}_2 & \mathbf{a}_3 & \cdots & \mathbf{a}_n \end{vmatrix}.$$

We can continue this expansion for the second column, third column, ..., and obtain

$$\det(A) = \sum_{i_1=1}^{n} \sum_{i_2=1}^{n} \cdots \sum_{i_n=1}^{n} a_{i_1 1} a_{i_2 2} \cdots a_{i_n n} \begin{vmatrix} \mathbf{e}_{i_1} & \mathbf{e}_{i_2} & \cdots & \mathbf{e}_{i_n} \end{vmatrix}.$$

Noting that $\begin{bmatrix} \mathbf{e}_{i_1} & \mathbf{e}_{i_2} & \cdots & \mathbf{e}_{i_n} \end{bmatrix}$ is a permutation matrix if $i_1 \neq i_2 \neq \cdots \neq i_n$, and has determinant 0 if two columns are the same, we obtain the Leibniz formula.

(*The following part of this section is not required for this course.*) The formula in (13.1) is derived using the three basic properties. We can also do this another way

round by verifying that this formula satisfy these three basic properties. Then we can treat this formula as the definition of determinants. It is straightforward to show that (13.1) satisfies the requirement of triangular matrices and type I elementary matrices. Let

$$\mathrm{Leib}(A) \triangleq \sum_{\tau \in S_n} \mathrm{sgn}(\tau) a_{1\tau(1)} a_{2\tau(2)} \cdots a_{n\tau(n)},$$

Lemma 13.1 For square matrices A, B, C of the same size, if $A = BC$, then

$$\mathrm{Leib}(C) = \mathrm{Leib}(A)\mathrm{Leib}(B).$$

Proof outline.

$$\begin{aligned}
\mathrm{Leib}(C) &= \sum_{\tau \in S_n} \mathrm{sgn}(\tau) c_{1\tau(1)} c_{2\tau(2)} \cdots c_{n\tau(n)} \\
&= \sum_{\tau \in S_n} \mathrm{sgn}(\tau) \prod_{i=1}^{n} \left(\sum_{k=1}^{n} a_{ik} b_{kj} \right) \\
&= \sum_{\tau \in S_n} \mathrm{sgn}(\tau) \sum_{k_1=1}^{n} \cdots \sum_{k_n=1}^{n} a_{1k_1} a_{2k_2} \ldots a_{nk_n} b_{k_1\tau_1} b_{k_2\tau_2} \ldots b_{k_n\tau_n} \\
&= \sum_{k_1=1}^{n} \cdots \sum_{k_n=1}^{n} a_{1k_1} a_{2k_2} \ldots a_{nk_n} \sum_{\tau \in S_n} \mathrm{sgn}(\tau) b_{k_1\tau_1} b_{k_2\tau_2} \ldots b_{k_n\tau_n}.
\end{aligned}$$

For a sequence $k_1, \ldots, k_n$, with $k_i \in \{1, \ldots, n\}$, if at least two numbers are the same, $\sum_{\tau \in S_n} \mathrm{sgn}(\tau) b_{k_1\tau_1} b_{k_2\tau_2} \ldots b_{k_n\tau_n} = 0$. Therefore,

$$\begin{aligned}
\mathrm{Leib}(C) &= \sum_{k \in S_n} a_{1k_1} a_{2k_2} \ldots a_{nk_n} \sum_{\tau \in S_n} \mathrm{sgn}(\tau) b_{k_1\tau_1} b_{k_2\tau_2} \ldots b_{k_n\tau_n} \\
&= \sum_{k \in S_n} a_{1k_1} a_{2k_2} \ldots a_{nk_n} \mathrm{sgn}(k) \sum_{\tau \in S_n} \mathrm{sgn}(\tau) b_{1\tau_1} b_{2\tau_2} \ldots b_{n\tau_n} \\
&= \mathrm{Leib}(A)\mathrm{Leib}(B).
\end{aligned}$$

■

The above verification shows that there exists a function satisfies the three basic properties of determinants. (We must have existence before showing the uniqueness.) The uniqueness of determinant functions can then be shown as follows: Suppose both det and det′ satisfy the three basic properties. For any matrix A with $A = PLU$, we have $\det(A) = \det(P)\det(L)\det(U) = \det'(P)\det'(L)\det'(U) = \det'(A)$.

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Part IV

Linear Transformations

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Nirtam Yaacov Nadav

Lecture 14

Linear Transformation, Kernel and Image

14.1 Definition and Examples

A mapping here means a function.

Definition 14.1 A mapping L from a vector space $\mathcal{V}$ into a vector space $\mathcal{W}$, denoted by $L : \mathcal{V} \to \mathcal{W}$, is said to be a *linear transformation* if

$$L(\alpha \mathbf{v}_1 + \beta \mathbf{v}_2) = \alpha L(\mathbf{v}_1) + \beta L(\mathbf{v}_2)$$

for all $\mathbf{v}_1, \mathbf{v}_2 \in V$ and all scalars α and β.

In other words, a linear mapping preserves the operations of addition and scalar multiplication:

- $L(\mathbf{v}_1 + \mathbf{v}_2) = L(\mathbf{v}_1) + L(\mathbf{v}_2)$,
- $L(\alpha \mathbf{v}) = \alpha L(\mathbf{v})$.

■ **Example 14.1** Verify that the following mappings $L : \mathbb{R}^2 \to \mathbb{R}^2$ are linear transformations and illustrate them.

- $L(\mathbf{x}) = 3\mathbf{x}$.
- $L(\mathbf{x}) = x_1 \mathbf{e}_1$.
- $L(\mathbf{x}) = \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix}$
- $L(\mathbf{x}) = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$.

■

■ **Example 14.2** Are the following mappings linear transformations?

- The mapping $L_1 : \mathbb{R}^2 \to \mathbb{R}$ defined by $L_1(\mathbf{x}) = x_1 + x_2$.
- The mapping $M : \mathbb{R}^2 \to \mathbb{R}$ defined by $M(\mathbf{x}) = (x_1^2 + x_2^2)^{1/2}$.
- The mapping $L_2 : \mathbb{R}^2 \to \mathbb{R}^3$ defined by $L_2(\mathbf{x}) = \begin{bmatrix} x_2 \\ x_1 \\ x_1 + x_2 \end{bmatrix}$.

■

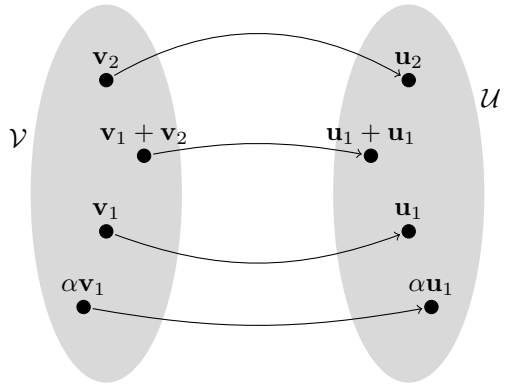


Figure 14.1: Illustration of Linear Transformation

Properties of Linear Transformations

If $L : \mathcal{V} \to \mathcal{W}$ is a linear transformation, then

1. $L(\mathbf{0}_V) = \mathbf{0}_W$.
2. $L(-\mathbf{v}) = -L(\mathbf{v})$.
3. $L(\alpha_1\mathbf{v}_1 + \cdots + \alpha_n\mathbf{v}_n) = \alpha_1 L(\mathbf{v}_1) + \cdots + \alpha_n L(\mathbf{v}_n)$, where $v_1, \ldots, v_n$ are in $\mathcal{V}$ and $\alpha_1, \ldots, \alpha_n$ are scalars.

■ **Example 14.3**

- Define $L : C[a, b] \to \mathbb{R}$ as $L(f) = \int_a^b f(x)dx$. In other words, L is the integral from a to b. Show that L is a linear transformation.
- Define $D : C^1[a, b] \to C[a, b]$ as $D(f) = f'$. Show that D is a linear transformation.

■

Determinant as a Linear Function of Vectors

- Fix $\mathbf{c}_2, \ldots, \mathbf{c}_n \in \mathbb{R}^n$.
- Define a mapping $L : \mathbb{R}^n \to \mathbb{R}$ as

$$L(\mathbf{x}) = \begin{vmatrix} \mathbf{x} & \mathbf{c}_2 & \cdots & \mathbf{c}_n \end{vmatrix}.$$

- Verify that $L(x)$ is a linear transformation.

14.2 Kernel and Image

Consider a linear transformation $L : \mathcal{V} \to \mathcal{W}$.

- If $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are linearly dependent in $\mathcal{V}$, then $L(\mathbf{v}_1), \ldots, L(\mathbf{v}_n)$ are linearly dependent in $\mathcal{W}$.
- However, if $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are linearly independent in $\mathcal{V}$, then $L(\mathbf{v}_1), \ldots, L(\mathbf{v}_n)$ may not be linearly independent in $\mathcal{W}$.

Definition 14.2 Let $L : \mathcal{V} \to \mathcal{W}$ be a linear transformation. The *kernel* of L, denoted by $\ker(L)$ is defined by

$$\ker(L) = \{\mathbf{v} \in V | L(\mathbf{v}) = \mathbf{0}_{\mathcal{W}}\}.$$

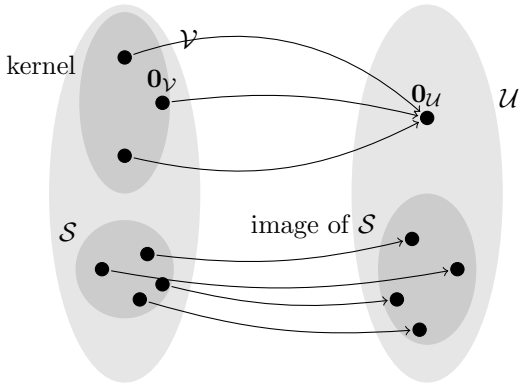


Figure 14.2: Illustration of Linear Transformation

Definition 14.3 Let $L : \mathcal{V} \to \mathcal{W}$ be a linear transformation and let S be a subspace of $\mathcal{V}$. The *image* of S, denoted by $L(S)$, is defined by

$$L(S) = \{\mathbf{w} \in W | \mathbf{w} = L(\mathbf{v}) \text{ for certain } \mathbf{v} \in S\}.$$

The image of the entire vector space $\mathcal{V}$ is called the **range** of L.

Theorem 14.1 Let $L : \mathcal{V} \to \mathcal{W}$ be a linear transformation and let S be a subspace of $\mathcal{V}$, then

1. $\ker(L)$ is a subspace of $\mathcal{V}$.
2. $L(S)$ is a subspace of $\mathcal{W}$.

Proof. Verify the conditions of the definition of subspace. First, as $\mathbf{0}_\mathcal{V}$ is in $\ker(L)$ and $\mathbf{0}_\mathcal{W}$ is $L(S)$, both $\ker(L)$ and $L(S)$ are not empty. We then verify that they are closed under scalar multiplication and vector addition. ■

■ **Example 14.4** Let $L : \mathbb{R}^3 \to \mathbb{R}^2$ be the linear transformation defined by

$$L(\mathbf{x}) = \begin{bmatrix} x_1 + x_2 \\ x_2 + x_3 \end{bmatrix}.$$

Let S be the subspace of $\mathbb{R}^3$ spanned by $\mathbf{e}_1$ and $\mathbf{e}_2$. Find $\ker(L)$ and $L(S)$. ■

In general, for $S = \text{Span}(B)$, where $B \subset \mathcal{V}$, we have $L(S) = \text{Span}(L(B))$.

■ **Example 14.5** Let $D : P_3 \to P_3$ be the linear transformation defined by $D(p(x)) = p'(x)$. Find $\ker(D)$ and $D(P_3)$. ■

Consider a linear transformation $L : \mathcal{V} \to \mathcal{W}$.

- What is the image of $\ker(L)$ in $\mathcal{W}$?
- For any $\mathbf{x}, \mathbf{v} \in V$, $\mathbf{x} \in \mathbf{v} + \ker(L)$, $L(\mathbf{x}) = L(\mathbf{v})$.
- For any linearly independent $\mathbf{v}, \mathbf{v}' \in V$, $L(\mathbf{v})$ and $L(\mathbf{v}')$ are linearly independent if and only if $\text{Span}\{\mathbf{v}, \mathbf{v}'\} \cap \ker(L) = \{\mathbf{0}_\mathcal{V}\}$

Proof. We first prove the sufficiency. Consider scalars c and c' such that $cL(\mathbf{v}) + c'L(\mathbf{v}') = \mathbf{0}$. As $L(c\mathbf{v} + c'\mathbf{v}') = cL(\mathbf{v}) + c'L(\mathbf{v}') = \mathbf{0}$, we have $c\mathbf{v} + c'\mathbf{v}' \in \ker(L)$ and hence $c\mathbf{v} + c'\mathbf{v}' = 0$. Due to the linear independence of $\mathbf{v}$ and $\mathbf{v}'$, we know $c = c' = 0$. Therefore, $L(\mathbf{v})$ and $L(\mathbf{v}')$ are linearly independent.

To prove the necessity, suppose there exists a nonzero vector $\mathbf{w} = c\mathbf{v} + c'\mathbf{v}' \in \ker(L)$, which implies c and c' are not all zero. As $\mathbf{0} = L(\mathbf{w}) = cL(\mathbf{v}) + c'L(\mathbf{v}')$, we have $L(\mathbf{v})$ and $L(\mathbf{v}')$ are linearly dependent. ■

14.3 Linear Transformation from $\mathbb{R}^n$ into $\mathbb{R}^m$

Linear Transformation formed by Matrices

- Let A be an $m \times n$ matrix.
- $L_A(\mathbf{x}) = A\mathbf{x}$ is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

In fact, any linear transformations from $\mathbb{R}^n$ into $\mathbb{R}^m$ can be represented in this matrix form.

Theorem 14.2 For each linear transformation $L : \mathbb{R}^n \to \mathbb{R}^m$, there exists an $m \times n$ matrix A such that

$$L(\mathbf{x}) = A\mathbf{x}$$

for each $\mathbf{x} \in \mathbb{R}^n$, where the jth column of A is $L(\mathbf{e}_j)$, $j = 1, 2, \ldots, n$.

Proof. Let $A = [L(\mathbf{e}_1), \ldots, L(\mathbf{e}_n)]$. For any $\mathbf{x} \in \mathbb{R}^n$, we can write

$$\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n,$$

and hence

$$L(\mathbf{x}) = x_1L(\mathbf{e}_1) + \cdots + x_nL(\mathbf{e}_n) = \begin{bmatrix} L(\mathbf{e}_1) & \cdots & L(\mathbf{e}_n) \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = A\mathbf{x}.$$

■

■ **Example 14.6** Define $L : \mathbb{R}^3 \to \mathbb{R}^2$ by

$$L(\mathbf{x}) = \begin{bmatrix} x_1 + x_2 \\ x_2 + x_3 \end{bmatrix}.$$

Show that L is a linear transformation and find the corresponding matrix A such that $L(\mathbf{x}) = A\mathbf{x}$. ■

A linear transformation from a vectors on itself is also called a *linear operator*. The matrix representation of a linear operator is a square matrix.

■ **Example 14.7** Let L be the linear operator on $\mathbb{R}^2$ that rotates each vector by an angle θ in the counter clockwise direction. Show that $L(\mathbf{x}) = \begin{bmatrix} x_1\cos(\theta) - x_2\sin(\theta) \\ x_1\sin(\theta) + x_2\cos(\theta) \end{bmatrix}$.
■

With the help of this matrix form of a linear transformations L from $\mathbb{R}^n$ into $\mathbb{R}^m$, the kernel and range of L can be better understood:

$$\ker(L) = \{\mathbf{x} : L(\mathbf{x}) = \mathbf{0}\} = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$$
$$L(R^n) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} = \text{Col}(A).$$

In other words, $\ker(L)$ is the solution set of $A\mathbf{x} = \mathbf{0}$ and the null space of A, and the range of L is the column space of $\text{Col}(A)$.

R The above claim does not hold when L is a linear transformations form a subspace of $\mathbb{R}^n$ to $\mathbb{R}^m$.

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Lecture 15

Matrix Representations and Similarity

15.1 Matrix Representation of Vector Spaces

For linear transformations between general vector spaces, we can use coordinates to characterize the similar matrix representation as $L : \mathbb{R}^n \to \mathbb{R}^m$.

Matrix Representation Theorem

- Consider vector spaces $\mathcal{V}$ and $\mathcal{W}$ with bases $\mathcal{B}$ and $\mathcal{C}$, respectively, where

$$\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}, \quad \mathcal{C} = \{\mathbf{w}_1, \ldots, \mathbf{w}_m\}.$$

- Let $L : \mathcal{V} \to \mathcal{W}$ be a linear transformation.
- Let A be an $m \times n$ matrix with $\mathbf{a}_j = [L(\mathbf{v}_j)]_{\mathcal{C}}$.
- For any vector $\mathbf{u} \in \mathcal{V}$, we have

$$[L(\mathbf{u})]_{\mathcal{C}} = A[\mathbf{u}]_{\mathcal{B}}.$$

Proof. Let $\mathbf{x} = [\mathbf{u}]_{\mathcal{B}}$. As

$$\begin{aligned} L(\mathbf{u}) &= L(x_1\mathbf{v}_1 + \cdots + x_n\mathbf{v}_n) \\ &= x_1L(\mathbf{v}_1) + \cdots + x_nL(\mathbf{v}_n), \end{aligned}$$

we have

$$[L(\mathbf{u})]_{\mathcal{C}} = x_1[L(\mathbf{v}_1)]_{\mathcal{C}} + \cdots + x_n[L(\mathbf{v}_n)]_{\mathcal{C}} = A[\mathbf{u}]_{\mathcal{B}}.$$

■

■ **Example 15.1** Let $L : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation defined by

$$L(\mathbf{x}) = x_1\mathbf{b}_1 + (x_2 + x_3)\mathbf{b}_2,$$

where

$$\mathbf{b}_1 = \begin{bmatrix} 3.7 \\ 4.8 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} -5.6 \\ 7.2 \end{bmatrix}.$$

Find the matrix A representing L with respect to the standard basis of $\mathbb{R}^3$ and the basis $\{\mathbf{b}_1, \mathbf{b}_2\}$ of $\mathbb{R}^2$.

We know A is a 2×3 matrix, where

$$\mathbf{a}_1 = [L(\mathbf{e}_1)]_{\{\mathbf{b}_1,\mathbf{b}_2\}} = [\mathbf{b}_1]_{\{\mathbf{b}_1,\mathbf{b}_2\}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$
$$\mathbf{a}_2 = [L(\mathbf{e}_2)]_{\{\mathbf{b}_1,\mathbf{b}_2\}} = [\mathbf{b}_2]_{\{\mathbf{b}_1,\mathbf{b}_2\}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$
$$\mathbf{a}_3 = [L(\mathbf{e}_3)]_{\{\mathbf{b}_1,\mathbf{b}_2\}} = [\mathbf{b}_2]_{\{\mathbf{b}_1,\mathbf{b}_2\}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

■

Coordinate Mapping is Linear

- Let $\mathcal{V}$ be an n-dimensional vector space with a basis $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$.
- Coordinate mapping $[\cdot]_\mathcal{B} : \mathcal{V} \to \mathbb{R}^n$: $[\mathbf{v}]_\mathcal{B}$ is the unique vector $\mathbf{x} \in \mathbb{R}^n$ such that $\mathbf{v} = x_1\mathbf{v}_1 + \cdots + x_n\mathbf{v}_n$.
- The coordinate mapping is linear:
  - kernel: $\{\mathbf{0}_\mathcal{V}\}$
  - range: $\mathbb{R}^n$
  - $[\mathbf{v}]_\mathcal{B} = \begin{bmatrix} \cdots & [\mathbf{v}_j]_\mathcal{B} & \cdots \end{bmatrix} [\mathbf{v}]_\mathcal{B} = I[\mathbf{v}]_\mathcal{B}$.
- The coordinate mapping has the identity matrix representation and is a bijection (one-to-one and onto mapping).

When the basis can be known from the context, we just write $\mathrm{cv}(\mathbf{v}) = [\mathbf{v}]$.

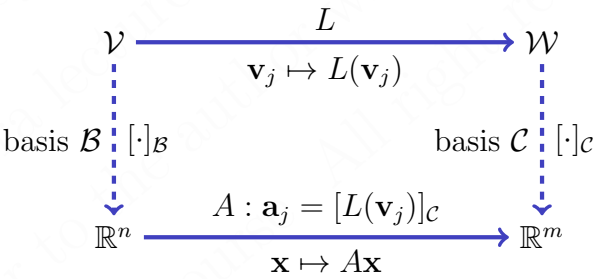


Figure 15.1: Matrix representation of a linear transformation.

■ **Example 15.2** Let L be a linear operator on $\mathbb{R}^2$ defined by

$$L(\alpha\mathbf{b}_1 + \beta\mathbf{b}_2) = (\alpha + \beta)\mathbf{b}_1 + 2\beta\mathbf{b}_2$$

where

$$\mathbf{b}_1 = \begin{bmatrix} 3.7 \\ 4.8 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} -5.6 \\ 7.2 \end{bmatrix}.$$

Find the matrix A representing L with respect to $\{\mathbf{b}_1, \mathbf{b}_2\}$.

We know A is a 2×2 matrix, where

$$\mathbf{a}_1 = [L(\mathbf{b}_1)] = [\mathbf{b}_1] = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$
$$\mathbf{a}_2 = [L(\mathbf{b}_2)] = [\mathbf{b}_1 + 2\mathbf{b}_2] = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

■

■ **Example 15.3** Let $L : \mathbb{R}^2 \to \mathbb{R}^3$ be the linear transformation defined by

$$L(\mathbf{x}) = \begin{bmatrix} x_2 \\ x_1 + x_2 \\ x_1 - x_2 \end{bmatrix}.$$

Find the matrix representation of L with respect to the basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ of $\mathbb{R}^2$ and the basis $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ of $\mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix},$$

and

$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{w}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{w}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

We know that A is a 3×2 matrix, where

$$\mathbf{a}_1 = [L(\mathbf{v}_1)]_{\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}}, \quad \mathbf{a}_2 = [L(\mathbf{v}_2)]_{\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}}.$$

We can solve A directly, or use the following relation:

$$L(\mathbf{v}_1) = W\mathbf{a}_1, \quad L(\mathbf{v}_2) = W\mathbf{a}_2.$$

So $A = W^{-1}[L(\mathbf{v}_1)\ L(\mathbf{v}_2)]$. ■

15.2 Rank-Nullity Theorem Revisited

Consider linear transformation $L : \mathbb{R}^n \to \mathbb{R}^m$. We know that there exists a matrix A such that

$$L(\mathbf{x}) = A\mathbf{x},$$

and hence $\ker(L) = \text{Null}(A)$ and $L(\mathbb{R}^n) = \text{Col}(A)$. By the rank-nullity theorem, we have

$$\dim(\ker(L)) + \dim(L(\mathbb{R}^n)) = n.$$

The same holds for linear transformations over general vector spaces.

Theorem 15.1 For a linear transformation $L : \mathcal{V} \to \mathcal{W}$,

$$\dim(\ker(L)) + \dim(L(\mathcal{V})) = \dim(\mathcal{V}).$$

Lemma 15.2 For a one-to-one linear transformation $L : \mathcal{V} \to \mathcal{W}$, the dimension of $\mathcal{V}$ is the same as the dimension of $L(\mathcal{V})$.

Exercise 15.1 Prove the above lemma ■

Proof of Theorem 15.1. Suppose that A is the matrix representation of $L : \mathcal{V} \to \mathcal{W}$ with basis $\mathcal{B}$ of $\mathcal{V}$ and basis $\mathcal{C}$ of $\mathcal{W}$. Let $n = \dim(\mathcal{V})$. We can check that

$$[\ker(L)]_{\mathcal{B}} = \text{Null}(A), \quad [L(\mathcal{V})]_{\mathcal{C}} = \text{Col}(A). \tag{15.1}$$

As the coordinate mapping is one-to-one, by the above lemma,

$$\begin{aligned} \dim(\ker(L)) &= \dim([\ker(L)]_{\mathcal{B}}) = \dim(\text{Null}(A)), \\ \dim(L(\mathcal{V})) &= \dim([L(\mathcal{V})]_{\mathcal{C}}) = \dim(\text{Col}(A)) = \text{rank}(A). \end{aligned}$$

By the rank-nullity theorem,

$$\dim(\ker(L)) + \dim(L(\mathcal{V})) = \dim(\text{Null}(A)) + \text{rank}(A) = n = \dim(\mathcal{V}).$$

■

Exercise 15.2 Prove the two equalities in (15.1). ■

■ **Example 15.4** Consider the linear transformation $D : P_3 \to P_2$ defined by $D(p) = p'$. Use the basis $\mathcal{B} = \{x^2, x, 1\}$ for P_3 and the basis $\mathcal{C} = \{x, 1\}$ for P_2.

The matrix representation for D with respect to the above bases is a 2×3 matrix A where

$$\begin{aligned} \mathbf{a}_1 &= [D(x^2)]_{\mathcal{C}} = [2x]_{\mathcal{C}} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \\ \mathbf{a}_2 &= [D(x)]_{\mathcal{C}} = [1]_{\mathcal{C}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \\ \mathbf{a}_3 &= [D(1)]_{\mathcal{C}} = [0]_{\mathcal{C}} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \end{aligned}$$

So, $\dim(\ker(L)) = \dim(\text{Null}(A)) = 1$ and $\dim(D(P_3)) = \text{rank}(A) = 2$. ■

15.3 Similarity

Linear Mapping induced by Change of Basis

- Consider an n-dimensional vector space $\mathcal{V}$ with two bases $\mathcal{B}$ and $\mathcal{C}$ be two bases.
- The transition matrix S from $\mathcal{B}$ to $\mathcal{C}$ induces a linear transformation $L_S : \mathbb{R}^n \to \mathbb{R}^n$,

$$L_S(\mathbf{x}) = S\mathbf{x}.$$

Theorem 15.3 Consider a linear operator L on a vector space $\mathcal{V}$. If B is the matrix representation of L with respect to the basis $\mathcal{B}$ of $\mathcal{V}$, then $C = S^{-1}BS$ is the matrix representation of L with respect to the basis $\mathcal{C}$ of $\mathcal{V}$, where S is the transition matrix from $\mathcal{C}$ to $\mathcal{B}$.

Proof. Consider a vector $\mathbf{v} \in \mathcal{V}$. As S is the transition matrix from $\mathcal{C}$ to $\mathcal{B}$, we have

$$[\mathbf{v}]_{\mathcal{B}} = S[\mathbf{v}]_{\mathcal{C}}. \tag{15.2}$$

As $L(\mathbf{v}) \in V$, taking $L(\mathbf{v})$ in place of $\mathbf{v}$ in (15.2), we get

$$[L(\mathbf{v})]_{\mathcal{B}} = S[L(\mathbf{v})]_{\mathcal{C}}.$$

Hence

$$\begin{aligned}[L(\mathbf{v})]_{\mathcal{C}} &= S^{-1}[L(\mathbf{v})]_{\mathcal{B}} \\ &= S^{-1}B[\mathbf{v}]_{\mathcal{B}} \\ &= S^{-1}BS[\mathbf{v}]_{\mathcal{C}},\end{aligned}$$

where the second equality follows that B is the matrix representation of L with respect to the basis $\mathcal{B}$ of $\mathcal{V}$, and the third equality follows from (15.2). Therefore, $C = S^{-1}BS$ the matrix representation of L with respect to the basis $\mathcal{C}$ of $\mathcal{V}$. ■

The theorem can be illustrated by Fig. 15.2.

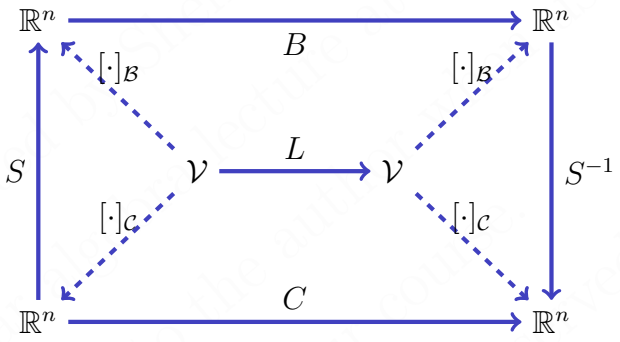


Figure 15.2: A linear operator in different bases.

Definition 15.1 Two $n \times n$ matrices A and B are said to be *similar* if there exits an invertible matrix S such that $B = S^{-1}AS$.

■ **Example 15.5** Let L be the linear operator on $\mathbb{R}^3$ defined by $L(\mathbf{x}) = A\mathbf{x}$, where

$$A = \begin{bmatrix} 2 & 2 & 0 \\ 1 & 1 & 2 \\ 1 & 1 & 2 \end{bmatrix}.$$

Find the matrix representing L with respect to $\mathcal{B} = \{\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3\}$, where

$$\mathbf{y}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{y}_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{y}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

The transition matrix S from $\mathcal{B}$ to the standard basis is

$$S = Y = \begin{bmatrix} \mathbf{y}_1 & \mathbf{y}_2 & \mathbf{y}_3 \end{bmatrix}.$$

Hence, $B = S^{-1}AS = Y^{-1}AY$. ■

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Part V

Orthogonality

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Lecture 16

Inner Product Spaces

16.1 Inner Product in $\mathbb{R}^n$

- Recall the inner product of two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ defined as

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^{\mathrm{T}}\mathbf{y} = x_1y_1 + x_2y_2 + \cdots + x_ny_n.$$

- An inner product in $\mathbb{R}^n$ is also called a scalar product.

Properties of Inner Product of $\mathbb{R}^n$

- $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$
- $\langle a\mathbf{x} + b\mathbf{y}, \mathbf{z} \rangle = a \langle \mathbf{x}, \mathbf{z} \rangle + b \langle \mathbf{y}, \mathbf{z} \rangle$
- $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ with equality if and only if $\mathbf{x} = 0$.
- Binomial expansion:

$$\begin{aligned} \langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y} \rangle &= \langle \mathbf{x}, \mathbf{x} \rangle + \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{x} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle \\ &= \langle \mathbf{x}, \mathbf{x} \rangle + 2 \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle . \end{aligned}$$

- The *(Euclidean) length* of $\mathbf{x} \in \mathbb{R}^n$ is defined as

$$\|\mathbf{x}\| = (\mathbf{x}^{\mathrm{T}}\mathbf{x})^{1/2} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

- The *(Euclidean) distance* between $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$ is defined to be $\|\mathbf{x} - \mathbf{y}\|$. We can normalize $\mathbf{x}$ as $\frac{\mathbf{x}}{\|\mathbf{x}\|}$ which always has length 1.

Cauchy-Schwarz Inequality

Theorem 16.1 If $\mathbf{x}$ and $\mathbf{y}$ are vectors in $\mathbb{R}^n$, then

$$| \langle \mathbf{x}, \mathbf{y} \rangle | \leq \|\mathbf{x}\| \|\mathbf{y}\|$$

with equality holding if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent.

Proof. If $\mathbf{y} = \mathbf{0} = 0\mathbf{x}$, both sides are $\mathbf{0}$ and hence the equality holds.

Consider $\mathbf{y} \neq \mathbf{0}$ in the following proof. For a scalar α, by the binomial expansion,

$$\|\mathbf{x} - \alpha\mathbf{y}\|^2 = \|\mathbf{x}\|^2 - 2\alpha \langle \mathbf{y}, \mathbf{x} \rangle + \alpha^2 \|\mathbf{y}\|^2.$$

So $\|\mathbf{x} - \alpha\mathbf{y}\|^2$ is a quadratic function of α and is minimized when

$$\alpha = \alpha^* \triangleq \frac{\langle \mathbf{y}, \mathbf{x} \rangle}{\|\mathbf{y}\|^2}.$$

Hence,

$$\|\mathbf{x}\|^2 - \frac{\langle \mathbf{y}, \mathbf{x} \rangle^2}{\|\mathbf{y}\|^2} = \|\mathbf{x} - \alpha^*\mathbf{y}\|^2 \geq 0, \tag{16.1}$$

i.e., $|\mathbf{x}\|^2\|\mathbf{y}\|^2 \geq \langle \mathbf{y}, \mathbf{x} \rangle^2$.

If $\mathbf{x} = a\mathbf{y}$, we can check that the equality holds. On the other hand, if the equality in (16.1) holds, i.e., $\|\mathbf{x} - \alpha^*\mathbf{y}\|^2$, by the positive definite property of inner product, $\mathbf{x} = \alpha^*\mathbf{y}$.

The proof is completed as the condition $\mathbf{y} = \mathbf{0}$, or $\mathbf{y} \neq \mathbf{0}$ and $\mathbf{x} = a\mathbf{y}$ can be more concisely expressed as $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent. ■

Angle between two Vectors

- Let $\mathbf{x}, \mathbf{y}$ be in $\mathbb{R}^2$ or $\mathbb{R}^3$, and let θ be the angle between them.
  - By the law of cosines,
  $$\|\mathbf{y} - \mathbf{x}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 - 2\|\mathbf{x}\|\|\mathbf{y}\| \cos(\theta).$$
  - Hence,
  $$\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\|\|\mathbf{y}\| \cos(\theta).$$
- In general, the angle θ between nonzero vectors $\mathbf{x}, \mathbf{y}$ in $\mathbb{R}^n$ is given by
$$\cos(\theta) = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\|\|\mathbf{y}\|}.$$
- It is well defined due to Cauchy-Schwarz inequality.

Definition 16.1 Two vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$ are said to be *orthogonal* if $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^{\mathrm{T}}\mathbf{y} = 0$, denoted by $\mathbf{x} \perp \mathbf{y}$.

Pythagorean Law

- For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,
$$\|\mathbf{x} + \mathbf{y}\|^2 = (\mathbf{x} + \mathbf{y})^{\mathrm{T}}(\mathbf{x} + \mathbf{y}) = \|\mathbf{x}\|^2 + 2\langle \mathbf{x}, \mathbf{y} \rangle + \|\mathbf{y}\|^2.$$
- If $\mathbf{x} \perp \mathbf{y}$, then
$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2.$$

16.2 Inner Product Spaces

Definition 16.2 An *inner product* on a vector space $\mathcal{V}$ is a mapping that assigns each pair of vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathcal{V}$ a *real number* $\langle \mathbf{x}, \mathbf{y} \rangle$ satisfying the following conditions:

1. (positive-definite) $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ with equality if and only if $\mathbf{x} = 0$.
2. (symmetry) $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$ for all $\mathbf{x}, \mathbf{y} \in \mathcal{V}$.
3. (linearity) $\langle \alpha\mathbf{x} + \beta\mathbf{y}, \mathbf{z} \rangle = \alpha \langle \mathbf{x}, \mathbf{z} \rangle + \beta \langle \mathbf{y}, \mathbf{z} \rangle$ for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{V}$ and all scalars

α and β.

A vector space with an inner product is called an *inner product space*.

- For any $n \times n$ diagonal matrix D with positive diagonal entries, $\mathbf{x}^{\mathrm{T}} D\mathbf{y}$ is an inner product on $\mathbb{R}^n$.
- $\langle A, B\rangle = \sum_{i,j} a_{ij} b_{ij}$ is an inner product on $\mathbb{R}^{m\times n}$.
- $\langle f, g\rangle = \int_a^b w(x) f(x) g(x) dx$ is an inner product on $C[a, b]$, where $w(x) \in C[a, b]$ is a positive function.
- $\langle p, q\rangle = \sum_{i=1}^n p(x_i) q(x_i)$ is an inner product on P_n, where $x_1, \ldots, x_n$ are distinct real numbers.

Norm and Orthogonal Vectors

- For $\mathbf{v}$ in an inner product space $\mathcal{V}$, the *norm* of $\mathbf{v}$ is give by

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v}\rangle}.$$

- Two vectors $\mathbf{v}$ and $\mathbf{u}$ in an inner product space is said to be *orthogonal* if $\langle \mathbf{v}, \mathbf{u}\rangle = 0$.

Cauchy-Schwart Inequality

Theorem 16.2 If $\mathbf{x}$ and $\mathbf{y}$ are vectors in an inner product space, then

$$|\langle \mathbf{x}, \mathbf{y}\rangle| \le \|\mathbf{x}\|\|\mathbf{y}\|$$

with equality holding if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent.

Proof. If $\mathbf{y} = \mathbf{0}$, $|\langle \mathbf{x}, \mathbf{y}\rangle| = \mathbf{0} = \|\mathbf{x}\|\|\mathbf{y}\|$.

If $\mathbf{y} \neq \mathbf{0}$, for a scalar α, write

$$\begin{aligned}\|\mathbf{x} - \alpha\mathbf{y}\|^2 &= \langle \mathbf{x} - \alpha\mathbf{y}, \mathbf{x} - \alpha\mathbf{y}\rangle \\ &= \langle \mathbf{x}, \mathbf{x}\rangle - 2\alpha \langle \mathbf{y}, \mathbf{x}\rangle + \alpha^2 \langle \mathbf{y}, \mathbf{y}\rangle \\ &= \|\mathbf{x}\|^2 - 2\alpha \langle \mathbf{y}, \mathbf{x}\rangle + \alpha^2 \|\mathbf{y}\|^2.\end{aligned}$$

So $\|\mathbf{x} - \alpha\mathbf{y}\|^2$ is a quadratic function of α and is minimized when

$$\alpha = \alpha^* \triangleq \frac{\langle \mathbf{y}, \mathbf{x}\rangle}{\|\mathbf{y}\|^2}.$$

Hence,

$$\|\mathbf{x} - \alpha^*\mathbf{y}\|^2 = \|\mathbf{x}\|^2 - \frac{\langle \mathbf{y}, \mathbf{x}\rangle^2}{\|\mathbf{y}\|^2} \ge 0,$$

i.e., $|\mathbf{x}\|^2\|\mathbf{y}\|^2 \ge \langle \mathbf{y}, \mathbf{x}\rangle^2$.

The following argument is same as that in the proof of Theorem 16.1. ■

Angle

- If $\mathbf{x}$ and $\mathbf{y}$ are nonzero vectors in an inner product space, then

$$-1 \leq \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\|\|\mathbf{y}\|} \leq 1.$$

- There exists a unique angle θ in $[0, \pi]$ such that

$$\cos(\theta) = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\|\|\mathbf{y}\|}.$$

- We call θ the angle between $\mathbf{x}$ and $\mathbf{y}$.

Pythagorean Law

- For $\mathbf{x}, \mathbf{y}$ in an inner product space,

$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2\langle \mathbf{x}, \mathbf{y} \rangle + \|\mathbf{y}\|^2.$$

- If $\mathbf{x} \perp \mathbf{y}$ (i.e., $\langle \mathbf{x}, \mathbf{y} \rangle = 0$), then

$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2.$$

16.3 Norm

The term *norm* can be defined independent of an inner product.

A vector space $\mathcal{V}$ is said to be a *normed linear space* if for each $\mathbf{v} \in \mathcal{V}$, there is associated a real number $\|\mathbf{v}\|$ such that

1. $\|\mathbf{v}\| \geq 0$ with equality if and only if $\mathbf{v} = \mathbf{0}$.
2. $\|\alpha \mathbf{v}\| = |\alpha| \|\mathbf{v}\|$ for any scalar α.
3. (triangle inequality) $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$ for all $\mathbf{v}, \mathbf{w} \in \mathcal{V}$.

We call $\|\mathbf{v}\|$ the norm of $\mathbf{v}$.

Theorem 16.3 If $\mathcal{V}$ is an inner product space, then $\|\mathbf{x}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$ is a norm on $\mathcal{V}$.

Proof. The first two conditions are easy to verify. Let us check the triangle inequality. Using Cauchy-Schwarz inequality,

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|^2 &= \|\mathbf{x}\|^2 + 2\langle \mathbf{x}, \mathbf{y} \rangle + \|\mathbf{y}\|^2 \\ &\leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\|\|\mathbf{y}\| + \|\mathbf{y}\|^2 \\ &= (\|\mathbf{x}\| + \|\mathbf{y}\|)^2, \end{aligned}$$

which implies the triangle inequality. ∎

Examples of Norm for $\mathbb{R}^n$

- (1-norm) $\|\mathbf{x}\|_1 = \sum_{i=1}^{n} |x_i|$
- (infinity norm) $\|\mathbf{x}\|_\infty = \max_{1 \le i \le n} |x_i|$
- (p-norm) $\|\mathbf{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{1/p}$

The norm $\|\mathbf{x}\|_2 = \sqrt{\mathbf{x}^{\mathrm{T}}\mathbf{x}}$ is induced by the inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^{\mathrm{T}}\mathbf{y}$. When $p \neq 2$, the norm may not correspond to an inner product.

■ **Example 16.1** Let $\mathbf{x} = \begin{bmatrix} 2 & -1, & 3, 1 \end{bmatrix}^{\mathrm{T}}$ be a vector in $\mathbb{R}^4$. Compute $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_2$ and $\|\mathbf{x}\|_\infty$. ■

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Lecture 17

Orthogonal Subspaces

17.1 More Properties of Orthogonality

Theorem 17.1 If two nonzero vectors $\mathbf{x}, \mathbf{y}$ in an inner product space are orthogonal, then $\mathbf{x}$ and $\mathbf{y}$ are linearly independent.

Proof. Suppose $\alpha\mathbf{x} + \beta\mathbf{y} = \mathbf{0}$. As $\|\mathbf{x}\|^2 > 0$ and

$$\begin{aligned}\alpha\|\mathbf{x}\|^2 &= \alpha\|\mathbf{x}\|^2 + \beta\langle\mathbf{x}, \mathbf{y}\rangle \\ &= \langle\mathbf{x}, \alpha\mathbf{x} + \beta\mathbf{y}\rangle \\ &= \langle\mathbf{x}, \mathbf{0}\rangle \\ &= 0,\end{aligned}$$

we have $\alpha = 0$. Similarly, we have also $\beta = 0$, and hence conclude that $\mathbf{x}$ and $\mathbf{y}$ are linearly independent. ■

Projection

- The proof of the Cauchy-Schwarz inequality can be interpretated as follows:
  - Let l be a line passing through $\mathbf{0}$ and $\mathbf{y} \neq \mathbf{0}$.
  - Find a point $\mathbf{p}$ on l that is closest to $\mathbf{x}$.
- The solution is

$$\mathbf{p} = \alpha^*\mathbf{y}, \quad \alpha^* = \frac{\langle\mathbf{x}, \mathbf{y}\rangle}{\|\mathbf{y}\|^2}$$

  - $\mathbf{p}$ is called the vector projection of $\mathbf{x}$ onto $\mathbf{y}$.
  - α^* is called the scalar projection of $\mathbf{x}$ onto $\mathbf{y}$.
- Check that $(\mathbf{x} - \mathbf{p}) \perp \mathbf{p}$.

Write

$$\begin{aligned}\langle\mathbf{x} - \mathbf{p}, \mathbf{p}\rangle &= \langle\mathbf{x}, \mathbf{p}\rangle - \langle\mathbf{p}, \mathbf{p}\rangle \\ &= \frac{\langle\mathbf{x}, \mathbf{y}\rangle}{\|\mathbf{y}\|^2}\langle\mathbf{x}, \mathbf{y}\rangle - \left(\frac{\langle\mathbf{x}, \mathbf{y}\rangle}{\|\mathbf{y}\|^2}\right)^2\langle\mathbf{y}, \mathbf{y}\rangle \\ &= \frac{\langle\mathbf{x}, \mathbf{y}\rangle^2}{\|\mathbf{y}\|^2} - \frac{\langle\mathbf{x}, \mathbf{y}\rangle^2}{\|\mathbf{y}\|^2} \\ &= 0.\end{aligned}$$

■ **Example 17.1** In $\mathbb{R}^2$, find the point Q on $y = \frac{1}{3}x$ that is closest to the point $(1, 4)$. Let $\mathbf{y} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, which is on $y = \frac{1}{3}x$. Let $\mathbf{x} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$. The point Q corresponds to the projection $\mathbf{q}$ of $\mathbf{x}$ onto $\mathbf{y}$:

$$\mathbf{q} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|^2} \mathbf{y} = \frac{7}{10} \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

■

17.2 Orthogonal Subspaces

Orthogonal Subspaces

Definition 17.1 Two subspaces $\mathcal{X}$ and $\mathcal{Y}$ of an inner product space are said to be *orthogonal*, denoted by $\mathcal{X} \perp \mathcal{Y}$, if $\mathbf{x} \perp \mathbf{y}$ for every $\mathbf{x} \in \mathcal{X}$ and $\mathbf{y} \in \mathcal{Y}$.

■ **Example 17.2** For the following subspaces $\mathcal{X}$ and $\mathcal{Y}$ of $\mathbb{R}^3$, check whether $\mathcal{X} \perp \mathcal{Y}$.

- $\mathcal{X} = \text{Span}\{\mathbf{e}_1\}$ and $\mathcal{Y} = \text{Span}\{\mathbf{e}_2\}$.
- $\mathcal{X} = \text{Span}\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\mathcal{Y} = \text{Span}\{\mathbf{e}_2, \mathbf{e}_3\}$.
- $\mathcal{X} = \text{Span}\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\mathcal{Y} = \text{Span}\{\mathbf{e}_3\}$.

■

From the above example, we see that it is not always intuitive that two subspaces are orthogonal or not.

Proposition 17.2 For orthogonal subspaces $\mathcal{X}$ and $\mathcal{Y}$ of an inner product space, $\mathcal{X} \cap \mathcal{Y} = \{\mathbf{0}\}$.

Proof. First, $\mathbf{0} \in X \cap \mathcal{Y}$ as $\mathcal{X}$ and $\mathcal{Y}$ are subspaces. For any $\mathbf{x} \in \mathcal{X} \cap \mathcal{Y}$, $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ and hence $\mathbf{x} = \mathbf{0}$. ■

Let $\mathcal{X} + \mathcal{Y} = \{\mathbf{x} + \mathbf{y} : \mathbf{x} \in \mathcal{X}, \mathbf{y} \in \mathcal{Y}\}$.

Theorem 17.3 For orthogonal subspaces $\mathcal{X}$ and $\mathcal{Y}$ of an inner product space,

1. The union of a basis of $\mathcal{X}$ and a basis of $\mathcal{Y}$ forms a basis of $\mathcal{X} + \mathcal{Y}$.
2. $\dim(\mathcal{X} + \mathcal{Y}) = \dim(\mathcal{X}) + \dim(\mathcal{Y})$;
3. Each $\mathbf{v} \in \mathcal{X} + \mathcal{Y}$ can be written uniquely as a sum $\mathbf{x} + \mathbf{y}$, where $\mathbf{x} \in \mathcal{X}$ and $\mathbf{y} \in \mathcal{Y}$.

Proof. Let $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ be a basis of $\mathcal{X}$ and let $\{\mathbf{y}_1, \ldots, \mathbf{y}_m\}$ be a basis of $\mathcal{Y}$.

1. We see that $\mathcal{X} + \mathcal{Y} = \text{Span}\{\mathbf{x}_1, \ldots, \mathbf{x}_n, \mathbf{y}_1, \ldots, \mathbf{y}_m\}$. Consider the linear combination

$$\sum_i \alpha_i \mathbf{x}_i + \sum_j \beta_j \mathbf{y}_j = \mathbf{0}.$$

As

$$\sum_i \alpha_i \mathbf{x}_i = -\sum_j \beta_j \mathbf{y}_j \in \mathcal{X} \cap Y,$$

by Proposition 17.2,

$$\sum_i \alpha_i \mathbf{x}_i = \sum_j \beta_j \mathbf{y}_j = \mathbf{0}.$$

Due to the linearly independence of $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$, we have $\alpha_i = 0$, and similarly, we have $\beta_i = 0$, which implies $\{\mathbf{x}_1, \ldots, \mathbf{x}_n, \mathbf{y}_1, \ldots, \mathbf{y}_m\}$ is linearly independent and hence form a basis of $\mathcal{X} + \mathcal{Y}$.

2. By 1), $\dim(\mathcal{X} + \mathcal{Y}) = m + n = \dim(\mathcal{X}) + \dim(\mathcal{Y})$.
3. Suppose we have $\mathbf{x} + \mathbf{y} = \mathbf{x}' + \mathbf{y}'$ for $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$ and $\mathbf{y}, \mathbf{y}' \in \mathcal{Y}$. As

$$\mathbf{x} - \mathbf{x}' = \mathbf{y}' - \mathbf{y} \in \mathcal{X} \cap Y,$$

by Proposition 17.2,

$$\mathbf{x} - \mathbf{x}' = \mathbf{y}' - \mathbf{y} = \mathbf{0}.$$

Therefore, $\mathbf{x} = \mathbf{x}'$ and $\mathbf{y}' = \mathbf{y}$. ■

17.3 Orthogonal Complement

Orthogonal Complement

Definition 17.2 For a subset $\mathcal{Y}$ of an inner product space $\mathcal{V}$, the set of all vectors in $\mathcal{V}$ that are orthogonal to every vector in $\mathcal{Y}$ is called the *orthogonal complement* of $\mathcal{Y}$, and is denoted by $\mathcal{Y}^\perp$. In other words,

$$\mathcal{Y}^\perp = \{\mathbf{x} \in \mathcal{V} : \langle \mathbf{x}, \mathbf{y} \rangle = 0 \text{ for every } \mathbf{y} \in \mathcal{Y}\}.$$

Proposition 17.4 If $\mathcal{Y}$ is a subspace of $\mathcal{V}$, then $\mathcal{Y}^\perp$ is also a subspace of $\mathcal{V}$.

■ **Example 17.3** Let $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ be the standard basis of $\mathbb{R}^3$. Let $\mathcal{X} = \text{Span}\{\mathbf{e}_1\}$ and $\mathcal{Y} = \text{Span}\{\mathbf{e}_2\}$.

- $\mathcal{X}$ and $\mathcal{Y}$ are orthogonal, but they are not orthogonal complements to each other.
- $\mathcal{X}^\perp$?
- $\mathcal{Y}^\perp$?

■

Orthogonal Complement of Column Space

- Consider $\mathcal{Y} = \text{Col}(A)$, where

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \ldots & \mathbf{a}_m \end{bmatrix} \in \mathbb{R}^{n \times m}.$$

- Let $\mathcal{X}$ be the solution set of

$$\mathbf{a}_i^{\mathrm{T}} \mathbf{x} = 0, \quad i = 1, \ldots, m.$$

  In other words, $\mathcal{X} = \text{Null}(A^{\mathrm{T}})$.
- Show that $\mathcal{Y}^\perp = \mathcal{X}$:
  - For $\mathbf{x} \in \mathcal{X}$, $\mathbf{x} \perp \mathbf{y}$ for any $\mathbf{y} \in \mathcal{Y}$. Therefore, $\mathcal{X} \subset \mathcal{Y}^\perp$.
  - For any $\mathbf{y} \in \mathcal{Y}^\perp$, as $\mathbf{a}_i \in \mathcal{Y}$, we have $\mathbf{a}_i^{\mathrm{T}} \mathbf{y} = 0$, $i = 1, \ldots, m$, and hence, $\mathcal{Y}^\perp \subset \mathcal{X}$.

Theorem 17.5 If A is an $m \times n$ matrix, then $\text{Null}(A^{\text{T}}) = \text{Col}(A)^{\perp}$ and $\text{Null}(A) = \text{Col}(A^{\text{T}})^{\perp} = \text{Row}(A)^{\perp}$.

Theorem 17.6 For a subspace $\mathcal{S}$ of $\mathbb{R}^n$,

1. $\dim(\mathcal{S}) + \dim(\mathcal{S}^{\perp}) = n$;
2. $\mathbb{R}^n = \mathcal{S} + \mathcal{S}^{\perp}$.
3. The union of a basis for $\mathcal{S}$ and a basis for $S^{\perp}$ forms a basis for $\mathcal{V}$.

Proof outline.

1. If $\mathcal{S} = \{\mathbf{0}\}$, then $\mathcal{S}^{\perp} = \mathbb{R}^n$. If $\mathcal{S} \neq \{\mathbf{0}\}$, let $\{\mathbf{a}_1, \dots \mathbf{a}_r\}$ be a basis for $\mathcal{S}$. Then $\mathcal{S}^{\perp} = \text{Null}(A^{\text{T}})$, where $A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_r \end{bmatrix}$. By the rank-nullity theorem, $\dim(\mathcal{S}^{\perp}) = n - r$.
2. As $\mathcal{S} + \mathcal{S}^{\perp} \subset \mathbb{R}^n$ and $\dim(\mathcal{S} + \mathcal{S}^{\perp}) = \dim(\mathcal{S}) + \dim(\mathcal{S}^{\perp}) = n$, we have $\mathbb{R}^n = \mathcal{S} + \mathcal{S}^{\perp}$.
3. The third part follows from Theorem 17.3.

■

Theorem 17.7 If $\mathcal{S}$ is a subspace of $\mathbb{R}^n$, then $(\mathcal{S}^{\perp})^{\perp} = \mathcal{S}$.

Proof.

- For any $\mathbf{x} \in \mathcal{S}$, we have $\{\mathbf{x}\} \perp \mathcal{S}^{\perp}$. So $\mathcal{S} \subset (\mathcal{S}^{\perp})^{\perp}$.
- For any $\mathbf{y} \in (\mathcal{S}^{\perp})^{\perp}$, write $\mathbf{y} = \mathbf{u} + \mathbf{v}$ with $\mathbf{u} \in \mathcal{S}$ and $\mathbf{v} \in \mathcal{S}^{\perp}$. As $\langle \mathbf{v}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{y}, \mathbf{v} \rangle = 0$ and $\langle \mathbf{v}, \mathbf{u} \rangle = 0$, we have $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ and hence $\mathbf{v} = \mathbf{0}$. Therefore $\mathbf{y} = \mathbf{u} \in \mathcal{S}$, and $(\mathcal{S}^{\perp})^{\perp} \subset \mathcal{S}$.

■

Lecture 18

Least Squares and Orthonormal Sets

18.1 Projection onto a Subspace of $\mathbb{R}^n$

We have studied the projection of a vector $\mathbf{x}$ onto $\mathbf{y}$, which is the vector $\mathbf{p}$ in Span$\{\mathbf{y}\}$ that is closest to $\mathbf{x}$. We know that $\mathbf{p}$ exists and is unique, and $\mathbf{p} \perp (\mathbf{x} - \mathbf{p})$. We obtain, by minimize a quadratic function, that $\mathbf{p} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|^2}\mathbf{y}$. Now we generalized the project of $\mathbf{x}$ onto a subspace $\mathcal{S}$.

Theorem 18.1 Let $\mathcal{S}$ be a subspace of $\mathbb{R}^n$. For each $\mathbf{x} \in \mathbb{R}^n$, there is a unique element $\mathbf{u} \in \mathcal{S}$ that is closest to $\mathbf{x}$, i.e.,

$$\|\mathbf{u} - \mathbf{x}\| < \|\mathbf{s} - \mathbf{x}\|, \quad \text{for any } \mathbf{s} \neq \mathbf{u} \in \mathcal{S}.$$

Moreover, a given vector $\mathbf{u} \in \mathcal{S}$ is closest to a give vector $\mathbf{x} \in \mathbb{R}^n$ if and only if $\mathbf{x} - \mathbf{u} \in \mathcal{S}^\perp$.

Proof. We know that $\mathbf{x}$ can be expressed uniquely as $\mathbf{x} = \mathbf{u} + \mathbf{v}$ where $\mathbf{u} \in \mathcal{S}$ and $\mathbf{v} \in \mathcal{S}^\perp$. For any $\mathbf{s} \neq \mathbf{u} \in \mathcal{S}$,

$$\begin{aligned} \|\mathbf{x} - \mathbf{s}\|^2 &= \|\mathbf{x} - \mathbf{u} + \mathbf{u} - \mathbf{s}\|^2 \\ &= \|\mathbf{x} - \mathbf{u}\|^2 + \|\mathbf{u} - \mathbf{s}\|^2 \\ &> \|\mathbf{x} - \mathbf{u}\|^2, \end{aligned}$$

where the second equality follows form the Pythagorean law.

The above argument also implies that if $\mathbf{u} \in \mathcal{S}$ and $\mathbf{x} - \mathbf{u} \in \mathcal{S}^\perp$, then $\mathbf{u}$ is a vector in $\mathcal{S}$ that is closest to $\mathbf{x}$. Conversely, if $\mathbf{s} \in \mathcal{S}$ and $\mathbf{x} - \mathbf{s} \notin S^\perp$, then $\mathbf{s} \neq \mathbf{u}$. The above argument implies $\|\mathbf{x} - \mathbf{s}\| > \|\mathbf{x} - \mathbf{u}\|$. ■

Projection onto a Subspace

- The projection of a vector $\mathbf{x}$ onto a subspace $\mathcal{S}$ is the vector $\mathbf{p} \in \mathcal{S}$ that is closest to $\mathbf{x}$.
- $\mathbf{p}$ exists and is unique.
- Let $\{\mathbf{a}_1, \ldots, \mathbf{a}_r\}$ be a basis of $\mathcal{S}$, and let $A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_r \end{bmatrix}$.

- The following three statements are equivalent to that $\mathbf{p} = A\mathbf{c}$ is the projection of $\mathbf{x}$ onto $\mathcal{S}$:
  - $\mathbf{x} - A\mathbf{c} \in \mathcal{S}^{\perp}$;
  - $\mathbf{a}_i^{\mathrm{T}}(\mathbf{x} - A\mathbf{c}) = 0$, $i = 1, \ldots, r$;
  - $A^{\mathrm{T}}(\mathbf{x} - A\mathbf{c}) = \mathbf{0}$.
- A necessary and sufficient condition that $\mathbf{p}$ is the projection of $\mathbf{x}$ onto $\mathcal{S}$:

$$A^{\mathrm{T}}A\mathbf{c} = A^{\mathrm{T}}\mathbf{x}. \tag{18.1}$$

As $\mathrm{rank}(A) = r$, by Theorem 11.9, $\mathrm{rank}(A^{\mathrm{T}}A) = r$ and hence the system (18.1) has a unique solution $(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}\mathbf{x}$.

The previous analysis can be summarized as follows.

Projection onto a Subspace

- Let A be the matrix formed by a basis of a subspace $\mathcal{S}$, which implies $A^{\mathrm{T}}A$ is invertible.
- The projection $\mathbf{p}$ of a vector $\mathbf{x}$ onto a subspace $\mathcal{S}$ is

$$\mathbf{p} = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}\mathbf{x}.$$

- The matrix $A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$ is called the projection matrix.

■ **Example 18.1** Let $\mathcal{S} = \mathrm{Span}\{\mathbf{e}_1, \mathbf{e}_4, \mathbf{e}_7\} \subset \mathbb{R}^7$. Find the projection of $\mathbf{x} = (1, 2, 3, 4, 5, 6, 7)^{\mathrm{T}}$ onto $\mathcal{S}$.

Let $A = \begin{bmatrix}\mathbf{e}_1 & \mathbf{e}_4 & \mathbf{e}_7\end{bmatrix}$. We have

$$(A^{\mathrm{T}}A)^{-1} = I.$$

So, the project $\mathbf{p}$ of $\mathbf{x}$ onto $\mathcal{S}$ is

$$\mathbf{p} = AA^{\mathrm{T}}\mathbf{x} = \mathbf{e}_1 + 4\mathbf{e}_4 + 7\mathbf{e}_7.$$

■

18.2 Least Squares Solutions

Least Squares Solutions

- A system of linear equations $A\mathbf{x} = \mathbf{b}$ may not have a solution.
- We want to find a value of $\mathbf{x}$ such that $A\mathbf{x}$ and $\mathbf{b}$ is "close".
- The value of $\mathbf{x}$ that minimize $\|A\mathbf{x} - \mathbf{b}\|$ is called a *least squares solution* of $A\mathbf{x} = \mathbf{b}$.
- To solve the least squares problem,
  - find the projection $\mathbf{p}$ of $\mathbf{b}$ onto $\mathrm{Col}(A)$,
  - solve $A\mathbf{x} = \mathbf{p}$.
- Similar to the analysis of projection

$$A^{\mathrm{T}}(\mathbf{b} - \mathbf{p}) = A^{\mathrm{T}}(\mathbf{b} - A\mathbf{x}) = \mathbf{0}.$$

- Hence, the least squares solutions can be found by solving

$$A^{\mathrm{T}}A\mathbf{x} = A^{\mathrm{T}}\mathbf{b}. \tag{18.2}$$

- Note that in general $A^{\mathrm{T}}A$ may not be of full rank!
- The solution $\hat{\mathbf{x}}$ of (18.2) is not unique, but $A\hat{\mathbf{x}}$ is the same for all solutions as $\mathbf{p} = A\hat{\mathbf{x}}$ is unique.

The system (18.2) with variable $\mathbf{x}$ must have a solution due to the existence of projection. We can also show that the above system has a solution: As $\mathrm{Null}(A) = \mathrm{Null}(A^{\mathrm{T}}A)$, we have $\mathrm{Col}(A^{\mathrm{T}}A) = \mathrm{Null}(A^{\mathrm{T}}A)^{\perp} = \mathrm{Null}(A)^{\perp} = \mathrm{Col}(A^{\mathrm{T}})$. Therefore, $A^{\mathrm{T}}\mathbf{x} \in \mathrm{Col}(A^{\mathrm{T}}) = \mathrm{Col}(A^{\mathrm{T}}A)$, which is equivalent to the system (18.1) has a solution.

■ **Example 18.2** Given a table of data

w	w_1	w_2	$\cdots$	w_m
x	x_1	x_2	$\cdots$	x_m
y	y_1	y_2	$\cdots$	y_m
z	z_1	z_2	$\cdots$	z_m

we wish to find a linear function $f(x, y, z) = ax + by + cz$ to best fit the data in the least squares sense, i.e., minimizing the *mean squared error* of the data fitting defined as

$$\frac{1}{m}\sum_{i=1}^{m}[f(x_i, y_i, z_i) - w_i]^2.$$

We get a system of m equations in three unknowns:

$$\begin{bmatrix} x_1 & y_1 & z_1 \\ \vdots & \vdots & \vdots \\ x_m & y_m & z_m \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}$$

The least squares solution is $\hat{a}, \hat{b}, \hat{c}$. The function $f(x, y, z) = \hat{a}x + \hat{b}y + \hat{c}z$ is the best least squares fit to the data in the table. ■

18.3 Orthonormal Sets

Definition 18.1 A set of *nonzero* vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ in an inner product space $\mathcal{V}$ is said to be an *orthogonal set* if $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ whenever $i \neq j$.

Theorem 18.2 If $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is an orthogonal set, then $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are linearly independent.

This theorem is a generalization of Theorem 17.1.

Exercise 18.1 Prove the theorem. ■

Definition 18.2 An *orthonormal* set of vectors is an othogonal set of *unit* vectors.

- The set $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ is orthonormal if and only if

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

- Given an orthogonal set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$,

$$\mathbf{u}_j = \frac{\mathbf{v}_j}{\|\mathbf{v}_j\|}, \quad j = 1, \ldots, n$$

  form an orthonormal set.

■ **Example 18.3** Check that the following set of vectors in $\mathbb{R}^3$ is orthogonal:

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 4 \\ -5 \\ 1 \end{bmatrix} \right\}.$$

Form an orthonormal set using the above set. ■

The standard basis of $\mathbb{R}^n$ is an orthonormal set of vectors.

Orthonormal Basis

- If $\mathcal{B} = \{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$ be a orthonormal set in an r-dimensional vector space $\mathcal{V}$, then $\mathcal{B}$ is a basis for $\mathcal{V}$.
- We say $\mathcal{B}$ is an *orthonormal basis* of $\mathcal{V}$.

An orthogonal basis has many desired properties of the standard basis.
Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$ be an orthonormal basis for an inner product space $\mathcal{V}$.

- If $\mathbf{x} = \sum_{i=1}^r c_i \mathbf{u}_i \in \mathcal{V}$, then $c_i = \langle \mathbf{x}, \mathbf{u}_i \rangle$.
- If $\mathbf{x} = \sum_{i=1}^r a_i \mathbf{u}_i \in \mathcal{V}$ and $\mathbf{y} = \sum_{i=1}^r b_i \mathbf{u}_i \in \mathcal{V}$, then $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^r a_i b_i$.
- If $\mathbf{x} = \sum_{i=1}^r c_i \mathbf{u}_i \in \mathcal{V}$, then $\|\mathbf{x}\| = \sqrt{\sum_{i=1}^r c_i^2}$.

■ **Example 18.4** Consider

$$\mathbf{u}_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}.$$

- Show that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthonormal basis of $\mathbb{R}^2$.
- Find the coordinates of $\mathbf{x} \in \mathbb{R}^2$ w.r.t. $\{\mathbf{u}_1, \mathbf{u}_2\}$.
- Evaluate the norm of $\mathbf{x}$ using the above coordinates.

■

Projection with Orthonormal Basis in $\mathbb{R}^n$

- Let A be the matrix formed by an *orthonormal basis* of a subspace $\mathcal{S}$ of $\mathbb{R}^n$, which implies $A^{\mathrm{T}}A = I$.
- The projection $\mathbf{p}$ of a vector $\mathbf{x}$ onto a subspace $\mathcal{S}$ is

$$\mathbf{p} = A(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}\mathbf{x} = AA^{\mathrm{T}}\mathbf{x} = \sum_i \langle \mathbf{a}_i, \mathbf{x} \rangle \, \mathbf{a}_i.$$

- The least squares solution of $A\mathbf{x} = \mathbf{b}$ is

$$\hat{\mathbf{x}} = A^{\mathrm{T}}\mathbf{b}.$$

Lecture 19

Gram-Schmidt Orthogonalization

19.1 Projection in Inner Product Spaces

Theorem 19.1 Let $\mathcal{S}$ be a subspace of an inner product space $\mathcal{V}$ and let $\mathbf{x} \in \mathcal{V}$. Let $\{\mathbf{a}_1, \ldots, \mathbf{a}_r\}$ be an orthonormal basis for $\mathcal{S}$, and let

$$\mathbf{p} = \sum_{i=1}^{r} \langle \mathbf{x}, \mathbf{a}_i \rangle \, \mathbf{a}_i$$

We have

1. $\mathbf{x} - \mathbf{p} \in S^{\perp}$;
2. $\mathbf{p}$ is the projection of $\mathbf{x}$ onto $\mathcal{S}$, that is $\mathbf{p}$ is the element of $\mathcal{S}$ that is closest to $\mathbf{x}$.

Proof. To show 1), it is sufficient to show $\mathbf{x} - \mathbf{p} \perp \mathbf{a}_j$ for any j (why?):

$$\begin{aligned} \langle \mathbf{x} - \mathbf{p}, \mathbf{a}_j \rangle &= \langle \mathbf{x}, \mathbf{a}_j \rangle - \langle \mathbf{p}, \mathbf{a}_j \rangle \\ &= \langle \mathbf{x}, \mathbf{a}_j \rangle - \sum_{i=1}^{r} \langle \mathbf{a}_i, \mathbf{x} \rangle \langle \mathbf{a}_j, \mathbf{a}_i \rangle \\ &= \langle \mathbf{x}, \mathbf{a}_j \rangle - \langle \mathbf{x}, \mathbf{a}_j \rangle \\ &= 0. \end{aligned}$$

To show 2), for any $\mathbf{u} \neq \mathbf{p} \in \mathcal{S}$,

$$\begin{aligned} \|\mathbf{u} - \mathbf{x}\|^2 &= \|\mathbf{u} - \mathbf{p} + \mathbf{p} - \mathbf{x}\|^2 \\ &= \|\mathbf{u} - \mathbf{p}\|^2 + \|\mathbf{p} - \mathbf{x}\|^2 \\ &> \|\mathbf{p} - \mathbf{x}\|^2, \end{aligned}$$

where the first equality follows from 1) and the Pythagorean law. ■

An Orthonormal Subset of $C[-\pi, \pi]$

For the vector space $C[-\pi, \pi]$, we define the inner product

$$\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) g(x) dx.$$

We have is an orthonormal set of vectors:

$$\mathcal{B} = \{1/\sqrt{2}, \cos x, \cos 2x, \ldots, \cos nx, \sin x, \sin 2x, \ldots, \sin nx\}$$

Verify that

$$\begin{aligned}
\left\langle 1/\sqrt{2}, 1/\sqrt{2} \right\rangle &= \frac{1}{\pi}\int_{-\pi}^{\pi} \frac{1}{2} dx = 1 \\
\left\langle 1/\sqrt{2}, \cos(kx) \right\rangle &= \frac{1}{\pi}\int_{-\pi}^{\pi} \frac{1}{\sqrt{2}} \cos(kx) dx = 0 \\
\left\langle 1/\sqrt{2}, \sin(kx) \right\rangle &= \frac{1}{\pi}\int_{-\pi}^{\pi} \frac{1}{\sqrt{2}} \sin(kx) dx = 0 \\
\langle \sin(kx), \cos(k'x) \rangle &= \frac{1}{\pi}\int_{-\pi}^{\pi} \sin(kx)\cos(k'x) dx = 0 \\
\langle \sin(kx), \sin(k'x) \rangle &= \frac{1}{\pi}\int_{-\pi}^{\pi} \sin(kx)\sin(k'x) dx = \begin{cases} 0 & k' \neq k \\ 1 & k' = k \end{cases} \\
\langle \cos(kx), \cos(k'x) \rangle &= \frac{1}{\pi}\int_{-\pi}^{\pi} \cos(kx)\cos(k'x) dx = \begin{cases} 0 & k' \neq k \\ 1 & k' = k \end{cases}
\end{aligned}$$

Let $\mathcal{S} = \text{Span}\{1/\sqrt{2}, \cos x, \cos 2x, \ldots, \cos nx, \sin x, \sin 2x, \ldots, \sin nx\}$. We know that $\mathcal{B}$ is an orthonormal basis of $\mathcal{S}$. For any function $f \in C[-\pi, \pi]$, we can use its projection f^* onto $\mathcal{S}$ as an approximation of f. By the above theorem, we know that

$$f^* = a_0/2 + \sum_{k=1}^{n} [a_k \cos(kx) + b_k \sin(kx)],$$

where

$$\begin{aligned}
a_0 &= \langle f, 1 \rangle \\
a_k &= \langle f, \cos(kx) \rangle \\
b_k &= \langle f, \sin(kx) \rangle .
\end{aligned}$$

This is the best approximation of f using a function g in $\mathcal{S}$ in terms of minimizing the squared error $\|f - g\|^2$.

19.2 Gram-Schmidt Orthogonalization

We have seen the advantages of using an orthonormal basis, but we have now shown the existence of an orthonormal basis. In an n-dimensional vector space, there exists a set of n linearly independent vectors. In this section, we show that a set of n linearly independent vectors can be transformed to an orthonormal set.

Orthogonalization Problem

- Let $\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}$ be a linearly independent set of vectors from an inner product space.

- We want to construct an orthonormal set $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ such that

$$\text{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} = \text{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_k\}$$

for $k = 1, \ldots, n$.

Lemma 19.2 Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be an orthonormal set of an inner product space $\mathcal{V}$ and $\mathbf{x}$ be a vector in $\mathcal{V}$, but not in $\text{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$. Let

$$\mathbf{v} = \mathbf{x} - \sum_{i=1}^{k} \langle \mathbf{x}, \mathbf{u}_i \rangle \mathbf{u}_i$$
$$\mathbf{u} = \mathbf{v}/\|\mathbf{v}\|.$$

Then we have
1. $\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{u}\}$ is an orthonormal set, and
2. $\text{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{u}\} = \text{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{x}\}$.

Proof. Let $\mathbf{S} = \text{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$. First, $\mathbf{u} \neq \mathbf{0}$ since otherwise, $\mathbf{x} \in S$, a contradiction to the condition of the lemma. Therefore $\mathbf{u}$ is well-defined and unit. By Theorem 19.1, $\mathbf{u}$ is in the orthogonal complement of $\mathbf{S}$, and hence $\mathbf{u} \perp \mathbf{u}_i$, $i = 1, \ldots k$.

Second, see $\text{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{u}\} \subset \text{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{x}\}$, and using 1), we have $\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{u}\}$ forms a basis of $\text{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{x}\}$. ■

Gram-Schmidt Process

$\mathbf{u}_k = \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|}$ for $k = 1, \ldots, n$, where

- $\mathbf{v}_1 = \mathbf{a}_1$;
- For $k = 2, \ldots, n$,

$$\begin{aligned}\mathbf{v}_k &= \mathbf{a}_k - \sum_{i=1}^{k-1} \langle \mathbf{a}_k, \mathbf{u}_i \rangle \mathbf{u}_i \\ &= \mathbf{a}_k - \sum_{i=1}^{k-1} \frac{\langle \mathbf{a}_k, \mathbf{v}_i \rangle}{\|\mathbf{v}_i\|^2} \mathbf{v}_i.\end{aligned}$$

For $k = 2, \ldots, n$, we can verify that $\{\mathbf{u}_1, \ldots, \mathbf{u}_{k-1}\}$ and $\mathbf{a}_k$ satisfy the condition of Lemma 19.2, and hence, $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ is orthonormal and

$$\begin{aligned}\text{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\} &= \text{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_{k-1}, \mathbf{a}_k\} \\ &= \text{Span}\{\mathbf{u}_1, \ldots, \mathbf{u}_{k-1}\} + \text{Span}\{\mathbf{a}_k\} \\ &= \text{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_{k-1}\} + \text{Span}\{\mathbf{a}_k\} \\ &= \text{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_k\}\end{aligned}$$

where the third equality follows from the induction hypothesis.

■ **Example 19.1** Let

$$A = \begin{bmatrix} 1 & -1 & 4 \\ 1 & 4 & -2 \\ 1 & 4 & 2 \\ 1 & -1 & 0 \end{bmatrix}.$$

Find an orthonormal basis for the column space of A. ■

19.3 Orthogonal Matrices

Change of basis between orthonormal bases is also simpler.

Definition 19.1 An $n \times n$ matrix Q is said to be an *orthogonal matrix* if the column vectors of Q form an orthonormal set in $\mathbb{R}^n$.

Theorem 19.3 An $n \times n$ matrix Q over $\mathbb{R}$ is orthogonal if and only if $Q^{\mathrm{T}}Q = I$.

Properties

If Q is an $n \times n$ orthogonal matrix, then

- the columns of Q form an orthonormal basis of $\mathbb{R}^n$;
- $Q^{\mathrm{T}} = Q^{-1}$;
- Q^{T} is also orthogonal;
- $\langle Q\mathbf{x}, Q\mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle$;
- $\|Q\mathbf{x}\| = \|\mathbf{x}\|$.

QR Decomposition

- From the G-S process,
  - $\mathbf{a}_1 = \|\mathbf{v}_1\|\mathbf{u}_1$
  - For $j = 2, \ldots, n$, $\mathbf{a}_j = \sum_{k=1}^{j-1} \langle \mathbf{a}_j, \mathbf{u}_k\rangle \mathbf{u}_k + \|\mathbf{v}_j\|\mathbf{u}_j$.
- Let $A = [\mathbf{a}_1, \ldots, \mathbf{a}_n]$, $U = [\mathbf{u}_1, \ldots, \mathbf{u}_n]$, and $R = (r_{ij})$ be an $n \times n$ upper triangular matrix with $r_{ii} = \|\mathbf{v}_i\|$, and for $j > i$, $r_{ij} = \langle \mathbf{a}_j, \mathbf{u}_i\rangle$. Then

$$A = UR.$$

Note that U here may not be squared, but all the columns of U form an orthonormal set and hence $U^{\mathrm{T}}U = I$. Matrix R is a upper triangular matrix will all the diagonal entries positive.

Find the QR decomposition of Example 19.1.

Solving Least Squares

- The least squares solution of $A\mathbf{x} = \mathbf{b}$ is the solution of the equations

$$A^{\mathrm{T}}A\mathbf{x} = A^{\mathrm{T}}\mathbf{b}.$$

- Substituting $A = QR$, we have $(QR)^{\mathrm{T}}QR\mathbf{x} = (QR)^{\mathrm{T}}\mathbf{b}$, or

$$R^{\mathrm{T}}R\mathbf{x} = R^{\mathrm{T}}Q^{\mathrm{T}}\mathbf{b}.$$

- Since R is invertible,

$$\mathbf{x} = R^{-1}Q^{\mathrm{T}}\mathbf{b}.$$

Part VI

Eigenvalues

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Lecture 20

Eigenvalues and Eigenvectors

Example

- In a certain town, 30% married women get divorced each year and 20% single women get married each year.
- There are 8000 married women and 2000 single women, and the total population remains constant.
- If these percentages of marriages and divorces continue indefinitely into the future, what are the long-range prospects of the marriage of women in the town?

Let $\mathbf{w}_0 = \begin{bmatrix} 8000 \\ 2000 \end{bmatrix}$ be the initial state. Let $\mathbf{w}_i$ be vector formed by the numbers of married and single women after i years. Let

$$A = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}.$$

Then $\mathbf{w}_n = A\mathbf{w}_{n-1} = A^n\mathbf{w}_0$ for $n = 1, \ldots$. After a certain number of years, we will see that $\mathbf{w}_n$ is very close to $\mathbf{x}_1 = \begin{bmatrix} 4000 \\ 6000 \end{bmatrix}$. We can check that

$$A\mathbf{x}_1 = \mathbf{x}_1. \tag{20.1}$$

So we call $\mathbf{x}_1$ the steady state.

It turns out that for different initial proportions of married and single women, the steady state is the same! To illustrate why this is the case, we can represent a general initial status as

$$\mathbf{w}_0 = \mathbf{x}_1 + c \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

where $-4000 \leq c \leq 6000$. This formula guarantee that the total population is fixed. We can check that

$$A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 0.5 \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Let

$$\mathbf{x}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Then

$$A^n\mathbf{w}_0 = A^n\mathbf{x}_1 + cA^n\mathbf{x}_2 = \mathbf{x}_1 + c0.5^n\mathbf{x}_2.$$

20.1 Eigenvalues and Eigenvectors

Definition 20.1 Let A be an $n \times n$ matrix. A scalar λ is said to be an *eigenvalue* of A if there exists a nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$. The vector $\mathbf{x}$ is said to be an eigenvector belonging to λ.

Some properties:

- An eigenvector $\mathbf{x}$ and $A\mathbf{x}$ have the same direction.
- If $\mathbf{x}$ is an eigenvector belonging to λ, so is $c\mathbf{x}$ for any $c \neq 0$.
- If λ is an eigenvalue of A, then λ^s is an eigenvalue of A^s.

■ **Example 20.1** Let

$$A = \begin{bmatrix} 0 & -3 \\ 1 & -4 \end{bmatrix}, \mathbf{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Then

$$A\mathbf{u} = \begin{bmatrix} 0 & -3 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ -3 \end{bmatrix} = (-3)\mathbf{u}.$$

Thus, -3 is the eigenvalue of A and the corresponding eigenvector is $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. ■

Equivalent Characterizations

- Note that $A\mathbf{x} = \lambda\mathbf{x}$ is equivalent to

$$(A - \lambda I)\mathbf{x} = 0.$$

- The following statements are equivalent:
  - λ is an eigenvalue of A.
  - $(A - \lambda I)\mathbf{x} = 0$ has a nontrivial solution.
  - $\text{Null}(A - \lambda I) \neq \{\mathbf{0}\}$.
  - $A - \lambda I$ is singular.
  - $\det(A - \lambda I) = 0$.

Eigenspace

- $\text{Null}(A - \lambda I)$ is called the *eigenspace* of eigenvalue λ.
- All nonzero vectors in $\text{Null}(A - \lambda I)$ are eigenvectors corresponding to λ.

■ **Example 20.2** Find the eigenvalues and the corresponding eigenvectors of

$$A = \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix}.$$

A scalar λ is an eigenvalue if and only if

$$\begin{vmatrix} 3 - \lambda & 2 \\ 3 & -2 - \lambda \end{vmatrix} = \lambda^2 - \lambda - 12 = 0.$$

Thus, the eigenvalues of A are $\lambda_1 = 4$ and $\lambda_3 = -3$.

The eigenvectors belonging to λ_1 are nonzero solutions of

$$(A - 4I)\mathbf{x} = \mathbf{0}.$$

The eigenvectors belonging to λ_2 are nonzero solutions of

$$(A + 3I)\mathbf{x} = \mathbf{0}.$$

■

■ **Example 20.3** Find the eigenvalues and the corresponding eigenvectors of

$$A = \begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 1 \\ 1 & -3 & 2 \end{bmatrix}.$$

■

Characteristic polynomial

- $p(\lambda) = \det(A - \lambda I)$ is an nth degree polynomial in λ.
- $p(\lambda)$ is called the *characteristic polynomial* of A.
- $p(\lambda) = 0$ is called the *characteristic equation* of A.

20.2 Complex Eigenvalues

■ **Example 20.4** Find the eigenvalues and the corresponding eigenvectors of

$$A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}.$$

The characteristic equation is

$$\begin{vmatrix} 1-\lambda & 2 \\ -2 & 1-\lambda \end{vmatrix} = (1-\lambda)^2 + 4 = 0.$$

Thus the eigenvalues of A is $\lambda_1 = 1 + 2i$ and $\lambda_2 = 1 - 2i$.

Write

$$A - \lambda_1 I = \begin{bmatrix} -2i & 2 \\ -2 & -2i \end{bmatrix} \xrightarrow{R_2 + iR_1} \begin{bmatrix} -2i & 2 \\ 0 & 0 \end{bmatrix}.$$

The eigenspace of λ_1 is $\text{Span}\{(1, i)^{\text{T}}\}$. Similarly, he eigenspace of λ_2 is $\text{Span}\{(1, -i)^{\text{T}}\}$.

■

Complex Matrices

- For a complex number $\lambda = a + bi$, the symbol $\bar{\lambda}$ is used to denote the complex conjugate of λ, i.e., $\bar{\lambda} = a - bi$.
- For a matrix $A = (a_{ij})$ of complex numbers, we denote $\bar{A} = (\bar{a}_{ij})$.
- A matrix is real if $\bar{A} = A$.

Exercise 20.1 For complex scalars α and β,

$$\overline{\alpha+\beta} = \overline{\alpha} + \overline{\beta} \quad \text{and} \quad \overline{\alpha\beta} = \overline{\alpha}\,\overline{\beta}$$

Moreover, for matrices A and B with complex entries, $\overline{AB} = \bar{A}\bar{B}$. ∎

Complex Eigenvalues of Real Matrices

- As $p(\lambda)$ has degree n, $p(\lambda)$ can be factored into the product of n linear terms:

$$p(\lambda) = c(\lambda_1 - \lambda)\cdots(\lambda_n - \lambda), \tag{20.2}$$

  where λ_i is the root of $p(\lambda)$.
- For real valued matrices,
  - Complex eigenvalues occur in *conjugate pairs*, i.e., if λ is an eigenvalue, so is $\bar{\lambda}$.
  - If $\mathbf{z}$ is an eigenvector belonging to a complex eigenvalue λ, then $\bar{\mathbf{z}}$ is an eigenvector belonging to $\bar{\lambda}$.

20.3 More Properties about Eigenvalues

Using the Leibniz formula of determinants, we have that the degree n and $n-1$ terms of $p(\lambda) = \det(A - \lambda I)$ appear only in $(a_{11} - \lambda)(a_{22} - \lambda)\cdots(a_{nn} - \lambda)$, as other additive terms involve at most $n-2$ diagonal entries of $A - \lambda I$. Consider

$$(a_{11} - \lambda)(a_{22} - \lambda)\cdots(a_{nn} - \lambda) = (-\lambda)^n + \sum_{i=1}^{n} a_{ii}(-\lambda)^{n-1} + \cdots .$$

Comparing the $(-\lambda)^n$ term with (20.2), we get $c = 1$; and comparing the $(-\lambda)^{n-1}$ term with (20.2), we get $\sum_{i=1}^n a_{ii} = \sum_{i=1}^n \lambda_i$. We further have, $\det(A) = p(0) = \prod_{i=1}^n \lambda_i$.

Product and Sum of the Eigenvalues

Let A be an $n \times n$ matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$.

- $\det(A) = \lambda_1 \cdots \lambda_n$
- $\sum_{i=1}^n a_{ii} = \sum_{i=1}^n \lambda_i$

We know that $\lambda_1, \ldots, \lambda_n$ may not be all distinct. Suppose there are p distinct eigenvalues. Let $\lambda_1, \ldots, \lambda_p$ be the p distinct eigenvalues. The characteristic polynomial can be written as

$$p(\lambda) = (\lambda_1 - \lambda)^{m_1} \cdots (\lambda_p - \lambda)^{m_p}, \tag{20.3}$$

where m_k is called the *multiplicity* of eigenvalue λ_k. We know that $\sum_k m_k = n$.

Recall that a matrix B is said to be similar to a matrix A if there exists a nonsingular matrix S such that $B = S^{-1}AS$. More properties about eigenvalues:

- A and A^{T} have same characteristic polynomial, and hence the same eigenvalues.

- Let A and B be $n \times n$ matrices similar to each other. Then, they have the same characteristic polynomial and, consequently, the same eigenvalues.
- For an invertible matrix A, λ is an eigenvalue of A if and only if λ^{-1} is an eigenvalue of A^{-1}.

Exercise 20.2 Prove the above properties. ■

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Lecture 21

Diagonalization

21.1 Diagonalization

Definition 21.1 An $n \times n$ matrix A is said to be *diagonalizable* if A is similar to a diagonal matrix (in other words, there exists a nonsingular matrix X and a diagonal matrix D such that $X^{-1}AX = D$).

Theorem 21.1 An $n \times n$ matrix A is diagnolizable if and only if A has n linearly independent eigenvectors.

Proof. Suppose that A has n linearly independent eigenvectors $\mathbf{x}_1, \ldots, \mathbf{x}_n$. Let λ_i be the eigenvalue of A corresponding to $\mathbf{x}_i$. Let X be the matrix with $\mathbf{x}_j$ as the jth column, and let $D = \text{diag}(\lambda_1, \ldots, \lambda_n)$. Then,

$$AX = XD.$$

Conversely, suppose that A is diagonlizable, i.e., $A = XDX^{-1}$, where D is a diagonal matrix. As

$$A\mathbf{x}_j = d_{jj}\mathbf{x}_j, \quad j = 1, \ldots, n,$$

d_{jj} is an eigenvalue of A with a corresponding eigenvector $\mathbf{x}_j$. ■

R

- If A is diagonalizable, then the column vectors of the X are eigenvectors of A and the diagonal elements of D are the corresponding eigenvalues of A.
- Matrix X is not unique.

Are eigenvectors linearly independent in general?

Theorem 21.2 Let $\lambda_1, \lambda_2, \ldots, \lambda_k$ are distinct eigenvalues of an $n \times n$ matrix with corresponding eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$, then $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$ are linearly independent.

Proof. Suppose the maximum linearly independent subset of $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k\}$ has i vectors. If $i = k$, we theorem is proved. Assume $i < k$. WLOG, suppose $\mathbf{x}_1, \ldots, \mathbf{x}_i$ are linearly independent. Then $\mathbf{x}_{i+1}$ can be written as a linear combination of $\mathbf{x}_1, \ldots, \mathbf{x}_i$, i.e.,

$$\mathbf{x}_{i+1} = c_1\mathbf{x}_1 + \cdots + c_i\mathbf{x}_i. \tag{21.1}$$

Multiplying A on both sides of the above equation, we get

$$\lambda_{i+1}\mathbf{x}_{i+1} = c_1\lambda_1\mathbf{x}_1 + \cdots + c_i\lambda_i\mathbf{x}_i.$$

Substituting $\mathbf{x}_{i+1}$ in (21.1), we obtain

$$c_1(\lambda_1 - \lambda_{i+1})\mathbf{x}_1 + \cdots + c_i(\lambda_i - \lambda_{i+1})\mathbf{x}_i = \mathbf{0}.$$

As $\mathbf{x}_1, \ldots, \mathbf{x}_i$ are linearly independent,

$$c_1(\lambda_1 - \lambda_{i+1}) = \cdots = c_i(\lambda_i - \lambda_{i+1}).$$

As $\lambda_1, \lambda_2, \ldots, \lambda_k$ are distinct, we further have $c_1 = \cdots = c_i = 0$, which implies $\mathbf{x}_{i+1} = 0$, a contradiction to that $\mathbf{x}_{i+1}$ is an eigenvector. ■

So if an $n \times n$ matrix A has n distinct eigenvalues, then A is diagonalizable. How about A has duplicated eigenvalues?

■ **Example 21.1** Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Find the eigenvalues and the corresponding eigenvectors of A. ■

■ **Example 21.2** Let

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 1 & 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 4 & 0 \\ -3 & 6 & 2 \end{bmatrix}.$$

A and B have the same eigenvalues:

$$\lambda_1 = 4, \lambda_2 = \lambda_3 = 2.$$

Check their eigenvectors, respectively. ■

Theorem 21.3 Let A be an $n \times n$ matrix whose distinct eigenvalues are $\lambda_1, \ldots, \lambda_p$.

1. For $1 \le k \le p$, the dimension of the eigenspace for λ_k is less than or equal to the multiplicity of λ_k.
2. The matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces equals n.
3. If A is diagonalizable and $\mathcal{B}_k$ is a basis for the eigenspace corresponding to λ_k for each k, then the total collection of vectors in the sets $\mathcal{B}_1, \ldots, \mathcal{B}_p$ forms an eigenvector basis for $\mathbb{R}^n$.

21.2 Applications

If $A = XDX^{-1}$ then $A^k = XD^kX^{-1}$.

■ **Example 21.3** Let

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}.$$

Compute A^8. ■

Fibonacci Numbers

- The Fibonacci sequence $\{F_n\}$ is defined as

$$F_0 = 0, F_1 = 1, \text{ and } F_n = F_{n-1} + F_{n-2}, n \geq 2.$$

- We can write a recursive matrix equation

$$\begin{bmatrix} F_n \\ F_{n-1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_{n-1} \\ F_{n-2} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^{n-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

- We have the diagnolization $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = S \begin{bmatrix} \lambda & 0 \\ 0 & -\lambda^{-1} \end{bmatrix} S^{-1}$, where

$$\lambda = \frac{1+\sqrt{5}}{2}, \quad S = \begin{bmatrix} 1 & -1 \\ \lambda^{-1} & \lambda \end{bmatrix}, \text{ and } S^{-1} = \frac{1}{\lambda + \lambda^{-1}} \begin{bmatrix} \lambda & 1 \\ -\lambda^{-1} & 1 \end{bmatrix}.$$

With the decomposition, we have

$$\begin{aligned} \begin{bmatrix} F_n \\ F_{n-1} \end{bmatrix} &= S \begin{bmatrix} \lambda^{n-1} & 0 \\ 0 & (-\lambda^{-1})^{n-1} \end{bmatrix} S^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= \frac{1}{\lambda + \lambda^{-1}} S \begin{bmatrix} \lambda^{n-1} & 0 \\ 0 & (-\lambda^{-1})^{n-1} \end{bmatrix} \begin{bmatrix} \lambda \\ -\lambda^{-1} \end{bmatrix} \\ &= \frac{1}{\lambda + \lambda^{-1}} \begin{bmatrix} \lambda^n - (-\lambda^{-1})^n \\ \lambda^{n-1} - (-\lambda^{-1})^{n-1} \end{bmatrix}. \end{aligned}$$

So, we have $F_n = \frac{\lambda^n - (-\lambda^{-1})^n}{\lambda + \lambda^{-1}}$.

21.3 Complex Inner Product Spaces

Let $\mathbb{C}$ be the set of all complex numbers and $\mathbb{C}^n$ be the vector space of all n-tuples of complex numbers.

Complex Inner Product

Definition 21.2 An *inner product* on a vector space $\mathcal{V}$ with complex scalars is a mapping that assigns each pair of vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathcal{V}$ a *complex number* $\langle \mathbf{x}, \mathbf{y} \rangle$ satisfying the following conditions:

1. (positive-definite) $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ with equality if and only if $\mathbf{x} = 0$.
2. (conjugate symmetry) $\langle \mathbf{x}, \mathbf{y} \rangle = \overline{\langle \mathbf{y}, \mathbf{x} \rangle}$ for all $\mathbf{x}, \mathbf{y} \in \mathcal{V}$.

3. (linearity) $\langle \alpha\mathbf{x} + \beta\mathbf{y}, \mathbf{z}\rangle = \alpha\langle \mathbf{x}, \mathbf{z}\rangle + \beta\langle \mathbf{y}, \mathbf{z}\rangle$ for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{V}$ and all scalars α and β.

For a complex inner product space, if we make proper modifications, the theorems proved in Lecture 16, 17 for general inner product spaces can also hold.

Theorem 21.4 If $\mathbf{x}$ and $\mathbf{y}$ are vectors in a complex inner product space, then

$$|\langle \mathbf{x}, \mathbf{y}\rangle| \leq \|\mathbf{x}\|\|\mathbf{y}\|$$

with equality holding if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent.

proof outline. For a scalar α, write

$$\begin{aligned}\|\mathbf{x} - \alpha\mathbf{y}\|^2 &= \langle \mathbf{x} - \alpha\mathbf{y}, \mathbf{x} - \alpha\mathbf{y}\rangle \\ &= \langle \mathbf{x}, \mathbf{x} - \alpha\mathbf{y}\rangle - \alpha\langle \mathbf{y}, \mathbf{x} - \alpha\mathbf{y}\rangle \\ &= \overline{\langle \mathbf{x} - \alpha\mathbf{y}, \mathbf{x}\rangle} - \alpha\overline{\langle \mathbf{x} - \alpha\mathbf{y}, \mathbf{y}\rangle} \\ &= \|\mathbf{x}\|^2 - \alpha\overline{\langle \mathbf{x}, \mathbf{y}\rangle} - \bar{\alpha}\langle \mathbf{x}, \mathbf{y}\rangle + \alpha\bar{\alpha}\|\mathbf{y}\|^2.\end{aligned}$$

Let $\alpha = \langle \mathbf{x}, \mathbf{y}\rangle / \|\mathbf{y}\|$. ■

Moreover, If $\{w_1, \ldots, w_n\}$ is an orthonormal basis and $\mathbf{z} = \sum_{i=1}^n c_i\mathbf{w}_i$,

$$c_i = \langle \mathbf{z}, \mathbf{w}_i\rangle, \ \bar{c}_i = \langle \mathbf{w}_i, \mathbf{z}\rangle \ \text{ and } \ \|\mathbf{z}\|^2 = \sum_{i=1}^n c_i\bar{c}_i.$$

We write $\mathbf{z}^H$ for the transpose of $\bar{\mathbf{z}}$ in $\mathbb{C}^n$, i.e., $\mathbf{z}^H = \bar{\mathbf{z}}^{\mathrm{T}}$. Define an inner product on $\mathbb{C}^n$ by

$$\langle \mathbf{x}, \mathbf{y}\rangle = \mathbf{y}^H\mathbf{x}.$$

- $\langle \mathbf{y}, \mathbf{x}\rangle = \mathbf{x}^H\mathbf{y} = \overline{\mathbf{y}^H\mathbf{x}} = \overline{\langle \mathbf{x}, \mathbf{y}\rangle}$
- $\|\mathbf{y}\| = \sqrt{\langle \mathbf{y}, \mathbf{y}\rangle} = \sqrt{\mathbf{y}^H\mathbf{y}}$.

■ **Example 21.4** For

$$\mathbf{z} = \begin{bmatrix} 4+i \\ 1-2i \end{bmatrix} \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} 2+2i \\ -1+3i \end{bmatrix},$$

evaluate $\mathbf{w}^H\mathbf{z}$, $\mathbf{z}^H\mathbf{z}$, $\mathbf{w}^H\mathbf{w}$, $\|\mathbf{w}\|$ and $\|\mathbf{z}\|$. ■

Lecture 22

Hermitian Matrices

22.1 Hermitian Matrices

Complex Matrices

- Let $M = (m_{ij})$ with $m_{ij} = a_{ij} + ib_{ij}$.
- We may write
$$M = A + iB, \quad \overline{M} = A - iB.$$
- Denote
$$M^H = \overline{M}^{\mathrm{T}}.$$
- Properties of complexity matrices
  - $(A^H)^H = A$
  - $(\alpha A + \beta B)^H = \bar{\alpha} A^H + \bar{\beta} B^H$
  - $(AC)^H = C^H A^H$

Definition 22.1 A matrix M is said to be *Hermitian* if $M = M^H$.

Theorem 22.1 The eigenvalues of a Hermitian matrix are *real*. Moreover, eigenvectors belonging to distinct eigenvalues are orthogonal.

Note that for a Hermitian matrix $M = A + iB$, we have

$$A^T = A \quad \text{and} \quad B^T == B,$$

i.e., A is symmetric and B is skew symmetric.

Proof. Let λ be an eigenvalue of a Hermitian matrix A and let $\mathbf{x}$ be an eigenvector belonging to λ. As

$$\mathbf{x}^H A\mathbf{x} = \mathbf{x}^H \lambda \mathbf{x} = \lambda \|\mathbf{x}\|^2,$$

we have $\lambda = \frac{\mathbf{x}^H A\mathbf{x}}{\|\mathbf{x}\|^2}$. By

$$\overline{\mathbf{x}^H A\mathbf{x}} = (\mathbf{x}^H A\mathbf{x})^H = \mathbf{x}^H A^H \mathbf{x} = \mathbf{x}^H A\mathbf{x},$$

and hence $\bar{\lambda} = \lambda$, i.e., λ is a real number.

Let $\mathbf{x}_1$ and $\mathbf{x}_2$ are eigenvectors belonging to distinct eigenvalues λ_1 and λ_2. We have

$$\mathbf{x}_1^H A\mathbf{x}_2 = \lambda_2 \mathbf{x}_1^H \mathbf{x}_2$$
$$\mathbf{x}_1^H A\mathbf{x}_2 = \mathbf{x}_1^H A^H \mathbf{x}_2 = (\mathbf{x}_2^H A\mathbf{x}_1)^H = \lambda_1 \mathbf{x}_1^H \mathbf{x}_2.$$

Hence, $\lambda_2 \mathbf{x}_1^H \mathbf{x}_2 = \lambda_1 \mathbf{x}_1^H \mathbf{x}_2$. As $\lambda_1 \neq \lambda_2$, $\langle \mathbf{x}_2, \mathbf{x}_1 \rangle = \mathbf{x}_1^H \mathbf{x}_2 = 0$, i.e., $\mathbf{x}_1$ and $\mathbf{x}_2$ are orthogonal. ■

Definition 22.2 An $n \times n$ matrix is said to be *unitary* if its column vectors from an orthonormal set in $\mathbb{C}^n$.

Corollary 22.2 If the eigenvalues of a Hermitian matrix A are distinct, then there exists a unitary matrix U that diagonalizes A.

■ **Example 22.1** Find a unitary matrix that diagonalizes

$$A = \begin{bmatrix} 2 & 1-i \\ 1+i & 1 \end{bmatrix}.$$

■

Suppose a Hermitian matrix has eigenvalues $\lambda_1 = \lambda_2 = 1$. If we can find two linearly independent eigenvalues $\mathbf{x}_1$ and $\mathbf{x}_2$ belonging to 1, we perform Gram-Schmidt orthogonalization and obtain $\mathbf{u}_1$ and $\mathbf{u}_2$. We can argue that $\mathbf{u}_1$ and $\mathbf{u}_2$ are also eigenvectors belonging to 1 as $\text{Span}\{\mathbf{u}_1, \mathbf{u}_2\} = \text{Span}\{\mathbf{x}_1, \mathbf{x}_2\}$ is the eigenspace of 1. Then the union of all such orthonormal basis of all the eigen spaces are also orthonormal.

For a general matrix, the dimension of the eigenspace of an eigenvalue can be less than the multiplicity of the eigenvalue. We show using another approach that a Hermitian matrix can be diagonalized by an unitary matrix, and hence has the dimension of the eigenspace of an eigenvalue same the multiplicity of the eigenvalue.

22.2 Spectral Theorem

Theorem 22.3 — Schur's Theorem. For each $n \times n$ matrix A, there exists a unitary matrix U such that $U^H AU$ is upper triangular.

Proof. Prove by induction. When $n = 1$, $I^H AI = A$ is upper triangular.

Suppose that the theorem holds for $k \times k$ matrices. Let A be a $(k+1) \times (k+1)$ matrix with an eigenvalue λ_1 and a corresponding *unit* eigenvector $\mathbf{x}_1$. Extend $\mathbf{x}_1$ to a set of orthonormal basis $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{k+1}\}$ of $\mathbb{C}^{k+1}$. Then $\mathbf{X}$ is unitary. The first column of $X^H AX$ is

$$X^H A\mathbf{x}_1 = X^H \lambda_1 \mathbf{x}_1 = \lambda_1 \mathbf{e}_1.$$

Thus, $X^H AX$ is of the form

$$\begin{bmatrix} \lambda_1 & \vec{\mathbf{a}} \\ \mathbf{0} & M \end{bmatrix}$$

where M is the submatrix obtained by remove the first row and the first column of $X^H AX$.

By the induction hypothesis, there exists a unitary matrix V_1 such that $V_1^H M V_1$ is an upper triangular matrix. Let

$$V = \begin{bmatrix} 1 & \\ & V_1 \end{bmatrix} \quad \text{and} \quad U = XV.$$

The matrix U is unitary as

$$U^H U = (XV)^H XV = V^H X^H XV = V^H V = I.$$

To complete the proof, observe that

$$U^H AU = V^H (X^H AX) V = \begin{bmatrix} 1 & \\ & V_1^H \end{bmatrix} \begin{bmatrix} \lambda_1 & \vec{\mathbf{a}} \\ \mathbf{0} & M \end{bmatrix} \begin{bmatrix} 1 & \\ & V_1 \end{bmatrix} = \begin{bmatrix} \lambda_1 & \vec{\mathbf{a}} V_1 \\ \mathbf{0} & V_1^H M V_1 \end{bmatrix}$$

is upper triangular. ■

Theorem 22.4 — Spectral Theorem. If A is Hermitian, then there exists a unitary matrix U that diagonalizes A.

Proof. By Schur's theorem, there exists a unitary matrix such that $T = U^H AU$ is upper triangular. Moreover,

$$T^H = (U^H AU)^H = U^H A^H U = U^H AU = T.$$

Therefore, T is Hermitian and hence diagonal. ■

Diagonalization of Hermitian Matrices

- By the spectral theorem, each Hermitial matrix A can be factored into UDU^H where U is unitary and D is diagonal.
  - Find all distinct eigenvalues $\lambda_1, \dots \lambda_p$ of A.
  - Find an orthonormal basis $\mathcal{B}_k$ for each eigenvalue λ_k.
  - Use all the vectors in $\cup_k \mathbf{B}_k$ as columns to form U, and Form D using the corresponding eigenvalues as the diagonal entries.
- As the columns of U forms an orthonormal basis of $\mathbb{C}^n$, we can write $\mathbf{x} \in \mathbb{C}^n$ as
$$\mathbf{x} = c_1 \mathbf{u}_1 + \cdots + c_n \mathbf{u}_n$$
where $c_i = \langle \mathbf{x}, \mathbf{u}_i \rangle = \mathbf{u}_i^H \mathbf{x}$.
- The operation of A on $\mathbf{x}$ is $A\mathbf{x} = c_1 \lambda_1 \mathbf{u}_1 + \cdots + c_n \lambda_n \mathbf{u}_n$.

22.3 Real Symmetric Matrices

If A is a real symmetric matrix, the spectral theorem can only show that there exists a unitary matrix that diagonalize A. We would like to show a stronger result that A can be diagonalized by a *real* unitary matrix, i.e., an orthogonal matrix.

Theorem 22.5 Every real symmetric matrix A can be factored into $U\Lambda U^T$ where U is an orthogonal matrix and Λ is a real diagonal matrix.

- The diagonal entries λ_i of Λ are eigenvalues of A.
- The columns $\mathbf{u}_i$ of U are eigenvectors belonging to λ_i, respectively.

We can also write

$$\begin{aligned} A &= \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \begin{bmatrix} \mathbf{u}_1^T \\ \vdots \\ \mathbf{u}_n^T \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{bmatrix} \begin{bmatrix} \lambda_1 \mathbf{u}_1^T \\ \vdots \\ \lambda_n \mathbf{u}_n^T \end{bmatrix} \\ &= \sum_{i=1}^{n} \lambda_i \mathbf{u}_i \mathbf{u}_i^T \end{aligned}$$

Towards proving the above theorem, we can use a special Schur decomposition for real symmetric matrix. We first emphasize the following fact. For a real $n \times n$ matrix A with a real eigenvalue λ, the eigenspace, i.e., $\text{Null}(A - \lambda I)$ has a real basis, though we can regard $\text{Null}(A - \lambda I)$ as a subspace of $\mathbb{C}^n$ instead of $\mathbb{R}^n$.

Proof. We prove by induction. When $n = 1$, $I^{\mathrm{T}} A I = A$ is upper triangular. Suppose that the theorem holds for $k \times k$ matrices. Let A be a $(k+1) \times (k+1)$ real matrix with an eigenvalue λ_1, which is real by Theorem 22.1, and a corresponding unit real eigenvector $\mathbf{x}_1$. Extend $\mathbf{x}_1$ to a set of orthonormal basis $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{k+1}\}$ of $\mathbb{R}^{k+1}$. Then $\mathbf{X}$ is orthogonal. The first column of $X^{\mathrm{T}} A X$ is

$$X^{\mathrm{T}} A \mathbf{x}_1 = X^{\mathrm{T}} \lambda_1 \mathbf{x}_1 = \lambda_1 \mathbf{e}_1.$$

Due to symmetry, $X^{\mathrm{T}} A X$ is of the form

$$\begin{bmatrix} \lambda_1 & \vec{\mathbf{0}} \\ \mathbf{0} & M \end{bmatrix}$$

where M is the real symmetry submatrix obtained by remove the first row and the first column of $X^{\mathrm{T}} A X$.

By the induction hypothesis, there exists an orthogonal matrix V_1 such that $V_1^{\mathrm{T}} M V_1$ is a real diagonal matrix. Let

$$V = \begin{bmatrix} 1 & \\ & V_1 \end{bmatrix} \quad \text{and} \quad U = XV.$$

The matrix U is orthogonal as

$$U^{\mathrm{T}} U = (XV)^{\mathrm{T}} XV = V^{\mathrm{T}} X^{\mathrm{T}} XV = V^{\mathrm{T}} V = I.$$

To complete the proof, observe that

$$U^{\mathrm{T}} A U = V^{\mathrm{T}} (X^{\mathrm{T}} A X) V = \begin{bmatrix} 1 & \\ & V_1^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} \lambda_1 & \vec{\mathbf{0}} \\ \mathbf{0} & M \end{bmatrix} \begin{bmatrix} 1 & \\ & V_1 \end{bmatrix} = \begin{bmatrix} \lambda_1 & \vec{\mathbf{0}} \\ \mathbf{0} & V_1^{\mathrm{T}} M V_1 \end{bmatrix}$$

is upper triangular. ∎

22.4 Normal Matrix

There are non-Hermitian matrices that are diagonalizable by unitary matrices. In general, if $A = UDU^H$ where U is unitary and D is diagonal, the diagonal entries of D may be complex and $D^H \neq D$. However,

$$AA^H = UDU^HUD^HU^H = U(DD^H)U^H$$
$$A^HA = UD^HU^HUDU^H = U(D^HD)U^H.$$

Since $D^HD = DD^H$, we have $AA^H = A^HA$.

Definition 22.3 A matrix A is said to be *normal* if $AA^H = A^HA$.

Examples:
- unitary, Hermitian, and skew Hermitian
- orthogonal, symmetric, and skew symmetric

Matrix

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

is normal, but is not in the above classes.

Theorem 22.6 A matrix A is diagonalizable by a unitary matrix if and only if A is normal.

Proof. We have shown the necessary condition and now prove sufficiency. By Schur's theorem, there exists a unitary matrix U such that $T = U^HAU$ is upper triangular. Write

$$T^HT = U^HA^HUU^HAU = U^HA^HAU$$
$$TT^H = U^HAUU^HA^HU = U^HAA^HU$$

Since $A^HA = AA^H$, we have $T^HT = TT^H$ (i.e., T is normal). The proof is completed by noting that a normal triangular matrix is diagonal. ■

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Lecture 23

Singular Value Decomposition

Properties of $A^{\mathrm{T}}A$

Let A be any $m \times n$ matrix A of real numbers.

- $A^{\mathrm{T}}A$ is symmetric.
- $A^{\mathrm{T}}A$ is diagonalizable by an orthogonal matrix, and the eigenvalues of $A^{\mathrm{T}}A$ are real.
- $\mathrm{Null}(A^{\mathrm{T}}A) = \mathrm{Null}(A)$.
- $\mathrm{rank}(A) = \mathrm{rank}(A^{\mathrm{T}}A)$.

Lemma 23.1 The eigenvalues of $A^{\mathrm{T}}A$ are nonnegative.

Proof. Let λ be an eigenvalue of $A^{\mathrm{T}}A$ and $\mathbf{x}$ be an eigenvector belonging to λ. Then

$$\|A\mathbf{x}\|^2 = \mathbf{x}^{\mathrm{T}}A^{\mathrm{T}}A\mathbf{x} = \lambda\mathbf{x}^{\mathrm{T}}\mathbf{x} = \lambda\|\mathbf{x}\|^2.$$

Hence, $\lambda = \frac{\|A\mathbf{x}\|^2}{\|\mathbf{x}\|^2} \geq 0$. ■

Let $\mathbf{v}_1, \ldots \mathbf{v}_r$ be an orthonormal basis of a subspace $\mathcal{V}$ of $\mathbb{R}^n$.

- For any vector $\mathbf{a} \in \mathcal{V}$, we have

$$\mathbf{a} = \sum_i (\mathbf{v}_i^{\mathrm{T}}\mathbf{a})\mathbf{v}_i = V \begin{bmatrix} \mathbf{v}_1^{\mathrm{T}}\mathbf{a} \\ \vdots \\ \mathbf{v}_r^{\mathrm{T}}\mathbf{a} \end{bmatrix} = VV^{\mathrm{T}}\mathbf{a}.$$

- For any matrix A with $\mathrm{Col}(A) \subset \mathcal{V}$,

$$A = VV^{\mathrm{T}}A.$$

23.1 Singular Value Decomposition

Definition 23.1 The *singular-value decomposition (SVD)* of an $m \times n$ matrix A of real numbers is a factorization of the form $U\Sigma V^{\mathrm{T}}$ where

- U is an $m \times m$ orthogonal matrix;
- V is an $n \times n$ orthogonal matrix;
- Σ is an $m \times n$ matrix whose off-diagonal entries are all 0's, and whose

diagonal entries σ_i, $i = 1, \ldots, n$, called the *singular values* satisfy

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0.$$

Here we use the convention that singular values $\sigma_k = 0$ if $k > \min\{m, n\}$.

Proposition 23.2 If $A = U\Sigma V^{\mathrm{T}}$, then the rank of A is equal to the number of *nonzero singular values*.

- Let r be the number of nonzero singular values of A.
- Let U_r and V_r be the first r columns of U and V, respectively.
- Let $\Sigma_r = \mathrm{diag}(\sigma_r, \ldots, \sigma_r)$.

Proof. We have

$$A = U_r \Sigma_r V_r^{\mathrm{T}}. \tag{23.1}$$

As the column space of A is a subset of that of U_r, we have $\mathrm{rank}(A) \leq \mathrm{rank}(U_r) = r$. Eq (23.1) also implies

$$AV_r\Sigma_r^{-1} = U_r.$$

As the column space of U_r is a subset of that of A, we have $r = \mathrm{rank}(U_r) \leq \mathrm{rank}(A)$. ■

Eq (23.1) is also called the compact form of SVD, which also implies the following properties when SVD exists.

Proposition 23.3 If an $m \times n$ matrix A has a singular-value decomposition $U\Sigma V^{\mathrm{T}}$.

- $\mathbf{u}_1, \ldots, \mathbf{u}_r$ form an orthogonal basis of $\mathrm{Col}(A)$.
- $\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m$ form an orthogonal basis of $\mathrm{Null}(A^{\mathrm{T}})$.
- $\mathbf{v}_1, \ldots, \mathbf{v}_r$ form an orthogonal basis of $\mathrm{Row}(A)$, which implies $AV_rV_r^{\mathrm{T}} = A$.
- $\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n$ form an orthogonal basis of $\mathrm{Null}(A)$.

Proposition 23.4 If an $m \times n$ matrix A has a singular-value decomposition $U\Sigma V^{\mathrm{T}}$.

- $\sigma_i = \sqrt{\lambda_i}$, $i = 1, \ldots, n$, where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0$ are eigenvalues of $A^{\mathrm{T}}A$.
- V diagonalizes $A^{\mathrm{T}}A$, and hence $\mathbf{v}_j$'s are eigenvectors of $A^{\mathrm{T}}A$.
- U is formed by

$$\mathbf{u}_j = \frac{1}{\sigma_j} A\mathbf{v}_j, \quad j = 1, \ldots, r = \mathrm{rank}(A)$$
$$A^{\mathrm{T}}\mathbf{u}_j = \mathbf{0}, \quad j = r+1, \ldots, m.$$

Proof. If $A = U\Sigma V^{\mathrm{T}}$, we have

$$AA^{\mathrm{T}} = U\Sigma\Sigma^{\mathrm{T}}U^{\mathrm{T}},$$

which implies the first two claims.

To show the third claim, we write $AV = U\Sigma$, i.e.,

$$A\mathbf{v}_j = \sigma_j \mathbf{u}_j, j = 1, \ldots, m.$$

The proof is completed by $\sigma_j > 0$ for $j = 1, \ldots, r$ (Proposition 23.2), and $\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m$ forming an orthogonal basis of $\mathrm{Null}(A^{\mathrm{T}})$ (Proposition 23.3). ■

Theorem 23.5 SVD exists for any real matrix.

Proof. Assume $m \geq n$ so that $\min\{m, n\} = n$ to simplify the notations. The case with $m < n$ can be proved similarly. Let $r = \text{rank}(A) = \text{rank}(A^{\mathrm{T}}A)$.

As $A^{\mathrm{T}}A$ is symmetric, its eigenvalues λ_i's are real and is diagonalizable by an orthogonal matrix V. Moreover, λ_i's are nonnegative. Assume that the columns of V have been properly ordered so that

$$\lambda_1 \geq \cdots \geq \lambda_r \geq \lambda_{r+1} = \cdots = \lambda_n = 0.$$

For $i = 1, \ldots, r$, let $\sigma_i = \sqrt{\lambda_i}$.

Let V_r be the first r columns of V and let $V_r^{\perp}$ the other columns. For $j > r$, as $A^{\mathrm{T}}A\mathbf{v}_j = \lambda_j \mathbf{v}_j = \mathbf{0}$,

$$\text{Col}(V_r^{\perp}) = \text{Null}(A^{\mathrm{T}}A) = \text{Null}(A).$$

Hence, $\text{Col}(V_r) = \text{Col}(V_r^{\perp})^{\perp} = \text{Null}(A)^{\perp} = \text{Col}(A^{\mathrm{T}})$. Therefore,

$$A = AV_rV_r^{\mathrm{T}}. \tag{23.2}$$

Let $\Sigma_r = \text{diag}(\sigma_1, \ldots, \sigma_r)$. We need to find a matrix U_r with orthogonal columns such that $A = U_r\Sigma_rV_r^{\mathrm{T}}$, by (23.2), which is equivalent to

$$AV_r = U_r\Sigma_r. \tag{23.3}$$

To satisfy (23.3), let

$$U_r = AV_r\Sigma_r^{-1}.$$

To verify that the columns of U_r are orthogonal, we write

$$U_r^{\mathrm{T}}U_r = \Sigma_r^{-1}V_r^{\mathrm{T}}A^{\mathrm{T}}AV_r\Sigma_r^{-1} = \Sigma_r^{-1}V_r^{\mathrm{T}}V_r\Sigma_r^2\Sigma_r^{-1} = \Sigma_r^{-1}\Sigma_r^2\Sigma_r^{-1} = I.$$

To complete the proof, we extend the columns of U_r to an orthonormal basis of $\mathbb{R}^m$. ■

■ **Example 23.1** Compute the SVD of $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{bmatrix}$.

The matrix $A^{\mathrm{T}}A = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ has eigenvalues $\lambda_1 = 4$ and $\lambda_2 = 0$, and can be diagonalized by the orthogonal matrix $V = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. So $\sigma_1 = 2$ and $\sigma_2 = 0$.

The first column of U is

$$\mathbf{u}_1 = \frac{1}{\sigma_1}A\mathbf{v}_1.$$

The other two columns of U form an orthonormal basis of $\text{Null}(A^{\mathrm{T}})$. ■

Similar to diagonalization, a matrix $A = U\Sigma V^T$ can be written as the outer product expansion

$$\begin{aligned} A &= U_r \Sigma_r V_r^T \\ &= \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_r \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^{\mathrm{T}} \\ \vdots \\ \mathbf{v}_r^{\mathrm{T}} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_r \end{bmatrix} \begin{bmatrix} \sigma_1 \mathbf{v}_1^{\mathrm{T}} \\ \vdots \\ \sigma_r \mathbf{v}_r^{\mathrm{T}} \end{bmatrix} \\ &= \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^T \end{aligned}$$

23.2 Low Rank Approximation

Matrix Norm

- Matrix norm $\|A\|_F = \sqrt{\sum_{i,j} |a_{ij}|^2}$, which is also called *Frobenius* norm.
- If $A = U\Sigma V^{\mathrm{T}}$, then $\|A\|^2 = \sigma_1^2 + \cdots + \sigma_n^2$.

Lemma 23.6 If Q is an orthogonal matrix, then $\|QA\|_F = \|A\|_F$.

Exercise 23.1 Prove the lemma. ∎

Low rank approximation problem:
For a fixed $m \times n$ matrix A and an integer k, solve

$$\min_{\mathrm{rank}(S) \le k} \|A - S\|_F. \tag{23.4}$$

Theorem 23.7 Let $A = U\Sigma V^T$ be an $m \times n$ matrix and let $A_k = U\Sigma_k V^T$ where Σ_k is same as Σ except that the (j, j), $j = k+1, \ldots$ entries are 0. Then

$$\min_{\mathrm{rank}(S) \le k} \|A - S\|_F = \|A - A_k\|_F = (\sigma_{k+1}^2 + \cdots + \sigma_n^2)^{1/2}.$$

In other words, A_k is the best rank k approximation of A in Frobenius norm.

R The theorem above assumes that the minimum value of (23.4) is achievable, which is a fact not proved here.

Proof. Let $X = Q\Omega P^{\mathrm{T}}$ be a matrix, and let $B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = Q^{\mathrm{T}}AP$, where B_{11} is the upper-left $k \times k$ sub-matrix of B. We have

$$\|A - X\|^2 = \|Q^{\mathrm{T}}AP - \Omega\|^2 = \|B - \Omega\|^2 = \|B_{11} - \Omega_k\|^2 + \|B_{12}\|^2 + \|B_{21}\|^2 + \|B_{22}\|^2.$$

We claim the following necessary condition such that X achieves the minimum of (23.4): First, $B_{12} = \mathbf{0}$. Otherwise, let

$$Y = Q \begin{bmatrix} B_{11} & B_{12} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} P^{\mathrm{T}}$$

and we have

$$\|A - Y\|^2 = \|B_{21}\|^2 + \|B_{22}\|^2 < \|A - X\|^2.$$

Second, we can similarly obtain $B_{21} = \mathbf{0}$. Third, $B_{11} = \Omega_k$. Otherwise, let

$$Z = Q \begin{bmatrix} B_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} P^{\mathrm{T}}$$

and we get

$$\|A - Z\|^2 = \|B_{22}\|^2 < \|B_{11} - \Omega_k\|^2 + \|B_{22}\|^2 = \|A - X\|^2.$$

Let $B_{22} = U_{22}\Lambda V_{22}^{\mathrm{T}}$, a SVD. Now we have $\|A - X\| = \|B_{22}\| = \|\Lambda\|$. Let

$$U' = \mathrm{diag}(I, U_{22}), \quad V' = \mathrm{diag}(I, V_{22}).$$

Then

$$A = QBP^{\mathrm{T}} = Q \begin{bmatrix} \Omega_k & \mathbf{0} \\ \mathbf{0} & U_{22}\Lambda V_{22}^{\mathrm{T}} \end{bmatrix} P^{\mathrm{T}} = QU' \begin{bmatrix} \Omega_k & \mathbf{0} \\ \mathbf{0} & \Lambda \end{bmatrix} (PV')^{\mathrm{T}},$$

i.e., the diagonal entries of Λ are singular values of A (due to the uniqueness of singular values). Hence,

$$\|A - X\| = \|\Lambda\| \geq (\sigma_{k+1}^2 + \cdots + \sigma_n^2)^{1/2}.$$

The proof is completed by observing that the equality is achieved by $X = A_k$. ■

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Lecture 24

Quadratic Forms

Matrices also play an important role in quadratic equations.

24.1 Quadratic Form

Quadratic Form

- A function $Q : \mathbb{R}^n \to \mathbb{R}$ is called a *quadratic form* if there exists a *symmetric* matrix A such that $Q(\mathbf{x}) = \mathbf{x}^{\mathrm{T}} A\mathbf{x}$.
- When $n = 1$,
$$Q(x) = ax^2$$
- When $n = 2$,
$$Q(x_1, x_2) = ax_1^2 + 2bx_1x_2 + cx_2^2$$
- When $n = 3$,
$$Q(x_1, x_2, x_3) = ax_1^2 + bx_2^2 + cx_3^2 + 2dx_1x_2 + 2ex_1x_3 + 2fx_2x_3$$

Note that if A is square, but not symmetric, $\mathbf{x}^{\mathrm{T}} A\mathbf{x}$ is also a quadratic form as

$$\mathbf{x}^{\mathrm{T}} A\mathbf{x} = \frac{\mathbf{x}^{\mathrm{T}} A\mathbf{x} + \mathbf{x}^{\mathrm{T}} A^{\mathrm{T}}\mathbf{x}}{2} = \mathbf{x}^{\mathrm{T}} \frac{A + A^{\mathrm{T}}}{2}\mathbf{x}$$

where we use $\mathbf{x}^{\mathrm{T}} A\mathbf{x} = (\mathbf{x}^{\mathrm{T}} A\mathbf{x})^{\mathrm{T}} = \mathbf{x}^{\mathrm{T}} A^{\mathrm{T}}\mathbf{x}$.

Norm is a quadratic form as $\|\mathbf{x}\|^2 = \mathbf{x}^{\mathrm{T}} I\mathbf{x}$.

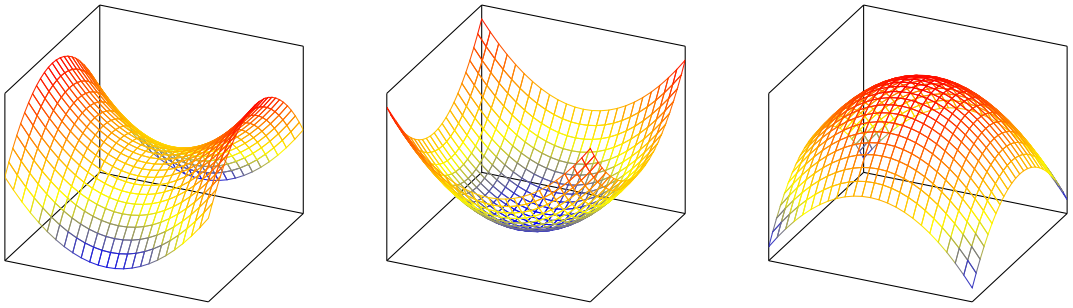
■ **Example 24.1** Write $Q(\mathbf{x}) = 5x_1^2 + 3x_2^2 + 4x_3^2 - 2x_1x_2 + 6x_2x_3$ as $\mathbf{x}^{\mathrm{T}} A\mathbf{x}$. ■

24.2 Change of Variables

Quadratic forms are easier to evaluate when A is diagonal.

Change of Variables

- A change of variables is given by an equation $\mathbf{x} = P\mathbf{y}$, where P is an invertible matrix.
- Then, $\mathbf{x}^{\mathrm{T}} A\mathbf{x} = (P\mathbf{y})^{\mathrm{T}} A(P\mathbf{y}) = \mathbf{y}^{\mathrm{T}}(P^{\mathrm{T}} AP)\mathbf{y}$.
- The matrix of the new quadratic form is $P^{\mathrm{T}} AP$.



(a) $x_1^2 - x_2^2$ (b) $x_1^2 + x_2^2$ (c) $-x_1^2 - x_2^2$

Figure 24.1: Classification of quadratic forms

Principal Axes

- When A is symmetric, there exists an orthogonal matrix U and a diagonal matrix D such that $A = UDU^{\mathrm{T}}$.
- Consider the change of variables $\mathbf{x} = U\mathbf{y}$, where $\mathbf{y}$ can be regarded as the coordinate vector of $\mathbf{x}$ under the basis formed by the columns of U.
- Then $\mathbf{x}^{\mathrm{T}}A\mathbf{x} = \mathbf{y}^{\mathrm{T}}(U^{\mathrm{T}}AU)\mathbf{y} = \mathbf{y}^{\mathrm{T}}D\mathbf{y}$.

Theorem 24.1 If A is a real symmetric matrix, then there is a change of variables $\mathbf{x} = U\mathbf{y}$ such that $\mathbf{x}^{\mathrm{T}}A\mathbf{x} = \mathbf{y}^{\mathrm{T}}D\mathbf{y}$, where D is diagonal.

24.3 Classification

- A quadratic form $Q(\mathbf{x}) = \mathbf{x}^{\mathrm{T}}A\mathbf{x}$ is said to be
  - *positive definite* if $Q(\mathbf{x}) > 0$ for all $\mathbf{x} \neq 0$,
  - *negative definite* if $Q(\mathbf{x}) < 0$ for all $\mathbf{x} \neq 0$,
  - *indefinite* if $Q(\mathbf{x})$ takes both positive and negative values.
- A real symmetric matrix A is said to be positive definite (resp. negative definite, indefinite) if $\mathbf{x}^{\mathrm{T}}A\mathbf{x}$ is positive definite (resp. negative definite, indefinite).

There exist other quadratic forms, e.g., those that can takes non-negative values, but not positive definite.

Theorem 24.2 Let A be a real symmetric matrix. Then

- A is positive definite if and only if all its eigenvalues are positive.
- A is negative definite if and only if all its eigenvalues are negative.
- A is indefinite if and only if A has both positive and negative eigenvalues.

Proof. We first prove the case of positive definite. If A is positive definite and λ is an eigenvalue of A, then for any eigenvector $\mathbf{x}$ belonging to λ,

$$\mathbf{x}^{\mathrm{T}}A\mathbf{x} = \mathbf{x}^{\mathrm{T}}\lambda\mathbf{x} = \lambda\|\mathbf{x}\|^2.$$

Hence, $\lambda = \mathbf{x}^{\mathrm{T}}A\mathbf{x}/\|\mathbf{x}\|^2 > 0$.

Conversely, suppose that all the eigenvalues λ_i of A are positive. Consider the diagonlization $A = UDU^{\mathrm{T}}$ where U is orthogonal and $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$. Let

$y_i = \mathbf{x}^{\mathrm{T}}\mathbf{u}_i$. Then,

$$\mathbf{x}^{\mathrm{T}}A\mathbf{x} = \mathbf{x}^{\mathrm{T}}UDU^{\mathrm{T}}\mathbf{x} = (\mathbf{x}^{\mathrm{T}}U)D(\mathbf{x}^{\mathrm{T}}U)^{\mathrm{T}} = \sum_{i=1}^{n} y_i^2\lambda_i > 0.$$

The proof of the positive definite case is completed.

To show the negative definite case, we only need the fact that i) A is positive definite if and only if $-A$ is negative definite and ii) if λ is an eigenvalue of A then $-\lambda$ is an eigenvalue of $-A$.

The last case can be proved by considering the corresponding negative statements. ■

24.4 Positive Definite Matrices

Proposition 24.3 If A is a symmetric positive definite matrix, then i) A is nonsingular and ii) $\det(A) > 0$.

Proof. If A is singular, then $\lambda = 0$ must be an eigenvalue, which is a contradition to A is positive definite.

The second follows from $\det(A) = \lambda_1 \cdots \lambda_n > 0$. ■

Definition 24.1 Given an $n \times n$ matrix A, the *leading principle submatrix* of A of order r, denoted by A_r, is formed by deleting the last $n - r$ rows and columns of A.

Proposition 24.4 If A is a symmetric positive definite matrix, then the leading principal submatrices $A_1, \ldots, A_n$ of A are all positive definite.

Proof. Let $\mathbf{x}_r$ be any non-zero vector in $\mathbb{R}^r$. Since

$$\mathbf{x}_r^{\mathrm{T}}A_r\mathbf{x}_r = \begin{bmatrix}\mathbf{x}_r^{\mathrm{T}} & \vec{\mathbf{0}}\end{bmatrix} A \begin{bmatrix}\mathbf{x}_r \\ \mathbf{0}\end{bmatrix} > 0$$

we have A_r is positive definite. ■

Proposition 24.5 If A is a symmetric positive definite matrix, then A can be reduced to upper triangular form using only row operation III, and the pivot elements are all positive.

$$\begin{bmatrix} a_{11} & * & * & * \\ * & a_{22} & * & * \\ * & * & a_{33} & * \\ * & * & * & a_{44} \end{bmatrix} \to \begin{bmatrix} a_{11} & * & * & * \\ 0 & a_{22}^{(1)} & * & * \\ 0 & * & a_{33}^{(1)} & * \\ 0 & * & * & a_{44}^{(1)} \end{bmatrix}$$

$$\to \begin{bmatrix} a_{11} & * & * & * \\ 0 & a_{22}^{(1)} & * & * \\ 0 & 0 & a_{33}^{(2)} & * \\ 0 & 0 & * & a_{44}^{(2)} \end{bmatrix} \to \begin{bmatrix} a_{11} & * & * & * \\ 0 & a_{22}^{(1)} & * & * \\ 0 & 0 & a_{33}^{(2)} & * \\ 0 & 0 & 0 & a_{44}^{(3)} \end{bmatrix}$$

Note that row operation III does not change determinant.

- First $a_{11} = \det(A_1) > 0$. So a_{11} is the pivot of the first row.
- As $\det(A_2) = a_{11}a_{22}^{(1)}$, $a_{22}^{(1)} = \det(A_2)/\det(A_1) > 0$.
- As $\det(A_3) = a_{11}a_{22}^{(1)}a_{33}^{(2)}$, $a_{33}^{(2)} = \det(A_3)/\det(A_2) > 0$.
- As $\det(A_4) = a_{11}a_{22}^{(1)}a_{33}^{(2)}a_{44}^{(3)}$, $a_{44}^{(3)} = \det(A_4)/\det(A_3) > 0$.

■ **Example 24.2** Let $A = \begin{bmatrix} 4 & 2 & -2 \\ 2 & 10 & 2 \\ -2 & 2 & 5 \end{bmatrix}$. LU decomposition of A is

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -1/2 & 1/3 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 & -2 \\ 0 & 9 & 3 \\ 0 & 0 & 3 \end{bmatrix}.$$

■

Proposition 24.6 If A is a symmetric positive definite matrix, then A can be factored into a product LDL^{T}, where L is unit lower triangular and D is a diagonal matrix with positive diagonal entries.

Proof. By the previous proposition, we have $A = LU$ with L as a lower triangular matrix and U an upper triangular matrix with all its diagonal entries positive. Let D be the matrix formed by the diagonal entries of U and let $U' = D^{-1}U$. So we have $A = LDU' = U'^{\mathrm{T}}DL^{\mathrm{T}}$. As LDU decomposition of a nonsingular matrix is unique (why?), we have $U' = L^{\mathrm{T}}$. ■

Exercise 24.1 Show that for a nonsigular square matrix A, the decomposition $A = LDU$ is unique, where L is unit lower triangular and U is unit upper triangular. ■

Following the previous example,

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -1/2 & 1/3 & 1 \end{bmatrix} \begin{bmatrix} 4 & & \\ & 9 & \\ & & 3 \end{bmatrix} \begin{bmatrix} 1 & 1/2 & -1/2 \\ 0 & 1 & 1/3 \\ 0 & 0 & 1 \end{bmatrix}.$$

Proposition 24.7 If A is a symmetric positive definite matrix, then A can be factored into a product LL^{T}, where L is a lower triangular matrix with positive diagonal entries.

Proof. By the previous proposition, $A = LDL^{\mathrm{T}}$. Let D' be the diagonal matrix with positive diagonal entries such that $D'^2 = D$. We have $A = LD'D'^{\mathrm{T}}L^{\mathrm{T}} = (LD')(LD')^{\mathrm{T}}$. ■

Following the previous example,

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ -1 & 1 & \sqrt{3} \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & \sqrt{3} \end{bmatrix}.$$

Theorem 24.8 The following are equivalent for an $n \times n$ symmetric matrix A:

1. A is positive definite.
2. The leading principal submatrices $A_1, \ldots, A_n$ all have positive determinants.
3. A can be reduced to upper triangular form using only row operation III, and the pivot elements are all positive.
4. A can be factored into a product LL^{T}, where L is a lower triangular matrix with positive diagonal entries.
5. A can be factored into a product $B^{\mathrm{T}}B$ for certain nonsingular matrix B.

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